Phenomena gallery: the arithmetic of $H^*(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^n)$ over $\mathbb{F}_\ell$, $\mathbb{Z}/\ell^K$ and $\mathbb{Z}$, in twenty-two console snippets and the project's runs

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Abstract

This gallery shows, one phenomenon at a time, what the project computes about the cohomology of $\mathrm{SL}_2(\mathbb{Z})$ (and of $\Gamma_0(N)$) with coefficients in the symmetric powers Sym^n of the standard representation, together with its Hecke operators. It is arranged by mathematics, in the order in which one would teach it:

I. *Cohomology and its dimensions*: $\dim H^1 = 2 \dim S_k + 1$ (Eichler–Shimura), and where small primes break the count;

II. *Eisenstein and cuspidal classes*: the involution ι^* and the sign of the Eisenstein class, the boundary, the compactly supported cohomology;

III. *Congruences modulo ℓ*: Ramanujan's 691 as an Eisenstein ideal, irregular primes and Herbrand–Ribet, visible and hidden congruences, a dihedral residual representation, multiplicity two, the θ-operator, Loewy length 4 — and what the congruence census found;

IV. *Level $\Gamma_0(N)$ and characters*: old forms, nebentypus, Mazur's Eisenstein ideal at $p = 2, 3$;

V. *Hida theory*: ordinary parts constant along a weight class, control modulo ℓ^m, to $n = 6910$ at $\ell = 691$;

VI. *Integral cohomology and torsion*: H^1 over $\mathbb{Z}$ by the integral descent, its torsion primes, the exponent law $\beta(n, \ell)$ (Callegaro–Cohen–Salvetti), checked to $n = 10^4$, a first non-split extension of Hecke modules between torsion and free part, and the Eisenstein congruence primes 691 and 5 read from the Hecke operators over $\mathbb{Z}$.

Each phenomenon is stated in a few sentences, cited to the binding contract [C] (and [E], [U]) by label, shown by the exact command and its *real* output — every listing and output is included from the files themselves and regenerated by a script, so the document and the program cannot diverge — and followed by what to look for in that output. Phenomena that need hours and tens of gigabytes are reported from the runs, with their sources, after the snippets of the same part. Nothing here is new mathematics unless it says so, and each run result is tagged as proved, certified computation, or empirical pattern; where a literature search found a source, it is cited.

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1 Introduction

The objects. For an even $n \geq 0$ and a prime ℓ the console computes $H^q = H^q(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^n \mathbb{F}_\ell^2)$, $q \in \{0, 1, 2\}$, on the seven-vertex orbifold model `[C, def:model7]` with the symmetric-power convention of `[C, prop:symconv]`, together with the Hecke operators T_p (and $U_\ell = T_\ell$ by the same code path), defined by the double-coset chain map of `[C, def:hecke]`, `[C, eq:heckechain]`. The weight of the cell (n, ℓ) is $k = n + 2$. For $\ell \geq 5$ and $0 < n < \ell$ (and in general, for $n > 0$, after the Eichler–Shimura isomorphism over $\mathbb{Q}$), H^1 has dimension $2 \dim S_k + 1$: two classes for each cusp eigenform of level one and weight k, one in each eigenspace of the involution ι^*, and one Eisenstein class ([LIT]). The eigenvalue of T_p on the Eisenstein class is $1 + p^{k-1}$; on the classes of a cusp eigenform $f = \sum a_n q^n$ it is a_p mod ℓ. All of this is visible in the snippets, and every congruence below is a statement about the $\mathbb{F}_\ell$-algebra generated by the T_p on H^1 (or its $\mathbb{Z}/\ell^K$-analogue), with the eigenforms in the background. The same verbs work over $\mathbb{Z}/\ell^K$ (`set coefficient_ring = "Z/ell^2"`; Parts III and V), and the integral descent computes over $\mathbb{Z}$, at level one and at level $\Gamma_0(N)$ (`set coefficient_ring = "Z"`; Part VI).

How the gallery is organised. The six parts follow the mathematics, not the history of the code. Part I computes the groups and their dimensions; Part II splits H^1 into its Eisenstein and cuspidal pieces; Part III studies how the two pieces (and different cusp forms) are glued modulo ℓ — the local Hecke algebras; Part IV changes the group to $\Gamma_0(N)$ and twists by characters; Part V moves the weight and follows the ordinary part along ℓ-adic families; Part VI returns to $\mathbb{Z}$-coefficients, where the torsion of the integral cohomology explains the jumps of Part I and the congruence primes of Part III reappear in the integral Hecke operators. Within a part, the snippets come first (seconds each) and the run results second (§2). Each section ends with a box *What to look for*. Read in order, the parts form a short course; each section can also be read alone, with its back references.

How to run a snippet. With a built binary `hacs` (`BUILDING.md`; `cohomology_app` is a link to it):

- *as a script:* `hacs -script notebook/examples/ramanujan_10_691.oc` (exit code 0; a failed `assert` exits 4);
- *in the console:* `hacs -i`, then paste the lines, or `hacs -script <file> -console` to run the snippet and keep the session (every variable stays bound; `?x` describes a value, `help <verb>` documents a verb);

- *in the notebook:* configure with `-DOC_BUILD_NOTEBOOK=ON` and start the gateway

  ```
  python3 notebook/gateway/oc_gateway.py --binary <build>/hacs
  ```

  (loopback by default; it prints a one-time token), open the page and paste a snippet into a cell. The notebook serves only verbs tagged `read_only` (`Console/serve/read_only_verbs.tsv`) and refuses `include`, `quit` and `exit`; `set` of a mathematical setting is allowed. All twenty-two snippets satisfy these rules (none uses `include`; the Hida and descent snippets use `set coefficient_ring`).

The language is documented in the CLI manual [CLI] (`docs/latex/cli_manual.pdf`) and in the generated reference `docs/console/console.md`: a handle such as `cohomology(...)` is lazy and computes on first access to a member; `print` prints its arguments separated by spaces; `/` on integers is exact (an error on a remainder), so the snippets write $\lfloor e/2 \rfloor$ as `(e - e % 2) / 2`; integer overflow is an error, so powers modulo ℓ are taken by a loop.

The drift guard. Each listing is read from `notebook/examples/<name>.oc` and each output from `docs/latex/phenomena_gallery/<name>.out`. The outputs and the cost line are written by `docs/latex/tools/gen_phenomena_outputs.sh <binary>`, which runs every snippet under a 4 GiB RSS watchdog; `gen_phenomena_outputs.sh <binary> -check` re-runs them and diffs against the committed outputs (costs are measured, never compared). The PDF is built by `docs/latex/build_phenomena_gallery.sh <outdir>`.

Tags. [PROVED] proved in a cited document; [LIT] classical literature, stated for orientation; [CMPD] computed by the snippet shown (and only that). No statement of this document is binding; [C] is.

§	phenomenon	cell(s) (n, ℓ)	snippet
I. Cohomology and its dimensions			
3	$\dim H^q$, Eichler–Shimura, small ℓ	$n \le 24$, $\ell = 691, 5$	`cohomology_dimensions`
4	cohomology against q-expansions	$(22, 7)$	`forms_vs_cohomology_24_7`
II. Eisenstein and cuspidal classes			
5	ι^*, "Eisenstein = +", boundary and cone	$(10, 691)$	`eisenstein_sign_cone_10_691`
III. Congruences modulo ℓ			
6	Ramanujan's congruence, the Eisenstein ideal	$(10, 691)$	`ramanujan_10_691`
7	irregular primes, $\zeta(1-k)$, Herbrand–Ribet	$(10, 37)$, $(30, 37)$	`kummer_ribet_30_37`
8	Kummer's test on cyclotomic units	$\ell \in \{37, 59\}$	`kummer_unit_37_59`
9	a visible congruence	$(48, 7)$	`visible_congruence_48_7`
10	a hidden congruence	$(300, 23)$	`hidden_congruence_300_23`
11	a dihedral $\bar{\rho}$: Wilton's congruence	$(10, 23)$	`dihedral_wilton_10_23`
12	multiplicity one fails mod 5	$(34, 5)$	`multiplicity_two_34_5`
13	the θ-operator and the Dickson sequence	$(16, 7) \to (24, 7)$	`dickson_theta_16_7`
14	Loewy length 4 mod 37	$(258, 37)$	`loewy4_258_37`
IV. Level $\Gamma_0(N)$ and characters			
18	old forms at level 2	$(10, 7)$	`level_n_old_forms_2_7`
19	a character, the parity obstruction	$(2, 13)$	`nebentypus_7_13`
V. Hida theory			
21	Hida's control theorem mod 13^2	$(10{+}12j, 13)$	`hida_control_13`
22	the ordinary rank in one call	$(10{+}12j, 13)$	`pow_ordinary_rank_13`
23	a Hida tower cell	$(10, 13)$	`hida_tower_13_10`
24	control on forms	$k = 12$, 122; $\ell = 11$	`hida_lambda_11_12`
VI. Integral cohomology and torsion			
27	H^1 over $\mathbb{Z}$, torsion primes, $\beta(n, \ell)$	$n = 4..60$, level 1	`descent_torsion_beta`
30	congruence primes, boundary pair	$n = 10$; $\Gamma_0(11)$, $n = 0$	`descent_congruence_primes`
31	non-split torsion, Galois	$\Gamma_0(8)$, 16, 27, 9	`nonsplit_torsion_16_2`

2 Snippets and runs: two kinds of evidence

A snippet reproduces its phenomenon in seconds, on any build. Each part then records the phenomena of the same kind found in the project's *runs*:

- **census edition 4.0** (`docs/matrix-engine/results/congruence_census/edition3/4.0/`), in four sections of different status:
  - the *main* section (`rows40.tsv.gz`; 279 093 rows in 13 505 cells: the 27 primes $5 \le \ell \le 103$, $\ell = 547$, $\ell = 691$, even $n \le 1000$) and *section b1* (`rows40_b1.tsv.gz`; 169 885 rows in 9500 cells: the 19 primes $107 \le \ell \le 199$, even $n \le 1000$) are the edition **signed** by the owner on 2026-10-27: 448 978 rows (one per maximal ideal of a cell's Hecke algebra) in 23 005 cells, 46 primes. Edition 3.4 (signed 2026-10-15, $n \le 700$) is contained in the main section row for row; its saturation columns were extended by the step SATC of 3.5. In 580 main cells (large n at $\ell \ge 73$, 547, 691) the record has no U_ℓ by design;
  - *section b2* (`rows40_b2.tsv.gz`, 4.0-pre.3; 152 760 rows in 11 911 cells, 35 primes $211 \le \ell \le 419$) is a **partial pre-release** addendum, not signed: complete for $\ell \le 347$, 411 of 6000 cells for $349 \le \ell \le 419$;
  - the *L23 section* (`README_l23.md`: the $\ell \in \{2, 3\}$ cells, 1052 cells, $n \le 700$, and the DEEP-1 tables, 12 220 cells, $n \le 10^4$) is a **pre-release** addendum, not signed; its tables are byte-identical to those of 3.5;
- the **integral census, edition 1** (`docs/integral_census/results/ed1/`; signed by the owner 2026-11-03): 1500 descent cells (N, n) of $H^q(\Gamma_0(N); \mathrm{Sym}^n \mathbb{Z}^2)$ (level one with even $n \le 400$, $2 \le N \le 100$ with even $n \le 20$, and the boundary-pair grid); the gallery quotes no number from it, its integral examples are the snippets of Part VI;
- the saturation runs, the second-cluster Hida run ACL2O, and the level-N runs at $p = 2, 3$.

Every census number in the run sections says which section it comes from; "the census" without qualification means the signed edition (main + b1). The numbers are recomputed from the rows files by `docs/latex/tools/pg_census_check.py`, which reads the printed literals out of this text and compares each with the data (`PG_CENSUS_CHECK ... verdict=OK`); ordinarity, which is not a census column, it reads from the cells' records. These cells take minutes to days and tens of gigabytes, so the run sections have no snippets: every figure is quoted from a committed report or data file, named in each section.

Tags in the run sections. [CMPD] there means *computed and certified on exactly the cells stated*. One more tag: [EMP] an empirical pattern on the cells stated, not proved and possibly false outside them. The phenomenology catalogue `docs/phenomenology/phenomena_catalogue.md` (entries PH-01 ... PH-48) lists every phenomenon with its evidence, sources and status, including those that are documented elsewhere. The run sections cover the ones that no LaTeX document described on 2026-10-19. The catalogue's version 2 adds the literature verdict of every entry, and `docs/phenomenology/FINDINGS_SUMMARY.md` ranks the items for which no prior source was found.

Notation.

- $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $V_n = \mathrm{Sym}^n \mathbb{Z}^2$, $M_n = V_n \otimes \mathbb{F}_\ell$, weight $k = n + 2$. A cell is written (n, ℓ), degree first; level-N cells are written (N, p) and say so.
- A *block* is a generalised eigenspace $H^1(\Gamma; M_n)_\mathfrak{m}$, of dimension D. $A = A_\mathfrak{m}$ is the image of the Hecke algebra in its endomorphisms (the *census algebra*).

Two ℓ-adic objects: the full localisation and its free quotient. Over $\mathbb{Z}_\ell$ the cohomology has torsion (Part VI), so there are two lattices to localise, and the gallery keeps them apart:

- $\mathbb{L}_\mathfrak{m} := H^1(\Gamma; V_n \otimes \mathbb{Z}_\ell)_\mathfrak{m}$, the *full* localisation, torsion included: as a $\mathbb{Z}_\ell$-module $\mathbb{L}_\mathfrak{m} \cong \mathbb{Z}_\ell^f \oplus \bigoplus_i \mathbb{Z}/\ell^{e_i}$. For $\ell \geq 5$ its reductions are the cohomology with finite coefficients, $\mathbb{L}_\mathfrak{m}/\ell^K = H^1(\Gamma; V_n \otimes \mathbb{Z}/\ell^K)_\mathfrak{m}$ and in particular $\mathbb{L}_\mathfrak{m}/\ell = H^1(\Gamma; M_n)_\mathfrak{m}$, the block (`[C, lem:freepart]`, with `[C, cl:bockstein]`: H^2 has no ℓ-torsion). Its torsion is non-zero only on the Dickson locus (§27).
- $\Lambda^{\mathrm{fr}}_\mathfrak{m} := \mathbb{L}_\mathfrak{m}/\mathrm{tors} \cong \mathbb{Z}_\ell^f$, its *free quotient* ($\mathbb{L}'_\mathfrak{m}$ of `[C, sec:satlevelone]`). On it, $\mathbb{T}_\mathfrak{m}$ is the $\mathbb{Z}_\ell$-algebra generated by the T_p (the image of the Hecke algebra in $\mathrm{End}(\Lambda^{\mathrm{fr}}_\mathfrak{m})$, [C]'s $\mathbb{T}^{\mathbb{L}}_\mathfrak{m}$), $\mathbb{T}^{\mathrm{sat}}_\mathfrak{m}$ its saturation and A_K its image in $\mathrm{End}(\Lambda^{\mathrm{fr}}_\mathfrak{m}/\ell^K)$ `[C, def:satdata]`, which is stated for a *free* lattice L; $\delta = \mathrm{length}(\mathbb{T}^{\mathrm{sat}}_\mathfrak{m}/\mathbb{T}_\mathfrak{m})$ is the saturation defect `[C, prop:satdefect]`.

When the block has no torsion ($\mathbb{L}_\mathfrak{m} = \Lambda^{\mathrm{fr}}_\mathfrak{m}$, the generic case) the two coincide, $A = \mathsf{A}_1$, and every statement below can be read on either. When it has torsion they differ, and the statistics live on different objects:

- the census algebra A, its Hilbert function, Loewy length and module structure live on the *full* mod-ℓ block $\mathbb{L}_\mathfrak{m}/\ell = H^1(\Gamma; M_n)_\mathfrak{m}$ (the extension record of the cell, `HeckeExt/HeckeExtDriver.cpp`);
- the $\mathbb{Z}/\ell^K$ module route (`hecke_ring_algebra`, §9) runs on the full $\mathbb{L}_\mathfrak{m}/\ell^K$, torsion summands included, reads A there (the console's `A1`; it equals A_1 only on a torsion-free block), and on a torsion block reads the free part separately, as $\ell^t(\mathbb{L}_\mathfrak{m}/\ell^K) \cong \Lambda^{\mathrm{fr}}_\mathfrak{m}/\ell^{K-t}$ (t the torsion exponent, `[C, lem:freepart]`, the verdict `torsion_free_stable`);
- the saturation statistics δ, $\dim \mathrm{Tor}_1(\mathbb{T}^{\mathrm{sat}}/\mathbb{T}, \mathbb{F}_\ell)$ and its Loewy length, and the census flag `hidden_congruence`, live on $\Lambda^{\mathrm{fr}}_\mathfrak{m}$ (δ and the saturation data as [C] defines them, the hidden flag as in [LG, def:hiddencong]), and the census computes them *only on torsion-free blocks*: a torsion block is refused by name (`not_computed:torsion_block`, `not_stable:torsion_block`), so no census value mixes the two objects;
- (Concordance of notation: [E] and [LG] write $\Lambda_\mathfrak{m}$ for the full lattice, our $\mathbb{L}_\mathfrak{m}$; [C] writes $\mathbb{L}'_\mathfrak{m}$ for our $\Lambda^{\mathrm{fr}}_\mathfrak{m}$.)
- the integral descent and the integral census (Part VI) compute the full group over $\mathbb{Z}$, with the Hecke operators on its free quotient (the integral characteristic polynomials) and on each ℓ-primary torsion part.

Since $\mathbb{L}_\mathfrak{m} \to \Lambda^{\mathrm{fr}}_\mathfrak{m}$ is onto and Hecke-equivariant, $\Lambda^{\mathrm{fr}}_\mathfrak{m}/\ell$ is a quotient of the block, so the algebra A_1 of the free part is a quotient of A: $\dim \mathsf{A}_1 \leq \dim A$, with equality on torsion-free blocks.

Part I
Cohomology and its dimensions

Before any arithmetic, the groups. The group $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ is the amalgam $\mathbb{Z}/4 *_{\mathbb{Z}/2} \mathbb{Z}/6$ `[C, cl:amalgam]`; the project computes its cohomology on a seven-vertex model of the orbifold $\Gamma\backslash\mathfrak{H}$ `[C, def:model7]`. This part shows the dimensions over a field, the classical count that explains them, and the first sign that small primes behave differently.

3 The dimensions of H^q and the Eichler–Shimura count

The centre $\{\pm I\}$ of Γ acts on Sym^n by $(-1)^n$, so for n odd and 2 invertible all cohomology vanishes `[C, rmk:negI]`. For n even and a field F in which 6 is invertible the finite stabilisers $\mathbb{Z}/4$, $\mathbb{Z}/6$

have no higher cohomology, so $H^q = 0$ for $q \geq 2$, H^0 is the space of invariants (0 for $n > 0$), and everything sits in H^1. Its size is the Eichler–Shimura theorem [C, thm:classical] ([LIT]): for even $n > 0$ and $k = n + 2$,

$$H^1(\Gamma; \mathrm{Sym}^n \mathbb{C}^2) \;\cong\; S_k \oplus \overline{S_k} \oplus \mathbb{C} \cdot E_k, \qquad \dim H^1 = 2 \dim S_k + 1,$$

$$\dim S_k = \lfloor k/12 \rfloor - [k \equiv 2 \bmod 12],$$

two classes for each cusp form and one for the Eisenstein series. Over $\mathbb{F}_\ell$ with $\ell > n+1$ the integral cohomology has no ℓ-torsion (§27), so the same count holds; part (1) of the snippet checks it for $n \leq 24$ at $\ell = 691$, computing $\dim S_k$ from the formula, not from a table.

Part (2) takes $\ell = 5$. Now $\mathbb{Z}/4$ and $\mathbb{Z}/6$ still have no 5-torsion, so H^2 stays 0, but $H^0(\Gamma; \mathrm{Sym}^n \mathbb{F}_5^2)$ is non-zero exactly when n is a *Dickson degree* at 5, i.e. $n \in 6\,\mathbb{Z}_{\geq 0} + 20\,\mathbb{Z}_{\geq 0}$: the invariants of $\mathrm{SL}_2(\mathbb{F}_5)$ are generated by the Dickson invariants of degrees $\ell+1 = 6$ and $\ell(\ell-1) = 20$ [C, def:dicksoninv]. By the universal coefficient theorem each such invariant is the shadow of a cyclic summand $\mathbb{Z}/5^a$ of $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2)$ [C, cl:bockstein], which raises $\dim H^0$ and $\dim H^1$ over $\mathbb{F}_5$ by one each: at $n = 12$, $\dim H^1 = 2$ against the characteristic-zero count 1. Part VI computes these torsion groups over $\mathbb{Z}$ directly.

The snippet `notebook/examples/cohomology_dimensions.oc`:

```
# cohomology_dimensions.oc -- the dimensions of H^q(SL_2(Z); Sym^n F_ell^2), against
↪  Eichler--Shimura.
# Phenomena gallery [PG] sec:pg-dims.  Cells (n, 691) for n = 0 .. 24 and (n, 5) for n = 0 ..
↪  24.
# Read_only verbs only (notebook-safe).  Cost: about 1 s, < 0.1 GB.
M = sl2z_model();

# (1) A large prime, ell = 691 > n + 1: dim H^1 = 2 dim S_k + 1 (k = n + 2), H^0 = H^2 = 0 for
↪  n > 0, and odd n gives 0.
print("ell = 691");
for n in range(0, 25) {
    k = n + 2;
    s = 0;
    if (n % 2 == 0) {
        s = (k - k % 12) / 12;
        if (k % 12 == 2 and s > 0) { s = s - 1; }
    }
    H = cohomology(symmetric_power(M, n, 691));
    e = 0;
    if (n % 2 == 0 and n > 0) { e = 2 * s + 1; }
    print("  n =", n, " k =", k, " dim H^q =", H.dims, " dim S_k =", s, " 2 dim S_k + 1 =", e);
    if (n > 0) { assert(H.dims[1] == e, "Eichler--Shimura count"); }
}

# (2) A small prime, ell = 5: H^0 and H^2 appear exactly at the Dickson degrees n in 6 Z + 20 Z
↪  (the shadow of 5-torsion over Z).
print("ell = 5");
for n in range(0, 25, 2) {
    H = cohomology(symmetric_power(M, n, 5));
    print("  n =", n, " dim H^q =", H.dims);
}
```

Expected output (generated file `phenomena_gallery/cohomology_dimensions.out`):

```
ell = 691
  n = 0  k = 2  dim H^q = [1, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 1  k = 3  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 2  k = 4  dim H^q = [0, 1, 0]  dim S_k = 0  2 dim S_k + 1 = 1
  n = 3  k = 5  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 4  k = 6  dim H^q = [0, 1, 0]  dim S_k = 0  2 dim S_k + 1 = 1
```

```
  n = 5  k = 7  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 6  k = 8  dim H^q = [0, 1, 0]  dim S_k = 0  2 dim S_k + 1 = 1
  n = 7  k = 9  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 8  k = 10  dim H^q = [0, 1, 0]  dim S_k = 0  2 dim S_k + 1 = 1
  n = 9  k = 11  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 10  k = 12  dim H^q = [0, 3, 0]  dim S_k = 1  2 dim S_k + 1 = 3
  n = 11  k = 13  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 12  k = 14  dim H^q = [0, 1, 0]  dim S_k = 0  2 dim S_k + 1 = 1
  n = 13  k = 15  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 14  k = 16  dim H^q = [0, 3, 0]  dim S_k = 1  2 dim S_k + 1 = 3
  n = 15  k = 17  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 16  k = 18  dim H^q = [0, 3, 0]  dim S_k = 1  2 dim S_k + 1 = 3
  n = 17  k = 19  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 18  k = 20  dim H^q = [0, 3, 0]  dim S_k = 1  2 dim S_k + 1 = 3
  n = 19  k = 21  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 20  k = 22  dim H^q = [0, 3, 0]  dim S_k = 1  2 dim S_k + 1 = 3
  n = 21  k = 23  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 22  k = 24  dim H^q = [0, 5, 0]  dim S_k = 2  2 dim S_k + 1 = 5
  n = 23  k = 25  dim H^q = [0, 0, 0]  dim S_k = 0  2 dim S_k + 1 = 0
  n = 24  k = 26  dim H^q = [0, 3, 0]  dim S_k = 1  2 dim S_k + 1 = 3
ell = 5
  n = 0  dim H^q = [1, 0, 0]
  n = 2  dim H^q = [0, 1, 0]
  n = 4  dim H^q = [0, 1, 0]
  n = 6  dim H^q = [1, 2, 0]
  n = 8  dim H^q = [0, 1, 0]
  n = 10  dim H^q = [0, 3, 0]
  n = 12  dim H^q = [1, 2, 0]
  n = 14  dim H^q = [0, 3, 0]
  n = 16  dim H^q = [0, 3, 0]
  n = 18  dim H^q = [1, 4, 0]
  n = 20  dim H^q = [1, 4, 0]
  n = 22  dim H^q = [0, 5, 0]
  n = 24  dim H^q = [1, 4, 0]
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.16 s, 26 MiB.

What to look for.

- odd n: `dim H^q = [0, 0, 0]`; $n = 0$: only H^0, the constants;
- even $n > 0$ at $\ell = 691$: `dim H^1 = 2 dim S_k + 1`, e.g. 3 at $k = 12$ (Δ), 5 at $k = 24$, 1 at $k = 14$ (no cusp form);
- at $\ell = 5$: `dim H^0 = 1` exactly at $n = 0, 6, 12, 18, 20, 24$, and for $n > 0$ among these $\dim H^1$ exceeds $2 \dim S_k + 1$ by one (at $n = 0$, $H^1 = 0$).

4 Cohomology against q-expansions (`forms_algebra`)

`forms_algebra(k, ell)` computes S_k over $\mathbb{F}_\ell$ from q-expansions (Miller basis) with the matrices of T_p. By Eichler–Shimura `[C, thm:classical]` (§3) the characteristic polynomial of T_p on $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^{k-2}\,\mathbb{Q}^2)$ is the square of the one on S_k times $x - (1 + p^{k-1})$; it reduces to $\mathbb{F}_\ell$ when H^1 over $\mathbb{F}_\ell$ is the reduction of the lattice (no ℓ-torsion in H^1 over $\mathbb{Z}$; here $\dim = 5 = 2 \cdot 2 + 1$ shows it). At $(k, \ell) = (24, 7)$ the snippet checks this identity for $p = 2$ as an `assert` ([CMPD]): two independent routes, q-expansions and the cohomology of the seven-vertex model, give the same polynomial.

The snippet `notebook/examples/forms_vs_cohomology_24_7.oc`:

```
# forms_vs_cohomology_24_7.oc -- the verb forms_algebra(k, ell): the forms side against the
↪  cohomology side.
# Phenomena gallery [PG] sec:pg-forms.  S_24 over F_7 (dim 2) with T_2 from q-expansions
↪  (Miller basis), against T_2 on H^1(SL_2(Z); Sym^22 F_7^2):
```

```
# by Eichler-Shimura ([C] thm:classical; no 7-torsion here, dim 5 = 2*2+1) the cohomology
↪  charpoly is (forms charpoly)^2 times (x - (1 + 2^23)), the Eisenstein class.
# Read_only verbs only (notebook-safe).  Cost: < 1 s, < 0.2 GB.
F = forms_algebra(24, 7, [2, 3], 6);
print("S_24 over F_7: d =", F.d, " charpoly T_2 =", F.ops[0].charpoly, " charpoly T_3 =",
↪  F.ops[1].charpoly);
M = sl2z_model();
H = cohomology(symmetric_power(M, 22, 7));
P = charpoly(hecke(H, 2).A(1));
print("H^1: dim =", H.dims[1], " charpoly T_2 =", P);
f = poly(7, F.ops[0].charpoly);
e = poly(7, [(7 - (1 + 2^23) % 7) % 7, 1]);
assert(P == mul(mul(f, f), e), "charpoly T_2 on H^1 = (forms charpoly)^2 (x - (1 + 2^23))");
print("Eisenstein eigenvalue 1 + 2^23 mod 7 =", (1 + 2^23) % 7);
```

Expected output (generated file `phenomena_gallery/forms_vs_cohomology_24_7.out`):

```
S_24 over F_7: d = 2  charpoly T_2 = [6, 5, 1]  charpoly T_3 = [0, 6, 1]
H^1: dim = 5  charpoly T_2 = X^5 + 5*X^4 + X^3 + X^2 + 2*X + 2 (mod 7)
Eisenstein eigenvalue 1 + 2^23 mod 7 = 5
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.01 s, 25 MiB.

What to look for.

- `d = 2`: $\dim S_{24} = 2$; the cohomology has dimension $5 = 2 \cdot 2 + 1$;
- the `assert` passes: the identity is checked, not displayed.

Part II
Eisenstein and cuspidal classes

H^1 contains one Eisenstein class and two classes per cusp form (§3). This part tells them apart: by an involution, and by the boundary of the Borel–Serre compactification, where only the Eisenstein class survives. The compactly supported cohomology H^1_c and the interior cohomology $H^1_!$, the cuspidal part, come out of the same long exact sequence.

5 The sign of the Eisenstein class, the boundary and the cone

The involution of the upper half-plane $z \mapsto -\bar{z}$ induces an involution on H^q; the project's production involution is $\iota^* := -\iota^*_{\mathrm{geo}}$ `[C, def:iota]`, and the sign was fixed so that the Eisenstein class lies in H^1_+: a convention by definition, not a theorem `[C, rmk:iotasign]`. For $2 \in R^\times$, $H^q = H^q_+ \oplus H^q_-$ `[C, prop:iotasplit]`, and for $6 \in R^\times$ both summands are Hecke-stable. Each cusp eigenform contributes one class to each sign.

At $(10, 691)$ the split is $\dim H^1_+ = 2$ (Eisenstein, Δ^+) and $\dim H^1_- = 1$ (Δ^-). The non-split extension of Ramanujan's congruence (§6, the first section of Part III) lives *entirely in* H^1_+: there the module is cyclic with Loewy series $[1, 1]$, while H^1_- is a single semisimple line. This is the shape that Harder's and Ribet's constructions predict for an Eisenstein congruence [E, cl:harder].

The boundary of the Borel–Serre compactification is computed by the circle model `[C, prop:circlemodel]`; T_p acts on H^*_∂ through the Eisenstein character, here the scalar $667 = 1 + 2^{11}$ for $p = 2$ in both degrees. The cone of the restriction r gives the compactly supported cohomology H^*_c with its long exact sequence `[C, sec:cone]`, `[C, eq:coneles]` (here $H^0 = 0$)

$$0 \to H^0_\partial \to H^1_c \to H^1 \xrightarrow{r^1} H^1_\partial \to H^2_c \to \cdots,$$

whose exactness the snippet reads; r^1 has rank 1 and the interior cohomology $H^1_! = \ker r^1$ has dimension 2. The Hecke action on H^*_c is defined at cochain level through a *pair homotopy* η over $\mathbb{Z}$ [C, def:pairhomotopy], [C, def:coneaction]; it is well defined and independent of η [C, thm:conewelldef] [PROVED]. The snippet prints η's certificate at $p = 2$ (solved over $\mathbb{Z}$, integral, re-derived) and the cone-action certificate.

The snippet notebook/examples/eisenstein_sign_cone_10_691.oc:

```
# eisenstein_sign_cone_10_691.oc -- the involution iota^*, "Eisenstein = +", the boundary and
↪  the cone at (10, 691).
# Phenomena gallery [PG] sec:pg-sign.  Read_only verbs only (notebook-safe).  Cost: < 1 s, <
↪  0.1 GB.
M = sl2z_model();
H = cohomology(symmetric_power(M, 10, 691));

# (1) The sign split H^1 = H^1_+ (+) H^1_-, and where the Eisenstein line lives.
J = iota(H);
print("dim H^q_+ =", J.dims_plus, " dim H^q_- =", J.dims_minus, " certificate", J.certificate);
S = J.sign_split(1);
print("Eisenstein line:", S.eisenstein);
L = J.eisenstein_line(1);
print("Eisenstein line in H^1_+:", contains(J.plus(1), L), "  in H^1_-:", contains(J.minus(1),
↪  L));
assert(contains(J.plus(1), L), "the Eisenstein line lies in H^1_+ (the production sign iota^* =
↪  -iota^*_geo)");
assert(J.dims_plus == [0, 2, 0] and J.dims_minus == [0, 1, 0], "Eisenstein + cusp in H^1_+,
↪  cusp alone in H^1_-");

# (2) The +/- records: the Eisenstein-cuspidal gluing happens in H^1_+ only.
R = ext_record(10, 691, cone = true, sign_split = true);
for m in R.ideals_plus { print("plus :", m, m.modules); }
for m in R.ideals_minus { print("minus:", m, m.modules); }

# (3) The boundary circle sees only the Eisenstein character 1 + p^11.
Bd = boundary(H);
print("dim H^q_bd =", Bd.dims, " T_2 on H^0_bd:", Bd.A(2, 0), " on H^1_bd:", Bd.A(2, 1));

# (4) The cone H^*_c = H^*(Cone(r)): its long exact sequence, H^1_! and the restriction r^1.
C = cone(H);
print("dim H^q_c =", C.dims, " rank r^1 =", rank(C.r(1)), " dim H^1_! =", C.H1_shriek, " LES
↪  exact:", C.les_exact);
assert(C.H1_shriek == 2 and rank(C.r(1)) == 1, "r^1 has rank 1: one Eisenstein class reaches
↪  the boundary");

# (5) The cone action A_c(T_2) is built from a certified pair homotopy eta over Z.
E = C.pair_homotopy(2);
print("eta at p = 2: regime", E.solve_regime, " integral", E.integral, " slots", E.slots, "
↪  certify", E.certify);
print("ConeActionCertificate(2) =", C.action_certificate(2));
```

Expected output (generated file phenomena_gallery/eisenstein_sign_cone_10_691.out):

```
dim H^q_+ = [0, 2, 0]  dim H^q_- = [0, 1, 0]  certificate 0
Eisenstein line: {status: "defined", defined: true, f: 1, dim_socle: 1, socle_equals_mV: true}
Eisenstein line in H^1_+: true   in H^1_-: false
plus : MaximalIdeal #0 (plus): dim 2, f = 1, local [LoewyProfile on H1_plus: loewy [1, 1], socle
↪  [1, 1], cyclic]
minus: MaximalIdeal #0 (minus): dim 1, f = 1, local [LoewyProfile on H1_minus: loewy [1], socle
↪  [1], cyclic]
dim H^q_bd = [1, 1]  T_2 on H^0_bd: [[667]]  on H^1_bd: [[667]]
dim H^q_c = [0, 3, 0]  rank r^1 = 1  dim H^1_! = 2  LES exact: true
eta at p = 2: regime Z  integral true  slots 21  certify 0
ConeActionCertificate(2) = 0
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.36 s, 49 MiB.

What to look for.

- `Eisenstein line in H^1_+: true`: the production sign; with ι_{geo}^{*} it would be `false`;
- `plus : ...loewy [1, 1] ...cyclic` against `minus : ...loewy [1]`: the extension is in the + part only;
- `rank r^1 = 1`, `dim H^1_! = 2`, `LES exact: true`.

Part III
Congruences modulo ℓ: local Hecke algebras

Over $\mathbb{Q}$ the Hecke algebra acting on H^1 is semisimple: a product of number fields, one per Galois orbit of eigenforms (and $\mathbb{Q}$ for the Eisenstein series). Modulo ℓ two eigensystems can coincide — a *congruence* — and then the generalised eigenspace of the common system $\mathfrak{m}$ is a module over a local Artin algebra $A_{\mathfrak{m}}$ which need not be a field. This part reads congruences as such local algebras: their Hilbert functions, their Loewy series, and the extensions of Hecke modules they encode [E, sec:ring]. It starts with the Eisenstein–cusp congruences (Ramanujan, Kummer, Ribet), goes on to cusp–cusp congruences (visible, hidden, dihedral, multiplicity two), the weight-raising operator θ that links congruences in different weights, and the longest local algebra found by snippet; then the census of all local algebras up to $n = 1000$ (edition 4.0).

6 Ramanujan's congruence as an Eisenstein ideal

The discriminant $\Delta = \sum \tau(n) q^n$ is the only normalised cusp eigenform of level one and weight 12, and $E_{12} = -\frac{B_{12}}{24} + \sum \sigma_{11}(n) q^n$ with $B_{12} = -\frac{691}{2730}$. Because 691 divides the numerator of B_{12}, the constant term of E_{12} vanishes modulo 691 and E_{12} is congruent to a cusp form, necessarily Δ: Ramanujan's $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ ([LIT]). This is the project's flagship example, [E, cl:ramanujan] and [EX, ex:691].

On cohomology, $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^{10} \mathbb{F}_{691}^2)$ has dimension 3: the two classes of Δ and the Eisenstein class. All three carry the same eigenvalue $a_p = 1 + p^{11} \bmod 691$ of every T_p, so H^1 is a single generalised eigenspace, and the question is whether the Hecke action on it is semisimple. It is not: $T_p - a_p$ is nilpotent of Jordan type $(2, 1)$ for every p tested, including U_{691} (eigenvalue $\tau(691) \equiv 1$). The congruence between the Eisenstein series and Δ is thereby a *non-split extension* of Hecke modules, and the $\mathbb{F}_{691}$-algebra generated by the T_p on the block is $\mathbb{F}_{691}[\varepsilon]/(\varepsilon^2)$: the maximal ideal $\mathfrak{m} = (T_2 + 24) = (T_2 - 667)$ is the *Eisenstein ideal* (Mazur's term at prime level; here at level one, weight 12), and ε measures the congruence.

The snippet derives the eigenvalues from σ_{11} (they are output, not input), checks the Jordan types, and reads the extension record of the cell [E, sec:ring]: one maximal ideal, residue field $\mathbb{F}_{691}$, local algebra with Hilbert function $(1, 1)$, Gorenstein, and the Loewy profile of $H^1_{\mathfrak{m}}$, $[2, 1]$: a free rank-one module over $\mathbb{F}_{691}[\varepsilon]/\varepsilon^2$ plus one more line.

The snippet `notebook/examples/ramanujan_10_691.oc`:

```
# ramanujan_10_691.oc -- Ramanujan's congruence tau(p) == 1 + p^11 (mod 691) as an Eisenstein
↪  ideal.
# Phenomena gallery [PG] sec:pg-ramanujan.  Cell (n, ell) = (10, 691), weight k = 12.
# Read_only verbs only (notebook-safe).  Cost: < 1 s, < 0.1 GB.
M = sl2z_model();
H = cohomology(symmetric_power(M, 10, 691));
print("dim H^q =", H.dims);

# (1) 691 divides the numerator of B_12: the Eisenstein series E_12 is congruent to a cusp form.
```

```
print("B_12 mod 691 =", bernoulli_mod(12, 691));
assert(bernoulli_mod(12, 691) == 0, "691 | B_12");

# (2) T_p - (1 + p^11) is nilpotent but NOT zero on H^1: Jordan type (2, 1).  At p = 691 this
↪  is U_691 - 1.
I3 = identity(691, 3);
for p in [2, 3, 5, 7, 691] {
    x = 1;
    for i in range(0, 11) { x = (x * p) % 691; }
    ap = (1 + x) % 691;
    Np = sub(hecke(H, p).A(1), scale(I3, ap));
    print("p =", p, " 1 + p^11 mod 691 =", ap, " Jordan type of T_p - a_p:", jordan_type(Np));
    assert(jordan_type(Np) == [2, 1], "one non-split Jordan block");
}

# (3) The extension record: one maximal ideal, its local Hecke algebra F_691[eps]/eps^2.
R = ext_record(10, 691);
print("maximal ideals:", R.num_ideals);
m = R.ideals[0];
print("m: dim", m.dim, " f =", m.residue_degree, " witness", m.witness_poly, " final", m.final);
T = local_algebra(m);
print("T_m: dim", T.dim, " Hilbert function", T.hilbert, " Gorenstein", T.gorenstein);
print("module H^1_m:", m.modules);
assert(T.hilbert == [1, 1], "T_m = F_691[eps]/eps^2");
```

Expected output (generated file `phenomena_gallery/ramanujan_10_691.out`):

```
dim H^q = [0, 3, 0]
B_12 mod 691 = 0
p = 2  1 + p^11 mod 691 = 667  Jordan type of T_p - a_p: [2, 1]
p = 3  1 + p^11 mod 691 = 252  Jordan type of T_p - a_p: [2, 1]
p = 5  1 + p^11 mod 691 = 684  Jordan type of T_p - a_p: [2, 1]
p = 7  1 + p^11 mod 691 = 531  Jordan type of T_p - a_p: [2, 1]
p = 691  1 + p^11 mod 691 = 1  Jordan type of T_p - a_p: [2, 1]
maximal ideals: 1
m: dim 3  f = 1  witness X + 24 (mod 691)  final true
T_m: dim 2  Hilbert function [1, 1]  Gorenstein true
module H^1_m: [LoewyProfile on H1: loewy [2, 1], socle [2, 1], not cyclic]
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.38 s, 58 MiB.

What to look for.

- `bernoulli_mod(12, 691) = 0`: the source of the congruence;
- the same Jordan type $(2, 1)$ for every p, i.e. one non-split 2×2 block and one 1×1 block, *not* a scalar matrix;
- `Hilbert function [1, 1]`: the local algebra is $\mathbb{F}_{691}[\varepsilon]/\varepsilon^2$; a reduced algebra would print `[1]`.

7 Irregular primes, $\zeta(1-k)$ and the Herbrand–Ribet congruence

The special values $\zeta(1-k) = -B_k/k$ ($k \geq 2$ even) control Eisenstein congruences: if $\ell \mid B_k$ with $2 \leq k \leq \ell - 3$ (an *irregular pair* (ℓ, k)), the constant term of E_k vanishes mod ℓ and E_k is congruent to a cusp form of weight k; Ribet used that cusp form to build an unramified $\mathbb{Z}/\ell$-extension of $\mathbb{Q}(\zeta_\ell)$ on which $\mathrm{Gal}(\mathbb{Q}(\zeta_\ell)/\mathbb{Q})$ acts by ω^{1-k}, proving the converse of Herbrand's theorem ([LIT]). Kummer's criterion says ℓ is irregular iff such a k exists.

The console computes B_w mod ℓ for $0 \leq w < \ell - 1$ by the $\mathbb{F}_\ell$-recursion (`bernoulli_mod`, von Staudt–Clausen bounds the range). The snippet lists the irregular indices of $37, 59, 67, 101, 103, 691$ — 32, 44, 58, 68, 24, and 12, 200 — and then compares the Eisenstein block of $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^n \mathbb{F}_{37}^2)$ at the regular weight $k = 12$ ($n = 10$) with the irregular weight $k = 32$ ($n = 30$). The Eisenstein

ideal is found by its character $a_2 = 1 + 2^{k-1}$. At $k = 12$ it is a one-dimensional block, nothing is glued; at $k = 32$ it is three-dimensional, with the Loewy profile $[2, 1]$ and local algebra $\mathbb{F}_{37}[x]/x^2$ of §6: Ribet's cusp form of weight 32 is glued to E_{32}.

The field-theoretic side of these two cases — the Kummer generators $X^{37} - \eta_{32}$ and $X^{691} - \eta_{12}$ over the cyclotomic fields, the $\mathbb{Z}/\ell^2$ layers — is in [L37, sec:fieldsribet] and [L37, app:ribetdata]; the ℓ-adic L-values (Kubota–Leopoldt, Coates–Wiles, the measure congruence) are [E, sec:lvalue], [E, sec:lvmeasure], [E, cl:lvmeascong], and the rungs of the Λ-adic Eisenstein ideal at 691 are [U, prop:u-691]. The Kummer test of the generators is the verb `kummer_unit` (§8); the snippet here shows the residue modulo ℓ of $\zeta(1-k)$ and its consequence on cohomology.

The snippet `notebook/examples/kummer_ribet_30_37.oc`:

```
# kummer_ribet_30_37.oc -- irregular primes, Bernoulli numbers and the Eisenstein ideal
↪  (Herbrand-Ribet, Mazur).
# Phenomena gallery [PG] sec:pg-kummer.  Read_only verbs only (notebook-safe).  Cost: ~1 s, <
↪  0.1 GB.
#
# zeta(1 - k) = -B_k / k.  An irregular pair (ell, k): ell | B_k, 2 <= k <= ell - 3, k even.
for ell in [37, 59, 67, 101, 103, 691] {
    irr = "";
    for k in range(2, ell - 2, 2) {
        if bernoulli_mod(k, ell) == 0 { irr = irr + " " + str(k); }
    }
    print("ell =", ell, " irregular indices k:", irr);
}

# At ell = 37 the irregular index is k = 32 (cell n = 30).  Compare with the regular weight k =
↪  12 (cell n = 10).
M = sl2z_model();
for n in [10, 30] {
    k = n + 2;
    x = 1;
    for i in range(0, k - 1) { x = (x * 2) % 37; }
    e2 = (1 + x) % 37;
    R = ext_record(n, 37);
    print("(n, ell) = (" + str(n) + ", 37), k =", k, " B_k mod 37 =", bernoulli_mod(k, 37), "
    ↪  Eisenstein a_2 = 1 + 2^(k-1) =", e2);
    for m in R.ideals {
        tag = "";
        if m.character(2) == poly(37, [e2]) { tag = "  <- Eisenstein ideal"; }
        print("   ideal: dim", m.dim, " a_2 =", m.character(2), " Loewy", m.modules[0].loewy, "
        ↪  T_m Hilbert", local_algebra(m).hilbert, tag);
    }
}
```

Expected output (generated file `phenomena_gallery/kummer_ribet_30_37.out`):

```
ell = 37  irregular indices k:  32
ell = 59  irregular indices k:  44
ell = 67  irregular indices k:  58
ell = 101  irregular indices k:  68
ell = 103  irregular indices k:  24
ell = 691  irregular indices k:  12 200
(n, ell) = (10, 37), k = 12  B_k mod 37 = 17  Eisenstein a_2 = 1 + 2^(k-1) = 14
   ideal: dim 1  a_2 = 14 (mod 37)  Loewy [1]  T_m Hilbert [1]   <- Eisenstein ideal
   ideal: dim 2  a_2 = 13 (mod 37)  Loewy [2]  T_m Hilbert [1]
(n, ell) = (30, 37), k = 32  B_k mod 37 = 0  Eisenstein a_2 = 1 + 2^(k-1) = 23
   ideal: dim 3  a_2 = 23 (mod 37)  Loewy [2, 1]  T_m Hilbert [1, 1]   <- Eisenstein ideal
   ideal: dim 2  a_2 = 14 (mod 37)  Loewy [2]  T_m Hilbert [1]
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.31 s, 32 MiB.

What to look for.

- `ell = 691 irregular indices k:  12 200`: 691 is irregular twice;
- at $(10, 37)$ the Eisenstein ideal has `dim 1, Loewy [1]`; at $(30, 37)$ it has `dim 3, Loewy [2, 1], T_m Hilbert [1, 1]`;
- the other ideal at each cell is a cusp eigenform not congruent to an Eisenstein series (Δ at $k = 12$, the second weight-32 form at $k = 32$), with a semisimple two-dimensional block.

8 Kummer's criterion on cyclotomic units (`kummer_unit`)

The cyclotomic unit η_i of the first layer of $\mathbb{Q}(\zeta_\ell)$ (its ω^i-component) satisfies $v_\pi(\eta_i^{\ell-1} - 1) > i$, i.e. it is closer to 1 π-adically than its eigencharacter forces, exactly when ℓ divides the Bernoulli number B_i (Kummer; [E, sec:lvnf]). It is not a global ℓ-th power at any index shown (the record's `not_lth_power`, tested at split primes); at an irregular index the congruence is what makes the Kummer extension $\mathbb{Q}(\zeta_\ell, \eta_i^{1/\ell})$ a candidate for an unramified one. The verb `kummer_unit(ell, i)` returns the unit on the power basis of $\mathbb{Z}[\zeta_\ell]$ with the valuation $v_\pi(\eta_i^{\ell-1} - 1)$; the test is $v_\pi > i$. The record also carries the comparison with the direct Bernoulli computation (`oracle_agrees`). At $\ell = 37$ the index $i = 32$ is the irregular one of §7, at $\ell = 59$ it is $i = 44$ ([LIT]; [CMPD]).

The snippet `notebook/examples/kummer_unit_37_59.oc`:

```
# kummer_unit_37_59.oc -- the verb kummer_unit(ell, i): the cyclotomic unit eta_i and Kummer's
↪  ell-th-power test.
# Phenomena gallery [PG] sec:pg-kummerunit.  v_pi(eta_i^(ell-1) - 1) > i exactly when ell
↪  divides the Bernoulli number B_i
# ([E] sec:lvnf); the record's oracle_agrees compares this with a direct Bernoulli computation.
↪   ell = 37 is irregular at i = 32,
# ell = 59 at i = 44; the other indices are regular.
# Read_only verbs only (notebook-safe).  Cost: ~1 s, < 0.2 GB.
for c in [[37, 12], [37, 30], [37, 32], [59, 20], [59, 44]] {
    ell = c[0];
    i = c[1];
    K = kummer_unit(ell, i);
    print("ell =", ell, " i =", i, " v_pi =", K.v_pi, " ell | B_i:", K.ell_divides_Bi, " eta
    ↪  not an ell-th power:", K.not_lth_power, " oracle agrees:", K.oracle_agrees);
    assert(K.oracle_agrees);
    assert((K.v_pi > i) == K.ell_divides_Bi);
}
```

Expected output (generated file `phenomena_gallery/kummer_unit_37_59.out`):

```
ell = 37  i = 12  v_pi = 12  ell | B_i: false  eta not an ell-th power: true  oracle agrees: true
ell = 37  i = 30  v_pi = 30  ell | B_i: false  eta not an ell-th power: true  oracle agrees: true
ell = 37  i = 32  v_pi = 38  ell | B_i: true  eta not an ell-th power: true  oracle agrees: true
ell = 59  i = 20  v_pi = 20  ell | B_i: false  eta not an ell-th power: true  oracle agrees: true
ell = 59  i = 44  v_pi = 60  ell | B_i: true  eta not an ell-th power: true  oracle agrees: true
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.03 s, 22 MiB.

What to look for.

- `v_pi` equals i at the regular indices and exceeds i at the irregular ones ($38 > 32$, $60 > 44$);
- `oracle agrees:  true` on every line.

9 A visible congruence between eigenforms

Two eigenforms $f \neq g$ of the same weight can be congruent modulo ℓ (all their Hecke eigenvalues agree mod a prime above ℓ). On cohomology this can show up in two ways. In the notation of §2

("Two ℓ-adic objects"), let $\Lambda_{\mathfrak{m}}^{\mathrm{fr}}$ be the free quotient of the localisation of $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^n \mathbb{Z}_\ell^2)$ at a residual system $\mathfrak{m}$, $\mathbb{T}_{\mathfrak{m}}$ the $\mathbb{Z}_\ell$-algebra generated by the T_p on it, $\mathbb{T}_{\mathfrak{m}}^{\mathrm{sat}}$ its saturation in $\mathrm{End}(\Lambda_{\mathfrak{m}}^{\mathrm{fr}})$ and A_K the image of $\mathbb{T}_{\mathfrak{m}}$ in $\mathrm{End}(\Lambda_{\mathfrak{m}}^{\mathrm{fr}}/\ell^K)$ `[C, def:satdata]`. The mod-ℓ algebra $\mathsf{A}_1 = \mathbb{T}_{\mathfrak{m}}/(\mathbb{T}_{\mathfrak{m}} \cap \ell\mathbb{T}_{\mathfrak{m}}^{\mathrm{sat}})$ is what H^1 with $\mathbb{F}_\ell$-coefficients sees when the block has no torsion (then $\Lambda_{\mathfrak{m}}^{\mathrm{fr}}/\ell$ is the block and $\mathsf{A}_1 = A$; on a torsion block A_1 is a quotient of A); A_K/ℓ (K large) is $\mathbb{T}_{\mathfrak{m}}/\ell$. The congruence is *visible* if already $\mathsf{A}_1 \neq \mathbb{F}_\ell$.

The console's `hecke_ring_algebra(n, ell, kmax)` runs the $\mathbb{Z}/\ell^K$ Hecke-algebra route with its certified reading `[C, cor:stablecert]`: per block it prints the Hilbert functions of A (`A1`, on the full block) and of A_K/ℓ (`AK/ell`, on the free part), the free rank, the torsion, and the verdict. On a torsion-free block (`stable`) the two are the objects of the previous paragraph; on a block with torsion and free rank $f > 0$ (`torsion_free_stable`) `AK/ell` is read on $\Lambda_{\mathfrak{m}}^{\mathrm{fr}}$ while `A1` stays the algebra A of the full block, and `hidden` compares the two. At $(48, 7)$ (weight 50) the block with system $(a_2, a_3, a_5, a_7) = (1, 0, 0, 0)$ has $\mathsf{A}_1 = \mathsf{A}_K/\ell = \mathbb{F}_7[x]/x^2$ (Hilbert function $(1, 1)$), verdict `stable` (certified): two eigenforms congruent mod 7, and H^1 with $\mathbb{F}_7$-coefficients already shows the nilpotent x. The second half reads the same block from the $\mathbb{F}_7$ extension record: a four-dimensional ideal with local algebra of Hilbert function $(1, 1)$ and nilpotency index 2. The cell is the first one of the console's frozen script `029_hecke_ring_algebra.oc`.

The snippet `notebook/examples/visible_congruence_48_7.oc`:

```
# visible_congruence_48_7.oc -- a congruence that the mod-ell Hecke algebra already sees
↪  ("visible"), for contrast
# with hidden_congruence_300_23.oc.  Phenomena gallery [PG] sec:pg-visible.  Cell (n, ell) =
↪  (48, 7), weight 50.
# Read_only verbs only (notebook-safe).  Cost: ~1 s, < 0.1 GB.
R = hecke_ring_algebra(48, 7, 2);
print("primes", R.primes, " K_cls", R.K_cls, " blocks", len(R.blocks));
for b in R.blocks {
    print("system", b.system, " D", b.D, " A1", b.A1_hilbert, " AK/ell", b.AK_hilbert, "
    ↪  hidden", b.hidden, " verdict", b.verdict);
}
B0 = R.blocks[0];
assert(B0.A1_hilbert == [1, 1] and B0.AK_hilbert == [1, 1] and not B0.hidden, "visible: A_1 =
↪  A_K/ell = F_7[x]/x^2");

# The same non-reduced algebra straight from H^1 mod 7: the decomposition by the anemic T_p.
R2 = ext_record(48, 7);
for m in R2.ideals {
    if m.group == "H1" {
        print("ideal: dim", m.dim, " a_2 =", m.character(2), " T_m Hilbert",
        ↪  local_algebra(m).hilbert, " nu", m.nilpotency);
    }
}
```

Expected output (generated file `phenomena_gallery/visible_congruence_48_7.out`):

```
primes [2, 3, 5, 7]  K_cls 2  blocks 4
system [1, 0, 0, 0]  D 4  A1 [1, 1]  AK/ell [1, 1]  hidden false  verdict stable
system [3, 3, 1, 0]  D 2  A1 [1]  AK/ell nil  hidden false  verdict reduced_rank1
system [3, 4, 6, 0]  D 1  A1 [1]  AK/ell [1]  hidden false  verdict torsion_stable
system [3, 4, 6, 1]  D 1  A1 [1]  AK/ell nil  hidden false  verdict reduced_rank1
ideal: dim 1  a_2 = 3 (mod 7)  T_m Hilbert [1]  nu 1
ideal: dim 1  a_2 = 3 (mod 7)  T_m Hilbert [1]  nu 1
ideal: dim 2  a_2 = 3 (mod 7)  T_m Hilbert [1]  nu 1
ideal: dim 4  a_2 = 1 (mod 7)  T_m Hilbert [1, 1]  nu 2
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.81 s, 71 MiB.

What to look for.

- `A1 [1, 1]` *equal to* `AK/ell [1, 1]`, `hidden false`: visible;
- `reduced_rank1` blocks are single eigenforms (rank-one Hecke algebra); `torsion_stable` is a *pure-torsion* block ($f = 0$: here $\mathbb{L}_{\mathfrak{m}} \cong \mathbb{Z}/7$, the Dickson class of $n = 48 = 6(7+1)$; $\Lambda^{\mathrm{fr}}_{\mathfrak{m}} = 0$ and `AK/ell` is read on $\mathbb{L}_{\mathfrak{m}}$ itself, `[C, lem:freepart]`(3)), not a block with a free part — that verdict is `torsion_free_stable`;
- `nu 2` in the $\mathbb{F}_7$ record for the same block.

10 A hidden congruence: $(300, 23)$

A congruence is *hidden* if at least two eigenforms share the residual system $\mathfrak{m}$ while $\mathsf{A}_1 = \mathbb{F}_\ell$ ([LG, def:hiddencong], restating [EX, s14:wp6]): $(T_p - a_p)\Lambda^{\mathrm{fr}}_{\mathfrak{m}} \subseteq \ell\Lambda^{\mathrm{fr}}_{\mathfrak{m}}$ for every p, but $T_p - a_p \notin \ell\mathbb{T}_{\mathfrak{m}}$ for some p. The congruence is then invisible on H^1 with $\mathbb{F}_\ell$-coefficients and appears only over $\mathbb{Z}/\ell^K$, $K \geq 2$; its marker is $\dim \mathsf{A}_1 < \dim \mathsf{A}_K/\ell$, its depth the saturation defect $\delta = \mathrm{length}(\mathbb{T}^{\mathrm{sat}}_{\mathfrak{m}}/\mathbb{T}_{\mathfrak{m}}) \geq 1$.

At $(n, \ell) = (300, 23)$ (weight 302) the block with system $(a_2, a_3, a_5, a_7, a_{23}) = (19, 14, 0, 0, 0)$ is $\bar{\rho}_\Delta \otimes \omega^2$, dihedral for $K = \mathbb{Q}(\sqrt{-23})$ (§11), non-ordinary ($a_{23} = 0$), of dimension $D = 4$: two weight-302 eigenforms. Mod 23, $\mathsf{A}_1 = \mathbb{F}_{23}$ (one eigensystem, Hilbert function (1)); over $\mathbb{Z}/23^2$, $\mathsf{A}_K/\ell = \mathbb{F}_{23}[x]/x^2$ (Hilbert function $(1, 1)$): the two forms are congruent mod 23 but not mod 23^2, depth $\delta = 1$. The record's `hidden` flag is true, and it is the only hidden block of the 14 at this cell. The dihedral pattern is checked on the system itself: $a_p = 0$ exactly at the primes inert in K. Hidden congruences off the dihedral locus, the "Frobenius layer", are [E, pr:frobeniuslayer]; the dihedral reading is [LG, sec:cm].

The snippet `notebook/examples/hidden_congruence_300_23.oc`:

```
# hidden_congruence_300_23.oc -- the (n = 300, ell = 23) "hidden congruence" [C] / [E] sec:ring,
# docs/latex/loewy_and_galois_cohomology.tex (sec. "The (300,23) block").
#
# n = 300 (weight k = 302), ell = 23.  The block with eigensystem (a_2, a_3, a_5, a_7, a_23) =
# (19, 14, 0, 0, 0) is rhobar_Delta (x) omega^2, dihedral for K = Q(sqrt(-23)), non-ordinary
↪  (a_23 = 0), D = 4.
#   * mod 23 its Hecke algebra A_1 = T_m/(T_m cap ell T_m^sat) is F_23: ONE eigensystem
↪  (Hilbert function [1]);
#   * over Z/23^2 the algebra A_K/ell has Hilbert function [1, 1], i.e. F_23[x]/x^2 -- two
↪  distinct
#     weight-302 level-one eigenforms, congruent mod 23 but not mod 23^2 (depth delta = 1).
#   dim A_1 = 1 < 2 = dim A_K/ell  is the "hidden" marker.   The record's `hidden` flag is true.
#
# Steps: (1) hecke_ring_algebra(300, 23, 2) -- the Z/ell^K Hecke-algebra route, all 14 blocks;
#        (2) pick the block with system (19,14,0,0,0) and show A1 vs A_K Hilbert functions;
#        (3) dihedral pattern: for the primes of the system, a_p = 0 exactly where p is inert
↪  in Q(sqrt(-23))
#            (-23 is a non-square mod p).  The other blocks of the same (n, ell) are shown for
↪  contrast:
#            none of them is hidden.
# Only read_only verbs are used (hecke_ring_algebra, print, keys, len, str, assert), so the
↪  script is
# notebook-safe.  Needs no full console.   Cost: ~97 s wall (shared binary, RelWithDebInfo),
↪  0.56 GB RSS (the dense route at n = 300).
# Not in Console/scripts/ (no frozen transcript).
#
# Expected output (excerpt):
#   BLOCKS 14 primes [2, 3, 5, 7, 23] K_cls 2 K_run [2]
#   BLOCK system=[19, 14, 0, 0, 0] D=4 verdict=stable
#     A1_hilbert (mod 23)     = [1]      -> one eigensystem mod 23
#     AK_hilbert (mod 23^2)/ell = [1, 1]   -> F_23[x]/x^2 over Z/23^2
#     hidden = true  (A_1 smaller than A_K/ell)
#   DIHEDRAL p=2 (-23|p)=+1 split a_p=19
#   DIHEDRAL p=3 (-23|p)=+1 split a_p=14
#   DIHEDRAL p=5 (-23|p)=-1 inert a_p=0
```

```
#   DIHEDRAL p=7 (-23|p)=-1 inert a_p=0
#   OTHER BLOCKS hidden: 1 of 14 ; D=4 with A1=[1,1]: 8

R = hecke_ring_algebra(300, 23, 2);
print("BLOCKS", len(R.blocks), "primes", R.primes, "K_cls", R.K_cls, "K_run", R.K_run);

TARGET = [19, 14, 0, 0, 0];
BI = -1;
i = 0;
nhid = 0;
nsplit = 0;
for b in R.blocks {
    if b.hidden { nhid = nhid + 1; }
    if b.A1_hilbert == [1, 1] and b.D == 4 { nsplit = nsplit + 1; }
    if b.system == TARGET { BI = i; }
    i = i + 1;
}
assert(BI >= 0, "the block (19,14,0,0,0) is present");
B = R.blocks[BI];

print("BLOCK system=" + str(B.system) + " D=" + str(B.D) + " verdict=" + B.verdict);
print("  A1_hilbert (mod 23)     = " + str(B.A1_hilbert) + "      -> one eigensystem mod 23");
print("  AK_hilbert (mod 23^2)/ell = " + str(B.AK_hilbert) + "   -> F_23[x]/x^2 over Z/23^2");
print("  hidden = " + str(B.hidden) + "  (A_1 smaller than A_K/ell)");
assert(B.A1_hilbert == [1] and B.AK_hilbert == [1, 1] and B.hidden, "hidden congruence at
↪  (300,23)");

# dihedral pattern: -23 is a square mod p iff p splits in K = Q(sqrt(-23))
i = 0;
for p in R.primes {
    if p != 23 {
        sq = false;
        for x in range(0, p) {
            if Fp(p, x * x + 23) == Fp(p, 0) { sq = true; }
        }
        if sq { tag = "split"; sgn = "+1"; } else { tag = "inert"; sgn = "-1"; }
        print("DIHEDRAL p=" + str(p) + " (-23|p)=" + sgn + " " + tag + " a_p=" +
        ↪  str(B.system[i]));
        if not sq { assert(B.system[i] == 0, "a_p = 0 at an inert prime"); }
    }
    i = i + 1;
}
print("OTHER BLOCKS hidden:", nhid, "of", len(R.blocks), "; D=4 with A1=[1,1]:", nsplit);
```

Expected output (generated file `phenomena_gallery/hidden_congruence_300_23.out`):

```
BLOCKS 14 primes [2, 3, 5, 7, 23] K_cls 2 K_run [2]
BLOCK system=[19, 14, 0, 0, 0] D=4 verdict=stable
  A1_hilbert (mod 23)     = [1]      -> one eigensystem mod 23
  AK_hilbert (mod 23^2)/ell = [1, 1]   -> F_23[x]/x^2 over Z/23^2
  hidden = true  (A_1 smaller than A_K/ell)
DIHEDRAL p=2 (-23|p)=+1 split a_p=19
DIHEDRAL p=3 (-23|p)=+1 split a_p=14
DIHEDRAL p=5 (-23|p)=-1 inert a_p=0
DIHEDRAL p=7 (-23|p)=-1 inert a_p=0
OTHER BLOCKS hidden: 1 of 14 ; D=4 with A1=[1,1]: 8
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 4.63 s, 370 MiB.

What to look for.

- `A1_hilbert = [1]` against `AK_hilbert = [1, 1]`: the congruence that $\mathbb{F}_\ell$-coefficients miss;
- compare §9, where both are `[1, 1]`;

- `OTHER BLOCKS hidden: 1 of 14`: hidden congruences are rare; eight other $D = 4$ blocks are visible.

11 A dihedral residual representation: Wilton's congruence mod 23

The mod-23 Galois representation $\bar{\rho}_{\Delta,23}$ is induced from $K = \mathbb{Q}(\sqrt{-23})$: K has class number 3, its Hilbert class field H is the splitting field of $x^3 - x - 1$, and $\bar{\rho}_{\Delta,23} \cong \mathrm{Ind}_K^{\mathbb{Q}} \chi$ for a character χ of $\mathrm{Gal}(H/K) \cong \mathbb{Z}/3$ ([LIT]: Wilton 1930; Serre, Swinnerton-Dyer). Taking traces of Frobenius:

$$\tau(p) \equiv \begin{cases} 0 & p \text{ inert in } K,\ (\frac{-23}{p}) = -1, \\ 2 & p = x^2 + xy + 6y^2 \text{ (split, principal prime ideals)}, \\ -1 & p = 2x^2 + xy + 3y^2 \text{ (split, non-principal)}, \end{cases} \pmod{23}.$$

The snippet reads $\tau(p) \bmod 23$ off $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^{10} \mathbb{F}_{23}^2)$ basis-independently: since $10 < 23$ the reduction of the characteristic-zero Hecke module is the $\mathbb{F}_{23}$-module, so $\mathrm{tr}\, T_p = 2\tau(p) + (1 + p^{11})$ (two cusp classes, one Eisenstein class), and $\tau(p) = (\mathrm{tr}\, T_p - 1 - p^{11})/2$ in $\mathbb{F}_{23}$. It classifies each $p < 175$, $p \neq 23$, by splitting and by the principal form (via $4p = (2x + y)^2 + 23y^2$) and asserts the three cases. The first principal primes are $59, 101, 167, 173$; note that $1 + p^{11} \equiv 1 + (\frac{p}{23})$ as well, so the Eisenstein eigenvalue also vanishes at the inert primes.

Dihedral residual representations are where hidden congruences concentrate (§10, [LG, sec:cm]); the dihedral ordinary blocks of the census are [EX, s14:dihedral].

The snippet `notebook/examples/dihedral_wilton_10_23.oc`:

```
# dihedral_wilton_10_23.oc -- tau(p) mod 23 is governed by K = Q(sqrt(-23)) (Wilton 1930):
↪  rhobar_Delta mod 23 is dihedral.
# Phenomena gallery [PG] sec:pg-dihedral.  Cell (n, ell) = (10, 23).  Read_only verbs only
↪  (notebook-safe).
# Cost: ~2 s, < 0.1 GB (the cosets of T_p grow like p).
#
#   p inert in K ((-23|p) = -1)                 :  tau(p) == 0   (mod 23)
#   p = x^2 + xy + 6y^2 (split, principal class) :  tau(p) == 2
#   p = 2x^2 + xy + 3y^2 (split, non-principal)  :  tau(p) == -1
# tau(p) is read off H^1 basis-independently: trace T_p = 2 tau(p) + (1 + p^11) (two cusp
↪  classes, one Eisenstein).
M = sl2z_model();
H = cohomology(symmetric_power(M, 10, 23));
print("dim H^q =", H.dims);
for p in [2, 3, 5, 7, 11, 13, 17, 19, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83,
↪  89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173] {
    # the principal form represents p iff 4p = (2x + y)^2 + 23 y^2 has a solution
    principal = false;
    for y in range(1, 6) {
        r = 4 * p - 23 * y * y;
        for s in range(0, 27) { if s * s == r { principal = true; } }
    }
    split = false;
    for x in range(0, p) { if (x * x + 23) % p == 0 { split = true; } }
    x11 = 1;
    for i in range(0, 11) { x11 = (x11 * p) % 23; }
    t = (trace(hecke(H, p).A(1)) - 1 - x11 + 3 * 23) % 23;
    tau = (t * 12) % 23;
    if not split { cls = "inert       "; want = 0; }
    else { if principal { cls = "split, x^2+xy+6y^2 "; want = 2; } else { cls = "split,
    ↪  2x^2+xy+3y^2"; want = 22; } }
    print("p =", p, cls, " tau(p) mod 23 =", tau);
    assert(tau == want, "Wilton's congruence");
}
```

Expected output (generated file `phenomena_gallery/dihedral_wilton_10_23.out`):

```
dim H^q = [0, 3, 0]
p = 2 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 3 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 5 inert         tau(p) mod 23 = 0
p = 7 inert         tau(p) mod 23 = 0
p = 11 inert         tau(p) mod 23 = 0
p = 13 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 17 inert         tau(p) mod 23 = 0
p = 19 inert         tau(p) mod 23 = 0
p = 29 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 31 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 37 inert         tau(p) mod 23 = 0
p = 41 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 43 inert         tau(p) mod 23 = 0
p = 47 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 53 inert         tau(p) mod 23 = 0
p = 59 split, x^2+xy+6y^2   tau(p) mod 23 = 2
p = 61 inert         tau(p) mod 23 = 0
p = 67 inert         tau(p) mod 23 = 0
p = 71 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 73 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 79 inert         tau(p) mod 23 = 0
p = 83 inert         tau(p) mod 23 = 0
p = 89 inert         tau(p) mod 23 = 0
p = 97 inert         tau(p) mod 23 = 0
p = 101 split, x^2+xy+6y^2   tau(p) mod 23 = 2
p = 103 inert         tau(p) mod 23 = 0
p = 107 inert         tau(p) mod 23 = 0
p = 109 inert         tau(p) mod 23 = 0
p = 113 inert         tau(p) mod 23 = 0
p = 127 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 131 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 137 inert         tau(p) mod 23 = 0
p = 139 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 149 inert         tau(p) mod 23 = 0
p = 151 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 157 inert         tau(p) mod 23 = 0
p = 163 split, 2x^2+xy+3y^2  tau(p) mod 23 = 22
p = 167 split, x^2+xy+6y^2   tau(p) mod 23 = 2
p = 173 split, x^2+xy+6y^2   tau(p) mod 23 = 2
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 3.14 s, 137 MiB.

What to look for.

- `tau(p) mod 23 = 0` on every inert line, `= 2` exactly at `p = 59, 101, 167, 173`, `= 22` ($= -1$) elsewhere;
- the eigenvalues come from the traces of the computed T_p, not from a table of τ.

12 Multiplicity one fails mod 5

Over $\mathbb{Q}$ every Hecke eigensystem of level one occurs in $H^1_\pm$ with multiplicity one. Mod ℓ this fails: distinct eigenforms can share a residual system, and they can share it *including* U_ℓ. At $\ell = 5$ every residual system of level one is reducible (a θ-twist of an Eisenstein system); the atlas $\Sigma(5)$ [E, sec:atlas] labels them by their characters $\chi = [a, b]$, $\bar\rho = \omega^a \oplus \omega^b$.

At weight $36 = 12 + 5^2 - 1$ ($n = 34$), H^1 has dimension 7 ($\dim S_{36} = 3$). The extension record finds three maximal ideals:

- atlas system 2, $\chi = [1, 2]$, i.e. $a_p \equiv p(1 + p) \equiv \tau(p)$ (Ramanujan's mod-5 system): one ideal of dimension 4, nilpotency 1 (the local algebra is $\mathbb{F}_5$), $U_5 = 0$ — two eigenforms congruent to

Δ mod 5 that no Hecke operator, U_5 included, separates, with the module $\mathbb{F}_5^4$ semisimple (2 in H^1_+). This is [EX, ex:mult4]: Glover's composition factors count the multiplicity and the θ-images from weight 30 account for it;

- atlas system 1, $\chi = [0, 3]$ (that of E_{36}): two ideals with the same a_2, a_3, separated by U_5 ($U_5 \equiv 1$ on the Eisenstein line, $U_5 \equiv 0$ on the two classes of a cusp form with the same anemic system).

So the anemic Hecke algebra (without U_ℓ) does not separate the second pair, and the full Hecke algebra does not separate the first.

The snippet `notebook/examples/multiplicity_two_34_5.oc`:

```
# multiplicity_two_34_5.oc -- multiplicity one fails mod 5 at weight 36 = 12 + ell^2 - 1: one
↪  residual eigensystem
# occupies two dimensions of H^1_+, and the anemic Hecke operators cannot separate two ideals
↪  that U_5 does separate.
# Phenomena gallery [PG] sec:pg-mult2.  Cell (n, ell) = (34, 5).  Read_only verbs only
↪  (notebook-safe).
# Cost: < 1 s, < 0.1 GB.
n = 34;
ell = 5;
assert(n + 2 == 12 + ell^2 - 1, "weight 12 + ell^2 - 1");
R = ext_record(n, ell, sign_split = true);
print("operators", R.operators, " separating set S_sep", R.s_sep, " ideals", R.num_ideals);
for m in R.ideals {
    print("ideal", m.id, " dim", m.dim, " atlas system", m.label.system, " chi", m.label.chi,
          " a_2", m.character(2), " a_3", m.character(3), " U_5", m.character(5), " nu",
          ↪  m.nilpotency);
}
for m in R.ideals_plus { print("H^1_+ ideal", m.id, " dim", m.dim, " system", m.label.system, "
↪  nu", m.nilpotency); }

# (a) Delta's residual system omega + omega^2 (atlas system 2) fills a 4-dimensional block,
↪  semisimple (nu = 1),
#     U_5 = 0: two weight-36 eigenforms congruent to Delta mod 5, which no Hecke operator
↪  separates.
# (b) The system 1 + omega^3 (atlas system 1, that of E_36) carries two ideals: T_2, T_3, T_7
↪  agree, U_5 does not.
m0 = R.ideals[0];
m1 = R.ideals[1];
m2 = R.ideals[2];
assert(m2.label.chi == [1, 2] and m2.dim == 4 and m2.nilpotency == 1, "(a)");
assert(m0.label.system == m1.label.system and m0.character(5) != m1.character(5), "(b)");
```

Expected output (generated file `phenomena_gallery/multiplicity_two_34_5.out`):

```
operators [2, 3, 5, 7]  separating set S_sep [2]  ideals 3
ideal 0  dim 2  atlas system 1  chi [0, 3]  a_2 4 (mod 5)  a_3 3 (mod 5)  U_5 0 (mod 5)  nu 1
ideal 1  dim 1  atlas system 1  chi [0, 3]  a_2 4 (mod 5)  a_3 3 (mod 5)  U_5 1 (mod 5)  nu 1
ideal 2  dim 4  atlas system 2  chi [1, 2]  a_2 1 (mod 5)  a_3 2 (mod 5)  U_5 0 (mod 5)  nu 1
H^1_+ ideal 0  dim 1  system 1  nu 1
H^1_+ ideal 1  dim 1  system 1  nu 1
H^1_+ ideal 2  dim 2  system 2  nu 1
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.55 s, 51 MiB.

What to look for.

- `ideal 2 dim 4 ...chi [1, 2] ...nu 1`: multiplicity two (four classes, two per sign) with a reduced algebra;
- `ideal 0` and `ideal 1`: identical `a_2`, `a_3`, different `U_5`;
- `S_sep [2]`: T_2 alone separates the residual systems of $\Sigma(5)$, not the ideals.

13 The θ-operator and the Dickson sequence

On mod-ℓ modular forms, $\theta = q\,\frac{d}{dq}$ raises the weight by $\ell+1$ and multiplies a_p by p ([LIT], Serre, Swinnerton-Dyer). On cohomology the same map is multiplication by the Dickson invariant $L = X^\ell Y - XY^\ell$ `[C, def:dicksoninv]`, which gives a short exact sequence of GL_2-modules over $\mathbb{F}_\ell$

$$0 \to \mathrm{Sym}^n \otimes \det \xrightarrow{\cdot L} \mathrm{Sym}^{n+\ell+1} \to Q_n \to 0$$

and the map θ_L on cohomology `[C, def:theta]`. Its long exact sequence `[C, thm:thetales]` [PROVED]

$$0 \to H^0_n \xrightarrow{\theta} H^0_\uparrow \to H^0(Q_n) \xrightarrow{\delta} H^1_n \xrightarrow{\theta} H^1_\uparrow \to H^1(Q_n) \to 0$$

is exact, so $\ker\theta^1 = \mathrm{im}\,\delta$; and θ is Hecke-equivariant only up to the determinant twist, $T_p\,\theta = p\,\theta\,T_p$, with $U_\ell\,\theta = 0$ `[C, prop:thetatwist]` [PROVED]. The console objects are [E, sec:thetaobj].

At $(16,7)$, $n+\ell+1 = 24$ (weights $18 \to 26$): the sequence is exact with δ of rank 1 (a Dickson class of $H^0(Q_{16})$ kills one class of H^1), θ^1 has rank 3, and the twisted identity holds at $p = 2,3,5$ while the untwisted one fails. The eigenvalues of T_p on the two H^1's show the twist directly: $\{4,5\}\cdot 2 = \{1,3\}$, $\{0,1,6\}\cdot 3 = \{0,3,4\}$, $\{0,3,4\}\cdot 5 = \{0,1,6\}$ in $\mathbb{F}_7$ (the images match because θ^1 has rank 3 and both spaces have dimension 4).

The snippet `notebook/examples/dickson_theta_16_7.oc`:

```
# dickson_theta_16_7.oc -- the theta operator q d/dq on cohomology: the Dickson map L_* and its
↪  long exact sequence.
# Phenomena gallery [PG] sec:pg-theta.  Cell (n, ell) = (16, 7) -> n + ell + 1 = 24.  Read_only
↪  verbs only (notebook-safe).
# Cost: < 1 s, < 0.1 GB.
#
#   0 -> Sym^n (x) det -> Sym^(n+ell+1) -> Q_n -> 0   (over F_ell; L = X^ell Y - X Y^ell, the
↪  Dickson invariant)
M = sl2z_model();
n = 16;
ell = 7;
D = dickson_les(n, ell);
print("dims", D.dims);
print("ranks", D.ranks, " exact:", D.exact, " ker theta^1 = im delta:", D.ker_equals_im, "
↪  certificate", D.certificate);
H = cohomology(symmetric_power(M, n, ell));
Hu = cohomology(symmetric_power(M, n + ell + 1, ell));
T = theta(H, Hu);
t = T.induced[1];
print("theta^1 : H^1(Sym^16 (x) det) -> H^1(Sym^24), rank", T.ranks[1], " twist check",
↪  T.twist_check, " U_7 theta = 0:", T.u_ell_zero);

# The twist: A(T_p) theta = p theta A(T_p).  So theta multiplies the eigenvalue a_p by p (the
↪  det twist).
for p in [2, 3, 5] {
    A = hecke(H, p).A(1);
    Au = hecke(Hu, p).A(1);
    print("p =", p, " A_up theta == p theta A:", mul(Au, t) == scale(mul(t, A), p), "   A_up
    ↪  theta == theta A:", mul(Au, t) == mul(t, A));
    ev = "";
    evu = "";
    for x in range(0, ell) {
        if eval(charpoly(A), x) == 0 { ev = ev + " " + str(x); }
        if eval(charpoly(Au), x) == 0 { evu = evu + " " + str(x); }
    }
    print("   eigenvalues of T_p on H^1(Sym^16):" + ev + "   on H^1(Sym^24):" + evu);
}
```

Expected output (generated file `phenomena_gallery/dickson_theta_16_7.out`):

```
dims {H0_n: 1, H0_up: 1, H0_Q: 1, H1_n: 4, H1_up: 4, H1_Q: 1}
ranks {theta0: 1, pi0: 0, delta: 1, theta1: 3, pi1: 1}  exact: true  ker theta^1 = im delta:
↪  true  certificate 0
theta^1 : H^1(Sym^16 (x) det) -> H^1(Sym^24), rank 3  twist check 0  U_7 theta = 0: true
p = 2  A_up theta == p theta A: true    A_up theta == theta A: false
   eigenvalues of T_p on H^1(Sym^16): 4 5   on H^1(Sym^24): 1 3
p = 3  A_up theta == p theta A: true    A_up theta == theta A: false
   eigenvalues of T_p on H^1(Sym^16): 0 1 6   on H^1(Sym^24): 0 3 4
p = 5  A_up theta == p theta A: true    A_up theta == theta A: false
   eigenvalues of T_p on H^1(Sym^16): 0 3 4   on H^1(Sym^24): 0 1 6
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.13 s, 35 MiB.

What to look for.

- `ranks {theta0: 1, pi0: 0, delta: 1, theta1: 3, pi1: 1}` and `exact: true`;
- `A_up theta == p theta A: true` against `A_up theta == theta A: false`;
- multiply the left eigenvalue set by p modulo 7 to get the right one.

14 A mod-37 Hecke algebra of Loewy length 4

The census found the first mod-ℓ Hecke algebras of Loewy length 4 on H^1 at $\ell = 37$ (edition 4.0 adds $\ell = 67$ and $\ell = 139$, §15), at the residual representations $\bar{\rho}_a = \omega^a \oplus \omega^{a+5}$ from weight $k = 222 + 38a$ on [L37]: the local algebra is $\mathbb{F}_{37}[x]/x^4$ and the 8-dimensional block is $A \oplus A/x^3 \oplus A/x$. The block is cut out by four cusp eigenforms reducing to $\theta^a E_6 \equiv \theta^{a+5} E_{32}$, and [L37]'s dimension formula traces the rank 4 to Ribet's cusp form of weight 32 (§7); the families and the fields are [L37, sec:families], [L37, sec:fieldsribet].

The snippet takes the first cell of the $a = 1$ family, $(258, 37)$, weight 260. Part (1) is the $\mathbb{F}_{37}$ extension record (two seconds): the 8-dimensional ideal with $a_2 = 2 + 2^6 = 29$, $U_{37} = 0$ (non-ordinary), local algebra of Hilbert function $(1, 1, 1, 1)$, Gorenstein, embedding dimension 1, Loewy series $[3, 2, 2, 1]$ and Jordan type $(4, 3, 1)$ of the nilpotent generator: $H^1_{\mathfrak{m}} \cong A \oplus A/x^3 \oplus A/x$. Part (2) is the $\mathbb{Z}/37^2$ route (about a minute): the block is torsion-free of rank 8, $\mathsf{A}_1 = \mathsf{A}_K/\ell$ with Hilbert function $(1, 1, 1, 1)$, so nothing is hidden and the census algebra is the saturated one, by the certified reading `[C, cor:stablecert]`.

The snippet `notebook/examples/loewy4_258_37.oc`:

```
# loewy4_258_37.oc -- a mod-37 Hecke algebra of Loewy length 4: T_m/37 = F_37[x]/x^4 at weight
↪  260.
# Phenomena gallery [PG] sec:pg-loewy4.  Cell (n, ell) = (258, 37), rhobar = omega (+) omega^6
↪  (a = 1 in [L37]).
# Read_only verbs only (notebook-safe).  Cost: ~65 s, 0.65 GB (part (2), the Z/37^2 route, is
↪  ~63 s of it).
n = 258;
ell = 37;

# (1) The F_37 extension record: the 8-dimensional block, its algebra and its module structure.
R = ext_record(n, ell);
print("ideals", R.num_ideals, " operators", R.operators);
BI = -1;
for m in R.ideals { if m.dim == 8 { BI = m.id; } }
m = R.ideal(BI);
a2 = (2 + 2^6) % ell;
print("block: dim", m.dim, " a_2 =", m.character(2), " (omega(2) + omega^6(2) =", a2, ")  U_37
↪  =", m.character(37));
T = local_algebra(m);
P = m.modules[0];
print("T_m/37: dim", T.dim, " Hilbert function", T.hilbert, " Gorenstein", T.gorenstein, "
↪  embdim", T.embdim);
```

```
print("H^1_m: Loewy", P.loewy, " socle", P.socle, " Jordan type of n_2", P.jordan(2), "
↪  cyclic", P.cyclic);
assert(T.hilbert == [1, 1, 1, 1], "T_m/37 = F_37[x]/x^4");
assert(P.jordan(2) == [4, 3, 1], "H^1_m = A + A/x^3 + A/x");

# (2) Saturation: the Z/37^2 Hecke-algebra route.  A_1 (the image mod 37) equals A_K/37:
↪  nothing is hidden.
A = hecke_ring_algebra(n, ell, 2);
for b in A.blocks {
    if b.D == 8 {
        print("Z/37^2 route: D", b.D, " free_rank", b.free_rank, " torsion", b.torsion, " A1",
        ↪  b.A1_hilbert, " AK/ell", b.AK_hilbert,
              " hidden", b.hidden, " verdict", b.verdict, " (", b.stable_cert, ")");
        assert(b.A1_hilbert == [1, 1, 1, 1] and not b.hidden, "saturated, Loewy length 4");
    }
}
```

Expected output (generated file `phenomena_gallery/loewy4_258_37.out`):

```
ideals 10  operators [2, 3, 5, 7, 37]
block: dim 8  a_2 = 29 (mod 37)  (omega(2) + omega^6(2) = 29 )  U_37 = 0 (mod 37)
T_m/37: dim 4  Hilbert function [1, 1, 1, 1]  Gorenstein true  embdim 1
H^1_m: Loewy [3, 2, 2, 1]  socle [3, 2, 2, 1]  Jordan type of n_2 [4, 3, 1]  cyclic false
Z/37^2 route: D 8  free_rank 8  torsion []  A1 [1, 1, 1, 1]  AK/ell [1, 1, 1, 1]  hidden false
↪  verdict stable  ( certified )
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 4.25 s, 725 MiB.

What to look for.

- `Hilbert function [1, 1, 1, 1]`: $\mathbb{F}_{37}[x]/x^4$, Loewy length 4;
- `Jordan type of n_2 [4, 3, 1]`: the module $A \oplus A/x^3 \oplus A/x$, not cyclic;
- `A1 [1, 1, 1, 1]` = `AK/ell [1, 1, 1, 1]`, `verdict stable (certified)`.

15 Census 4.0 at a glance: the shape of every non-reduced algebra (runs)

Six algebras. [CMPD] (each row certified) and [EMP] (as a pattern). Every one of the 49 176 non-reduced census algebras of the signed edition 4.0 (main + b1) is one of six algebras ($\mathbb{F}_q$ the residue field):

A	rows	Hilbert function	Gorenstein
$\mathbb{F}_q[x]/x^2$	46 185	$(1,1)$	yes
$\mathbb{F}_q[x]/x^3$	414	$(1,1,1)$	yes
$\mathbb{F}_q[x]/x^4$	248	$(1,1,1,1)$	yes
$\mathbb{F}_q[x,y]/(x,y)^2$	2016	$(1,2)$	no
$\mathbb{F}_q[x,y,z]/(x,y,z)^2$	303	$(1,3)$	no
$\mathbb{F}_q[x,y]/(x^3,xy,y^2)$ (embedding dimension 2, $\mathfrak{m}_A^3 = 0 \neq \mathfrak{m}_A^2$)	10	$(1,2,1)$	no

The shape $(1,2,1)$ is a single algebra: $\mathfrak{m}_A^2$ is a line, the multiplication form on $\mathfrak{m}_A/\mathfrak{m}_A^2$ is non-degenerate exactly when A is Gorenstein, and a non-Gorenstein one has a rank-one form, unique up to scaling, whence $\mathbb{F}_q[x,y]/(x^3,xy,y^2)$. Section b1 contributes only $(1,1)$ (2542 rows) and $(1,1,1,1)$ (15 rows); the partial section b2 (pre-release) has 825 non-reduced rows, all $\mathbb{F}_q[x]/x^2$. So, on the signed census:

- a census algebra is Gorenstein if and only if it is curvilinear (0 exceptions);
- the embedding dimension is at most 3, the tangent-space bound of [LG, prop:tangent];

- a non-curvilinear algebra has $\mathfrak{m}_A^2 = 0$, and an embedding dimension of at least 2 never comes with Loewy length ≥ 3 — *except* on the 10 rows of shape $(1,2,1)$, all at $n > 700$: eight dihedral rows at $\ell = 23$ ($n = 826, \ldots, 980$, $D = 16$, module Loewy series $[8,4,4]$; §16) and two at $\ell = 59$, $(942, 59)$ and $(1000, 59)$ ($D = 8$, $[4,2,2]$). On edition 3.4's window $n \leq 700$ both statements hold without exception.

Where each shape occurs.

- **Loewy length** ≥ 3 occurs on 672 rows, at $\ell \in \{23, 31, 37, 59, 67, 139, 547\}$:
  - 418 dihedral rows at 23 and 31, Loewy length 3 (§16);
  - 233 Kummer–Frobenius rows of Loewy length 4: 222 at 37 from $n = 258$ (§14, [L37]) and 11 at 67 from $n = 736$, at the residual representations $\omega^a \oplus \omega^{a+9}$ (67 is irregular at $k = 58$, and $9 = 67 - 58$ as $5 = 37 - 32$); all listed in `kummer_frobenius40.tsv`;
  - two Kummer rows at 547: $(484, 547)$, which is ordinary (§25), and $(608, 547)$;
  - 4 rows at 59 from $n = 884$, Loewy length 3, labelled $\theta^0(f_{16})$, $\theta^1(f_{16})$ (the label rests on evidence at 61 primes, not on a certificate);
  - 15 rows at 139 (section b1) from $n = 416$: $A = \mathbb{F}_{139}[x]/x^4$, module Loewy series $[2,2,2,2]$, $D = 8$, labelled $\theta^j(f_{138})$, $j = 2, \ldots, 6$; they are not Kummer–Frobenius.

  The rows at 59 and 139 are new in 4.0 (at $n > 700$, resp. at the new primes). A referee check against the census records (2026-11-06; four-prime comparison of the systems (a_2, a_3, a_5, a_7), not a certificate) gives: at $\ell = 139$ the class has a congruence in weight 138 (the ideal $(136, 139)$ id 2, $D = 4$, Hilbert function $(1,1)$; evidence on four primes), so it is not congruence-free and [E, pr:latticesplit] (R) says nothing about these 15 rows. At $\ell = 59$ the class is irreducible, of residue degree one, not dihedral ($(2/59) = -1$ and $a_2 \neq 0$) and congruence-free in the sense of §16 (the only low-weight matches are single forms, at weights 16 and 46), so the 4 rows at 59 are a *candidate* counterexample to (R), under independent verification and pending the owner's ruling; the gallery claims no more than this. (1.4.2: verified independently, `docs/implementation/slot25/R59_verification.md`, $(T_2 - a_2)^2 \neq 0$ recomputed at all four cells; the class is non-ordinary, $a_{59} \equiv 0$. By owner ruling of 2026-11-06 [E] 1.25.0-pre.1 restates (R): on irreducible classes it assumes "of one weight", exactly one pair (w, j) of the test with a non-zero eigenspace, which the class at 59, at $(16, 0)$ and $(46, 14)$, is not.)
- **The non-Gorenstein algebras** occur only at $\ell \in \{7, 11, 13, 17, 19, 23, 29, 59\}$. Their first cells are $n = 82$, 174, 206, 338, 416, 596, 926 at $\ell = 7$, 11, 13, 17, 19, 23, 29 (421, 680, 569, 282, 222, 148, 5 rows), roughly $n \approx \ell^2 + 30 \ldots 85$ ([EMP]); at 59 they are the two $(1,2,1)$ rows above, from $n = 942$, off that pattern.
- **Embedding dimension** 3 occurs only at $\ell = 7$, 11, 13, 17, from $n = 142$, 354, 496, 852.

In [EX] these first appear at $(82, 7)$ [EX, ex:nongor] and $(142, 7)$ [EX, ex:largest]. That document's "What we have not seen" list [EX, sec:notseen] is scoped to $n \leq 200$ and remains true there. "Only at $\ell \leq 29$" (non-Gorenstein, apart from the two rows at 59) and the list of primes with Loewy length ≥ 3 are statements about the window of the census (even $n \leq 1000$, $\ell \leq 199$): the onsets grow like ℓ^2 (non-Gorenstein, dihedral) or ℓw (Kummer–Frobenius), and the window of 4.0 already moved both lists.

The literature (search of 2026-10-19). Failure of the Gorenstein property for mod-p Hecke algebras is classical in weight 2 and weight p, through "unramified at p with scalar Frobenius" (Kilford–Wiese [KW08]); the embedding-dimension bound is the tangent-space bound (Bellaïche–Khare [BK15], Deo [Deo17]). Our A is the image on mod-ℓ cohomology; when that agrees with $\mathbb{T}_{\mathfrak{m}}/\ell$ is the subject of Wiese [Wie]. For the dichotomy "curvilinear or $\mathfrak{m}^2 = 0$" on edition 3.4 (which 4.0 breaks at $n > 700$ by the ten $(1,2,1)$ rows) and for the non-Gorenstein onset near $n \approx \ell^2$ in weights $k \gg \ell + 1$, *no prior was found in our search (2026-10-19).* A mechanism to test: the first n at which a Serre weight occurs twice among the composition factors of $\mathrm{Sym}^n \mathbb{F}_\ell^2$.

Hidden congruences ($\dim A < \dim \mathbb{T}_{\mathfrak{m}}/\ell$ on a torsion-free block; §10) are certified on 196, 338, 164, 144, 31, 10 rows at $\ell = 7, 11, 13, 17, 19, 23$ (883 rows; the flag is computed only where a weight-shift run exists, on 16 117 rows, and never on a block with torsion; a further 459 rows with a run read `not_stable`: 454 torsion blocks and 5 torsion-free blocks not stable by `KTOP`). None occurs on the computed rows at any $\ell \geq 29$, section b1 included (738 rows read `false`). At 17 and 19 they begin in "Frobenius layers" $n = \ell k_0 + j(\ell+1) - 2$ [E, pr:frobeniuslayer]. At 23 all ten are dihedral twists of Δ.
Sources: `docs/matrix-engine/results/congruence_census/edition3/4.0/` (`rows40.tsv.gz`, `rows40_b1.tsv.gz`, `rows40_b2.tsv.gz`, `category_table40*.txt`, `findings40*.txt`, `kummer_frobenius40.tsv`); every number recomputed by `docs/latex/tools/pg_census_check.py`; the onset analysis `docs/implementation/slot22/C33_followups_report.md` §F33-3.

16 Dihedral classes: Loewy length 3 on congruence-free classes (runs)

The conjecture. The conjecture [E, pr:latticesplit] ("nearly split lattice", (R)) predicts $\mathfrak{m}_A^2 = 0$ on every congruence-free class: a class whose Serre weights $w \leq \ell + 1$ carry no congruence (every generalised eigenspace in $M_w(\mathbb{F}_\ell)$ has dimension ≤ 1).

The counterexample. Census 3.3 contradicted it on 161 rows ([CMPD]; report F33-2, refereed, owner ruling of 2026-10-09):

- *The classes.* They are the twists of $\bar{\rho}_\Delta$ mod 23 and of $\bar{\rho}_{f_{16}}$ mod 31. Both are dihedral, induced from $\mathbb{Q}(\sqrt{-\ell})$, and congruence-free. Their eigensystems are $(a_2, a_3, a_5, a_7) \equiv (22, 22, 0, 0)$ mod 23 and $(30, 0, 30, 30)$ mod 31.
- *The algebra.* On those 161 rows of 3.3/3.4 ($n \leq 700$) the census algebra is $A \cong \mathbb{F}_\ell[x]/x^3$, with module Loewy series $[4, 2, 2]$ or $[6, 4, 2]$: 140 rows at $\ell = 23$ (from $n = 298$; $D = 8$ or 12) and 21 at $\ell = 31$ (from $n = 526$). The signed census 4.0 (main section, $n \leq 1000$) has 418 dihedral rows of Loewy length 3: 286 at $\ell = 23$ and 132 at $\ell = 31$ (onsets unchanged). Above $n = 700$ the blocks at 23 grow: $D = 12$ with module Loewy series $[8, 2, 2]$ (from $n = 782$) and $D = 16$ with $[8, 4, 4]$ (from $n = 826$), and on eight of the $D = 16$ rows ($n = 826, \ldots, 980$) A is no longer curvilinear but has Hilbert function $(1, 2, 1)$ (§15); every other dihedral row has $A \cong \mathbb{F}_\ell[x]/x^3$.
- *The mechanism.* $U_\ell = 0$ on these blocks. $(T_q - a_q)(T_{q'} - a_{q'}) \neq 0$ on $H^1_{\mathfrak{m}}$ exactly when q and q' both split in $\mathbb{Q}(\sqrt{-\ell})$. The inert primes and U_ℓ lie in $\mathfrak{m}_A^2$.
- *The forms side.* $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[x]/x^4$ at $(298, 23)$, $(540, 23)$ and $(526, 31)$, with $\delta = 1, 2, 1$.
- *The ℓ-adic signature* (mod ℓ^6). The Newton slope is 1/2 at split primes and 1 at inert primes; there is no $\mathbb{Z}_\ell$-rational eigenform.

After the counterexample. [E] now assumes "congruence-free *and not dihedral*". On the remaining 100 025 residue-degree-one, non-Dickson rows of 3.3 there is no row of Loewy length 3. The 84 rows of Loewy length 4 of 3.3 are all Kummer–Frobenius, hence never congruence-free. Census 4.0 adds Loewy-length-≥ 3 rows that are neither dihedral nor Kummer–Frobenius: 4 at $\ell = 59$ ($n \geq 884$) and 15 at $\ell = 139$ (section b1), §15. A referee check on the census records (§15; four-prime evidence, not a certificate) finds that the class at $\ell = 139$ has a congruence in weight 138, so (R) says nothing there, while the class at $\ell = 59$ is congruence-free and not dihedral: its 4 rows are a *candidate* counterexample to the restated hypothesis, under independent verification and pending the owner's ruling. Until that is settled, the restated (R) is tested with certainty only on 3.3. (1.4.2: settled. The four rows are a counterexample, verified from scratch, `docs/implementation/slot25/R59_verification.md`; the class is non-ordinary and has two cuspidal Serre weights. By owner ruling of 2026-11-06 [E] 1.25.0-pre.1 replaces "not dihedral" by "of

one weight" on irreducible classes; that excludes the dihedral, the non-ordinary and the companion classes, and no row of census 4.0 contradicts the new form, [E, pr:latticesplit].)

The onset. The onsets $n_0 = 298, 526$ fit $n_0 = \ell(\ell+3)/2 - 1$ ([EMP], two points). The next primes $\ell \equiv 3 \pmod 4$ with $h(-\ell) > 1$ are 47 ($h = 5$), 59, 71, 79, 83, 103, 107, so the fit predicts $n_0 = 1174$ at $\ell = 47$ and 1828 at $\ell = 59$, both beyond the census grid.

The literature (search of 2026-10-19): *no prior found.* The nearest results are Deo's non-reduced dihedral algebras [Deo17, Prop. 27–28], whose nilpotents come from diamond operators with p unramified in the CM field (here ℓ is ramified in $\mathbb{Q}(\sqrt{-\ell})$ and there are no diamond operators), and Bellaïche–Khare's $\mathbb{F}[[x, y]]$ for the algebra of all weights [BK15]. A fixed-weight quotient of $\mathbb{F}[[x, y]]$ can be non-reduced without any obstruction, so the phenomenon may still follow from [BK15, Deo17] and the CM geometry; the next check is $\dim H^1(G_{\mathbb{Q},\{\ell\}}, \mathrm{ad}^0\bar\rho)$ for $\bar\rho_\Delta$ mod 23.
Sources: `docs/implementation/slot22/C33_followups_report.md` §F33-2, `F33-2_referee.md`, `owner_ruling_F33-2_1009.md`.

17 The saturation defect in the census (runs)

$\delta > 0$ means that the Hecke algebra $\mathbb{T}_\mathfrak{m}$ on the free lattice $\Lambda^{\mathrm{fr}}_\mathfrak{m}$ is not saturated: $\mathsf{A}_1 = \mathbb{T}_\mathfrak{m}/\ell$ fails, and some congruence is hidden or deepened (§10). The census computes δ (column `sat_length`), together with $\dim \mathrm{Tor}_1^{\mathbb{Z}_\ell}(\mathbb{T}^{\mathrm{sat}}_\mathfrak{m}/\mathbb{T}_\mathfrak{m}, \mathbb{F}_\ell)$ and its Loewy length, by `HeckeSaturationDefectValidation` on the torsion-free blocks of the cells $n \le 700$ of the main section; each value is certified final (a plateau of ε_K, `[C, prop:satdefect]`(2)). The 1593 rows of blocks with torsion are refused by name (§2, "Two ℓ-adic objects"); the cells $n > 700$ and section b1 had no saturation pass. In the signed census 4.0 ([CMPD]):

- *Coverage.* δ is computed on 62 021 rows (edition 3.4: 56 287; the step SATC of 3.5 added the cells at $\ell = 5$, 7, 11, 13, 547, 691, at $\ell = 7$ up to $K = 16$).
- *Where $\delta > 0$.* 8954 rows have $\delta > 0$, all at $\ell \le 53$ except one: $(272, 547)$, ideal 0, $\delta = 1$. Only $\ell \le 23$ has $\delta \ge 2$.
- *The largest value.* The largest is $\delta = 190$ at $(698, 7)$ ideal 4, where $\mathrm{Tor}_1^{\mathbb{Z}_\ell}(\mathbb{T}^{\mathrm{sat}}/\mathbb{T}_\mathfrak{m}, \mathbb{F}_\ell)$ has dimension 26 and Loewy length 6. (Edition 3.4, whose $\ell = 7$ values stopped at $n = 300$, had its maximum $\delta = 29$ at $(298, 7)$ ideal 5, Tor of dimension 9 and Loewy length 4.)
- *The block of Δ at* $(700, 7)$ (ideal 5, $D = 58$). Over $\mathbb{Z}/7^K$ the defect is $\varepsilon_K = 26, 49, 70, 91, 109, 125$ for $K = 1, \ldots, 6$, still rising at $K = 6$ (the run of slot 23); the SATC run went to $K = 16$ and found the plateau: $\delta = 182$, with Tor of dimension 26 and Loewy length 6.
- *The second-workstation run.* On the new cells of 3.3 ($\ell = 17, \ldots, 59$, $n = 122, \ldots, 700$; rows of source `c1a_local`), 5060 rows in 723 cells are non-zero. The values (dimension, Loewy length, δ) reach $(4, 3, 6)$. The non-zero cells are 219, 195, 158, 75, 55, 15, 4, 2 at $\ell = 17$, 19, 23, 29, 31, 37, 41, 43, with none at $\ell \ge 47$. These rows are unchanged in 4.0.

On the forms side, [LG, sec:where] reports $\delta = 129$ for $\mathbb{T}^{\mathrm{f}}_\mathfrak{m}$ at $(604, 7)$; the census value of the same block (ideal 5, $D = 50$) on the cohomological lattice is $\delta = 128$ — two different algebras (`[C, prop:cohforms]`), so the two numbers need not agree. That the defect shrinks with ℓ is [EMP] and a statement about the window $n \le 700$. Context from the literature: Jochnowitz [Joc82] and Calegari–Emerton [CE, Thm. 2.1] (the index of $\mathbb{T}$ in its normalisation grows at least linearly in k). Our saturation lies between $\mathbb{T}$ and the normalisation, so their bound makes large δ unsurprising without bounding it below; for the census distribution itself no prior was found in our search (2026-10-19).
Sources: the census files (`rows40.tsv.gz`, columns `sat_defect_dim`, `sat_defect_loewy`, `sat_length`, `sat_kmax`; `../3.5/sat_kmax35_cells.tsv`), recomputed by `pg_census_check.py`;

the merge report of the second workstation's saturation run (§5), `slot22/census34_sat_ingest_report.md`, `slot23/KIT_redraft_report.md` §5.1 (the ε_K of $(700, 7)$ to $K = 6$).

Part IV
Level $\Gamma_0(N)$ and characters

Everything so far was level one. By Shapiro's lemma the cohomology of the congruence subgroup $\Gamma_0(N)$ is that of $\mathrm{SL}_2(\mathbb{Z})$ with coefficients induced from $\Gamma_0(N)$, so the same seven-vertex model computes it `[C, sec:levelN]`; characters χ of $(\mathbb{Z}/N)^\times$ twist the coefficients `[C, sec:nebentypus]`. New phenomena: old forms coming from lower level, a parity obstruction for characters, and Mazur's Eisenstein ideal of prime level — here at the primes 2 and 3, where it is largest. (Its integral face, the congruence prime 5 at level 11, is in §30.)

18 Level $\Gamma_0(N)$: old forms at level 2 (`level_n`)

`level_n(N, n, ell)` computes $H^1(\Gamma_0(N); \mathrm{Sym}^n \mathbb{F}_\ell^2)$ by Shapiro's lemma on the fixed model `[C, sec:levelN]`, with the Heilbronn operators T_p ($p \nmid N$) and U_p ($p \mid N$). For $N = 2$ the group has index 3, and the weight-12 cohomology at $\ell = 7$ has dimension $6 = 2 \cdot 3$: Δ contributes through its two old forms $f(z)$ and $f(2z)$ (two classes each), and the Eisenstein part is that of the two cusps ([CMPD]). On the old part the operators T_3 and T_5 of level 2 act as those of level 1.

The snippet `notebook/examples/level_n_old_forms_2_7.oc`:

```
# level_n_old_forms_2_7.oc -- the verb level_n(N, n, ell): level N by Shapiro's lemma; old
↪  forms and degeneracy.
# Phenomena gallery [PG] sec:pg-leveln.  H^1(Gamma_0(N); Sym^n F_ell) with the Heilbronn
↪  operators ([C] sec:levelN).  Delta of level 1 gives
# two old forms of level 2 (f(z) and f(2z)); Gamma_0(2) has index 3 in SL_2(Z) and h^1 = 6
↪  here, twice 3.  The operators T_p (p not dividing N)
# and U_p (p | N) are listed in the record's ops.
# Read_only verbs only (notebook-safe).  Cost: < 1 s, < 0.2 GB.
A = level_n(1, 10, 7);
B = level_n(2, 10, 7);
print("level 1: index", A.index, " h1 =", A.h1, " dim S_12 =", A.dim_cusp_forms, " cusps =",
↪  A.cusps);
print("level 1: operator", A.ops[0].label, " charpoly", A.ops[0].charpoly);
print("level 2: index", B.index, " h1 =", B.h1, " dim S_12(Gamma_0(2)) =", B.dim_cusp_forms, "
↪  cusps =", B.cusps);
for o in B.ops {
    print("level 2: operator", o.label, " bad:", o.bad, " charpoly", o.charpoly);
}
assert(B.index == 3);
assert(B.h1 == 2 * A.h1);
```

Expected output (generated file `phenomena_gallery/level_n_old_forms_2_7.out`):

```
level 1: index 1  h1 = 3  dim S_12 = 1  cusps = 1
level 1: operator T_2^(N)  charpoly [4, 0, 1, 1]
level 2: index 3  h1 = 6  dim S_12(Gamma_0(2)) = 2  cusps = 2
level 2: operator U_2^(N)  bad: true  charpoly [1, 2, 6, 5, 5, 1, 1]
level 2: operator T_3^(N)  bad: false  charpoly [0, 0, 0, 0, 1, 2, 1]
level 2: operator T_5^(N)  bad: false  charpoly [0, 0, 0, 0, 2, 6, 1]
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.00 s, 21 MiB.

What to look for.

- `h1 = 3` at level 1 and `h1 = 6` at level 2, `index 3`;
- the printed `charpoly` lists are the coefficients from low to high degree; the factor x^4 of T_3 and T_5 at level 2 is the four old classes of Δ, on which they act as $\tau(3) \equiv \tau(5) \equiv 0 \pmod 7$, and the rest is the Eisenstein part.

19 Characters and the parity obstruction (`nebentypus`)

`nebentypus(N, m, n, ell)` replaces Sym^n by $\mathrm{Sym}^n \otimes \chi$ for the Dirichlet character of Conrey label $N.m$ reduced into $\mathbb{F}_\ell$ (`[C, sec:nebentypus]`); $\ell \nmid N$ and $\operatorname{ord}\chi \mid \ell - 1$. Three checks at $N = 7$, $\ell = 13$, $n = 2$ ([CMPD]): the trivial character 7.1 reproduces `level_n` operator by operator; the cubic character 7.2 is even, so Sym^2 is allowed, and gives $h^1 = 4$; the sextic character 7.3 is odd and the cohomology vanishes ($\chi(-1) = (-1)^n$ is required).

The snippet `notebook/examples/nebentypus_7_13.oc`:

```
# nebentypus_7_13.oc -- the verb nebentypus(N, m, n, ell): cohomology of Gamma_0(N) with a
↪  Dirichlet character.
# Phenomena gallery [PG] sec:pg-nebentypus.  chi = Conrey character 7.m reduced into F_13 ([C]
↪  sec:nebentypus): m = 1 is trivial and reproduces level_n;
# m = 2 has order 3 (even), so Sym^2 is allowed; m = 3 has order 6 and is odd, so n = 2
↪  vanishes (parity chi(-1) = (-1)^n).
# Read_only verbs only (notebook-safe).  Cost: < 1 s, < 0.2 GB.
L = level_n(7, 2, 13);
T = nebentypus(7, 1, 2, 13);
print("trivial character: h1 =", T.h1, " level_n h1 =", L.h1, " same charpoly T_2:",
↪  T.ops[0].charpoly == L.ops[0].charpoly);
assert(T.h1 == L.h1);
assert(T.ops[0].charpoly == L.ops[0].charpoly);
C = nebentypus(7, 2, 2, 13);
print("7.2: order =", C.order, " parity_ok =", C.parity_ok, " h0 =", C.h0, " h1 =", C.h1, "
↪  charpoly T_2 =", C.ops[0].charpoly);
assert(C.order == 3);
D = nebentypus(7, 3, 2, 13);
print("7.3: order =", D.order, " parity_ok =", D.parity_ok, " h1 =", D.h1);
assert(D.parity_ok == false);
assert(D.h1 == 0);
```

Expected output (generated file `phenomena_gallery/nebentypus_7_13.out`):

```
trivial character: h1 = 4  level_n h1 = 4  same charpoly T_2: true
7.2: order = 3  parity_ok = true  h0 = 0  h1 = 4  charpoly T_2 = [8, 3, 4, 2, 1]
7.3: order = 6  parity_ok = false  h1 = 0
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.00 s, 21 MiB.

What to look for.

- `parity_ok: false` and `h1 = 0` for 7.3: the parity obstruction, not a computational accident;
- the characteristic polynomial of T_2 for 7.2 has degree 4 and differs from the trivial-character one.

20 Mazur's Eisenstein ideal at $p = 2, 3$, and a class-group correction

The setting. For a prime N and p prime, let W be the Eisenstein-localised Hecke module on $H_1(X_0(N), \text{cusps}; \mathbb{F}_p)$ in weight 2. Its Loewy length measures Mazur's Eisenstein ideal mod p.

The construction at $p = 2, 3$. The model's construction of W through Manin symbols is valid at $p = 2, 3$ as well `[C, prop:levelNcurve]` ([PROVED]).

- *The largest case.* At $(N, p) = (21\,929, 2)$: W has dimension 255 and Loewy length 128, with layers $255, 253, \ldots, 1$ (curvilinear), and $h(-4 \cdot 21\,929) = 256$ ([CMPD]). This is Calegari–Emerton's own example: $e_2 = 2^{m-1} - 1$ for $N \equiv 9 \pmod{16}$ [CE05, Thm. 1.1], and §6 there states $\mathrm{Cl}(\mathbb{Q}(\sqrt{-21929})) \cong \mathbb{Z}/256$ and $e_2 = 127$. Our computation is an independent check on cohomology, with the layer structure.
- *Large e_3.* For $N \leq 10^4$ the cells with $e_3 \geq 3$ (the Eisenstein-ideal exponent at $p = 3$) are:
  - $e_3 = 3$ at 1747, 2143, 2467, 3583, 4177, 4519, 4789, 5437, 5527, 8011, 8389, 8677;
  - $e_3 = 4$ at 6733, 7219;
  - $e_3 = 5$ at 3061.

  Calegari–Emerton mention only 2143; for this list no prior was found in our search (2026-10-19).
- *Generation without a good prime.* Mazur's good-prime criterion is sufficient, not necessary, at $p = 2$: at $(2473, 2)$ the operator η_2 generates although no prime ≤ 13 is good `[C, rmk:levelNcurvefamily]`.

A correction to Calegari–Emerton ([CMPD], exact h^- via $B_{1,\chi}$). Let $K^{(0)} = \mathbb{Q}(\sqrt{-3})K_3$ be the cyclic sextic CM field of conductor $3N$. Calegari–Emerton [CE05, §6] name $N = 2143$ as the smallest prime $N \equiv 1 \pmod 9$ for which $\mathrm{Cl}(K^{(0)})$ has an element of order 9 (their sentence was re-read verbatim, and 1297 re-checked independently, on 2026-10-19; no published correction was found). That is not so:

- $N = 1297$ already gives $\mathrm{Cl}(K^{(0)})[3^\infty] \cong \mathbb{Z}/9 \oplus \mathbb{Z}/9$;
- within $N \equiv 10, 19 \pmod{27}$ the first is 1747 ($\mathbb{Z}/9 \oplus \mathbb{Z}/3$).

A charitable reading reconciles the sentence: 2143 is the smallest $N \equiv 1 \pmod 9$, $N \not\equiv 1 \pmod{27}$, with $\mathrm{Cl}(K^{(0)})[3^\infty] \cong \mathbb{Z}/9 \oplus \mathbb{Z}/9$. The 3-part has the structure $\mathbb{Z}_3[\omega]/\lambda^m$ with $m = v_3(h^-)$ ([PROVED]). Two equivalences are observed ([EMP]):

- $m \geq 2 \iff (\frac{N-1}{3})!$ is a cube mod N, on all 203 primes $N \leq 10^4$, $N \equiv 1 \pmod 9$. The criterion is Calegari–Emerton's [CE05, Thm. 1.3(i)] for the 3-part of $\mathrm{Cl}(\mathbb{Q}(\sqrt{-3}, N^{1/3}))$ (with Elkies's formula $((N-1)/3)!^3 \equiv \pi \bmod \bar{\pi}$; see also Gerth [Ger05]); its form for $K^{(0)}$ was not found stated and is probably a reflection argument;
- $e_3 \geq 2 \iff m \geq 2$, on the 39 of them where e_3 is computed ($N \leq 2500$).

Class groups of $\mathbb{Q}(N^{1/5})$. Calegari–Emerton's Lemma 5.13 holds on all 12 cells with $g_5 \geq 2$ ($N \leq 800$). The 5-rank is 3 at $N = 211, 281, 691$. The theory is known (Chevalley, Gerth, Calegari–Emerton); these cells are still to be checked against the 5-rank criterion of Schaefer–Stubley [SS19].
Sources: `docs/implementation/slot22/LN23_report.md`, `LN23F_report.md`, `NFQ1_report.md`, `NF-PRANK_report.md`; `docs/phenomenology/literature_search_B.md` §4, §6.

Part V
Hida theory: ordinary families across weights

So far the weight was fixed. Hida theory moves it: the *ordinary* part of H^1 — where U_ℓ acts invertibly — has the same rank at all weights in a class modulo $\ell - 1$, and its Hecke eigenvalues vary ℓ-adically continuously, so that weights congruent modulo $(\ell-1)\ell^{m-1}$ give characteristic polynomials

congruent modulo ℓ^m [C, thm:hidacontrol], [C, thm:hidaweight]. The first section shows this on cohomology modulo 13 and 13^2; the next three show the same computation through the dedicated verbs, on cohomology and on forms; the run sections follow one family to $n = 6910$ and record the exceptional ordinary block at $\ell = 547$.

21 Hida's control theorem, mod 13 and mod 13^2

For an endomorphism U of a finite $\mathbb{Z}/\ell^e$-module M the *ordinary projector* is $e_{\mathrm{ord}} = \lim_j U^{j!}$ and $e_{\mathrm{ord}}M$ the ordinary part [C, def:ordinary], [C, prop:ordinary]; for H^1 and $U = U_\ell$ it is the ordinary part in Hida's sense. Hida's theorems [C, thm:hidacontrol], [C, thm:hidaweight] [PROVED] (from Hida's papers, locators verified there) give: for $\ell \geq 5$ the $\mathbb{Z}_\ell$-rank of the ordinary part depends only on the weight modulo $\ell - 1$, so $h^1_{\mathrm{ord}}(n) = 2\,\mathrm{rank}\, S^{\mathrm{ord}}_{n+2} + 1$ is constant along a weight class; and the ordinary Hecke eigenvalues are ℓ-adically analytic in the weight, so two weights $k \equiv k' \pmod{(\ell-1)\ell^{m-1}}$ have ordinary characteristic polynomials congruent modulo ℓ^m. The project's data show the congruence level — the minimum over the operators T_p, $p \leq 23$, and U_ℓ — is *exactly* $m = 1 + v_\ell(\Delta k/(\ell-1))$ on every pair tested [EX, s14:wshida]. (A single T_p can agree further: on the Eisenstein factor $1 + p^{k-1}$ whenever $p^{\ell-1} \equiv 1 \pmod{\ell^2}$, e.g. $p = 3$, $\ell = 11$.) the Λ-adic Hecke algebra itself is [U, sec:u-lambda].

Δ is 13-ordinary ($\tau(13) = -577738 \not\equiv 0$). Part (1) of the snippet runs along $n \equiv 10 \pmod{12}$, $n = 10, \ldots, 166$: $\dim H^1$ grows from 3 to 29, the ordinary rank (the rank of U_{13}^d, $d = \dim H^1$) stays 3 (Eisenstein and the two classes of the Δ-family), and the characteristic polynomial of T_2 on the ordinary part is the same at every weight. Part (2) switches the coefficient ring to $\mathbb{Z}/13^2$ (`set coefficient_ring = "Z/ell^2"`, a mathematical setting, [E, sec:ring]); H^1 is free at the three cells, the ordinary idempotent is computed as $E = (U^{13^8})^{13^{12}-1}$ and certified by $E^2 = E$ (on the non-ordinary part U is topologically nilpotent; on the ordinary part U^{13^8} is the Teichmüller part of a unit, of order dividing $13^f - 1$), and the snippet prints $\mathrm{charpoly}(T_2 E) = X^{d-3}\, f_{\mathrm{ord}}(X)$. Weights 12 and 168 differ by $156 = 12 \cdot 13$, so $m = 2$: the ordinary factors agree modulo 169. Weights 12 and 24 differ by 12, so $m = 1$: the factors agree modulo 13 ($27 \equiv 92$, $75 \equiv 153$, $72 \equiv 163$) but not modulo 169. [CMPD].

The snippet `notebook/examples/hida_control_13.oc`:

```
# hida_control_13.oc -- Hida's control theorem, seen on H^1(SL_2(Z); Sym^n): the ordinary part
↪  does not grow with the
# weight, and ordinary eigenvalues are congruent mod ell^m when the weights are congruent mod
↪  (ell - 1) ell^(m-1).
# Phenomena gallery [PG] sec:pg-hida.  ell = 13; the family of Delta (tau(13) = -577738 is a
↪  13-adic unit).
# Read_only verbs and `set` of a mathematical setting only (notebook-safe).  Cost: ~10 s, < 0.2
↪  GB.
M = sl2z_model();
ell = 13;

# (1) Over F_13: the ordinary rank (rank of U_13^d) along n == 10 (mod 12), against dim H^1.
for n in range(10, 167, 12) {
    H = cohomology(symmetric_power(M, n, ell));
    U = hecke(H, ell).A(1);
    P = U;
    for i in range(1, U.rows) { P = mul(P, U); }
    O = image(P);
    print("n =", n, " dim H^1 =", H.dims[1], " ordinary rank =", rank(P), " charpoly T_2 | ord
    ↪  =", charpoly(restrict(hecke(H, 2).A(1), O)));
}

# (2) Over Z/169: the ordinary idempotent e = lim U^(N!) and the characteristic polynomial of
↪  T_2 e.
#     n = 10 and n = 166 differ by 156 = 12 * 13: the ordinary charpolys agree mod 13^2.
#     n = 22 differs from n = 10 by 12 only: they agree mod 13 but not mod 13^2.
```

```
set coefficient_ring = "Z/ell^2";
q = ell * ell;
for n in [10, 22, 166] {
    H = cohomology(symmetric_power(M, n, ell));
    for e in H.invariant_factors(1) { assert(e == 2, "H^1 is free over Z/169"); }
    T = matrix(q, hecke(H, 2).A(1));
    U = matrix(q, hecke(H, ell).A(1));
    # P = U^(13^8) kills the non-ordinary part and the pro-13 part of the units ...
    P = U;
    for j in range(0, 8) { P2 = mul(P, P); P4 = mul(P2, P2); P8 = mul(P4, P4); P = mul(mul(P8,
    ↪  P4), P); }
    # ... and E = P^(13^12 - 1) is the ordinary idempotent.
    e = 13^12 - 1;
    E = identity(q, U.rows);
    Bm = P;
    for i in range(0, 45) { if e % 2 == 1 { E = mul(E, Bm); } Bm = mul(Bm, Bm); e = (e - e % 2)
    ↪  / 2; }
    assert(E == mul(E, E), "E is idempotent");
    print("n =", n, " H^1 = (Z/169)^" + str(U.rows), " charpoly(T_2 e) =", charpoly(mul(T, E)));
}
```

Expected output (generated file `phenomena_gallery/hida_control_13.out`):

```
n = 10  dim H^1 = 3  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 22  dim H^1 = 5  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 34  dim H^1 = 7  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 46  dim H^1 = 9  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 58  dim H^1 = 11  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 70  dim H^1 = 14  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 82  dim H^1 = 15  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 94  dim H^1 = 17  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 106  dim H^1 = 19  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 118  dim H^1 = 21  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 130  dim H^1 = 23  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 142  dim H^1 = 25  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 154  dim H^1 = 28  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 166  dim H^1 = 29  ordinary rank = 3  charpoly T_2 | ord = X^3 + X^2 + 10*X + 7 (mod 13)
n = 10  H^1 = (Z/169)^3  charpoly(T_2 e) = X^3 + 27*X^2 + 75*X + 72 (mod 169)
n = 22  H^1 = (Z/169)^5  charpoly(T_2 e) = X^5 + 92*X^4 + 153*X^3 + 163*X^2 (mod 169)
n = 166  H^1 = (Z/169)^29  charpoly(T_2 e) = X^29 + 27*X^28 + 75*X^27 + 72*X^26 (mod 169)
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 2.39 s, 278 MiB.

What to look for.

- `ordinary rank = 3` on every line of part (1), against a growing `dim H^1`;
- the ordinary factor $X^3+27X^2+75X+72$ at `n = 10` and `n = 166`, but $X^3+92X^2+153X+163$ at `n = 22`;
- the snippet ends with `set coefficient_ring = "F_ell"`, so a notebook session is left at the default.

22 The ordinary rank in one call (`pow`)

For the ordinary rank of §21 one needs the rank of U_ℓ^d, $d = \dim H^1$: the power U^d annihilates the nilpotent (non-ordinary) part and is invertible on the ordinary part, so $\operatorname{rank} U^d$ is the ordinary rank, and it equals $\operatorname{rank} U^{d+1}$. The verb `pow(A, e)` (square-and-multiply, $A^0 = I$) replaces the multiplication loop of `hida_control_13.oc`. At $\ell = 13$ and $n \equiv 10 \pmod{12}$ the dimension of H^1 grows by 2 with each step of 12 in n, while the ordinary rank stays 3: $\Delta^{\pm}$ and the Eisenstein class (Hida's control theorem, [U, sec:u-lambda]; [CMPD]).

The snippet `notebook/examples/pow_ordinary_rank_13.oc`:

```
# pow_ordinary_rank_13.oc -- the verb pow(A, e): the ordinary rank as rank(U^d), without a loop.
# Phenomena gallery [PG] sec:pg-pow.  U = U_13 on H^1(SL_2(Z); Sym^n F_13^2); U^d (d = dim H^1)
↪  kills the non-ordinary
# (nilpotent) part and is invertible on the ordinary part, so rank(U^d) is the ordinary rank
↪  (Hida: it stays 3 along n == 10 mod 12).
# Read_only verbs only (notebook-safe).  Cost: ~1 s, < 0.2 GB.
M = sl2z_model();
for n in [10, 22, 34, 46, 58] {
    H = cohomology(symmetric_power(M, n, 13));
    U = hecke(H, 13).A(1);
    d = U.rows;
    print("n =", n, " dim H^1 =", H.dims[1], " rank U =", rank(U), " ordinary rank = rank(U^d)
    ↪  =", rank(pow(U, d)));
    assert(rank(pow(U, d)) == rank(pow(U, d + 1)), "rank(U^d) is stable");
    assert(rank(pow(U, d)) == 3, "ordinary rank 3 along n = 10 mod 12");
}
```

Expected output (generated file `phenomena_gallery/pow_ordinary_rank_13.out`):

```
n = 10  dim H^1 = 3  rank U = 3  ordinary rank = rank(U^d) = 3
n = 22  dim H^1 = 5  rank U = 3  ordinary rank = rank(U^d) = 3
n = 34  dim H^1 = 7  rank U = 3  ordinary rank = rank(U^d) = 3
n = 46  dim H^1 = 9  rank U = 3  ordinary rank = rank(U^d) = 3
n = 58  dim H^1 = 11  rank U = 3  ordinary rank = rank(U^d) = 3
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.14 s, 46 MiB.

What to look for.

- `dim H^1` runs $3, 5, 7, 9, 11$ while `ordinary rank` is 3 throughout; `rank U` is already 3, i.e. U_{13} itself vanishes on the non-ordinary part in these cells (a longer Jordan block there would make `rank U` exceed `rank(U^d)`).

23 The Hida tower of a weight class (`hida_tower`)

The weight tower of `[C, def:hidatower]` at the class $n \equiv 10 \pmod{12}$, $\ell = 13$ is read in one call: the ordinary part of $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^n \cdot)$ over $\mathbb{Z}/13^e$ for $e = 1, 2$ on the reduction of the module over $\mathbb{Z}/13^K$ and on its free block (`[C, cor:ordprecision]`), their agreement, and the characteristic polynomial of T_2 on the ordinary part. This is the computation of `hida_control_13.oc` (§21) as a verb ([CMPD]).

The snippet `notebook/examples/hida_tower_13_10.oc`:

```
# hida_tower_13_10.oc -- the verb hida_tower(n, ell, m): one cell of Hida's weight tower, both
↪  readings at once.
# Phenomena gallery [PG] sec:pg-hidatower.  Class n = 10 mod 12, ell = 13, levels e = 1, 2 ([C]
↪  def:hidatower): the ordinary part of
# H^1(SL_2(Z); Sym^n) over Z/13^e is free of rank 3 (Delta^+, Delta^-, Eisenstein), read on the
↪  reduction of the module over Z/13^K and on its free block.
# Compare hida_control_13.oc, which computes the same characteristic polynomial by hand.
# Read_only verbs only (notebook-safe).  Cost: ~1 s, < 0.2 GB.
T = hida_tower(10, 13, 2);
print("h =", T.h, " e_max =", T.e_max, " K =", T.K, " exps =", T.exps);
for L in T.levels {
    print("e =", L.e, " ord_rank =", L.ord_rank, " free_ord_rank =", L.free_ord_rank, "
    ↪  image_free:", L.image_free, " readings agree:", L.readings_agree, " charpoly T_2 =",
    ↪  L.charpoly["2"]);
    assert(L.readings_agree);
    assert(L.ord_rank == 3);
}
```

Expected output (generated file `phenomena_gallery/hida_tower_13_10.out`):

```
h = 3  e_max = 0  K = 2  exps = [2, 2, 2]
e = 1  ord_rank = 3  free_ord_rank = 3  image_free: true  readings agree: true  charpoly T_2 =
↪  [7, 10, 1, 1]
e = 2  ord_rank = 3  free_ord_rank = 3  image_free: true  readings agree: true  charpoly T_2 =
↪  [72, 75, 27, 1]
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.02 s, 24 MiB.

What to look for.

- `ord_rank 3` at both levels, `readings agree: true`, `image_free: true`;
- the level-2 polynomial $x^3 + 27x^2 + 75x + 72$ reduces to the level-1 polynomial modulo 13.

24 Hida's control on forms (`hida_lambda`)

`hida_lambda(ell, k, W)` computes the ordinary part eS_k over $\mathbb{Z}/\ell^W$ and the characteristic polynomials of T_p and U_ℓ on it ([U, sec:u-lambda]). For $\ell = 11$, $W = 2$ the weights 12 and $122 = 12+110$ are congruent modulo $(\ell-1)\ell^{W-1} = 110$: although $\dim S_{122} = 9$ against $\dim S_{12} = 1$, both ordinary parts have rank 1 and the same polynomials for T_2 and U_{11} over $\mathbb{Z}/11^2$, as Hida's control theorem predicts ([CMPD]).

The snippet `notebook/examples/hida_lambda_11_12.oc`:

```
# hida_lambda_11_12.oc -- the verb hida_lambda(ell, k, W): the ordinary forms of weight k over
↪  Z/ell^W (the Lambda-adic forms side).
# Phenomena gallery [PG] sec:pg-hidalambda ([U] sec:u-lambda).  ell = 11, W = 2: the weights 12
↪  and 12 + 110 = 122 are congruent mod (ell - 1) ell^(W-1) = 110,
# so the ordinary parts e S_k have the same rank and the same characteristic polynomials of T_2
↪  and U_11 modulo 11^2 (Hida's control, on forms).
# Read_only verbs only (notebook-safe).  Cost: ~1 s, < 0.2 GB.
A = hida_lambda(11, 12, 2, [2]);
B = hida_lambda(11, 122, 2, [2]);
for R in [A, B] {
    print("k =", R.k, " dim S =", R.dim_S, " rank e S =", R.rank_S, " rank e M =", R.rank_M, "
    ↪  charpoly T_2 =", R.ops[0].charpoly, " charpoly U =", R.ops[1].charpoly, " comm_ok:",
    ↪  R.comm_ok, " stable_ok:", R.stable_ok);
}
assert(A.rank_S == B.rank_S);
assert(A.ops[0].charpoly == B.ops[0].charpoly);
assert(A.ops[1].charpoly == B.ops[1].charpoly);
```

Expected output (generated file `phenomena_gallery/hida_lambda_11_12.out`):

```
k = 12  dim S = 1  rank e S = 1  rank e M = 2  charpoly T_2 = [24, 1]  charpoly U = [87, 1]
↪  comm_ok: true  stable_ok: true
k = 122  dim S = 9  rank e S = 1  rank e M = 2  charpoly T_2 = [24, 1]  charpoly U = [87, 1]
↪  comm_ok: true  stable_ok: true
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.00 s, 21 MiB.

What to look for.

- `dim S` 1 and 9, `rank e S` 1 and 1;
- identical `charpoly T_2` and `charpoly U` for $k = 12$ and $k = 122$.

25 The first ordinary level-one block of Loewy length 3: $(484, 547)$, $k = 486$

Ordinary rows have period $\ell - 1$. By Hida's control theorem [C, thm:hidacontrol], [C, thm:hidaweight], ordinary data depend on the weight only modulo $\ell - 1$. The census shows this at the level of the full module. Ordinarity (U_ℓ a unit on the block) is not a census column; it is read from the cells' records. In the signed census 4.0, 57 728 rows are ordinary (27 539 main, 30 189 b1); the 18 124 rows of the 580 main cells without U_ℓ are left out. Pairing each cell $n \geq \ell$ with $n + \ell - 1$ and comparing the multiset of (residue degree, Loewy length) over the ordinary rows of H^1, the census is exactly periodic at every $\ell = 17, \ldots, 199$: 302–484 pairs per prime, 0 differing ([CMPD]).

- *Loewy length.* Exactly one ordinary row has Loewy length ≥ 3: $(484, 547)$, ideal 1, the exception below. (The investigation of 2026-10-10 found none among the 16 865 ordinary rows of 3.4-pre.1, which did not yet contain $\ell = 547$.)
- *The* 240 *ordinary rows of Loewy length* 2 occur only at the primes 37, 59, 67, 71, 73, 89, 101, 103, 547, 691 (main) and 113, 131, 139, 149, 157, 163, 173, 179, 181 (b1). 106 have a reducible residual representation and 134 an irreducible one. On 3.4-pre.1 (then 96 rows) the investigation identified the two mechanisms behind this split:
  - an Eisenstein–cusp congruence at an irregular pair (the reducible rows; 56 of the 96);
  - a cusp block over a ramified quadratic extension of $\mathbb{Q}_\ell$ (the irreducible rows; 40 of the 96, for instance $(66, 89)$, $k = 68$).

  The new rows of 4.0 have not been examined one by one; that they follow the same two mechanisms is [EMP].

The exception $\ell = 547$. 547 is irregular at $k \in \{270, 486\}$, with $v_{547}(B_k) = 1$ at both. At $k = 486$ ($n = 484$), *two* cusp eigenforms are congruent to E_{486}. They are conjugate over a ramified quadratic extension of $\mathbb{Q}_{547}$. The census cell $(484, 547)$ reads ([CMPD]):

- ideal 1 has $D = 5 = 2g + 1$;
- $A = \mathbb{F}_{547}[x]/x^3$ (Hilbert function $(1, 1, 1)$, Gorenstein), and $H^1_{\mathfrak{m}} \cong \mathbb{F}[x]/x^3 \oplus \mathbb{F}[x]/x^2$;
- the Eisenstein eigenvalues are $1 + p^{485} \equiv 147, 539, 3, 377$ at $p = 2, 3, 5, 7$.

What the forms side and Hida theory add. A forms-side computation with T_{547} itself shows that the block is ordinary. With Ohta's theorem and Hida control it also shows ([PROVED] in the review cited below):

- $\delta = 0$;
- the Loewy length is 3 at every weight $k \equiv 486 \pmod{546}$.

The argument is modulo ℓ (the same ring $\mathbb{T}_{k,\mathfrak{m}}/\ell$ and the same module at every such weight), so it does not depend on the Eisenstein depth $v_{547}(B_k/k)$, which is ≥ 2 on one class modulo $546 \cdot 547$; it rests on Hida's control at the level of modules mod ℓ, which the review cites from memory (not yet checked against the text).

How rare it is: a known exception. That two cusp forms of weight 486 are congruent to E_{486} mod 547 is **known**: Bellaïche and Pollack [BP19, §1.1, §3.9] checked their hypothesis (Cuspidal rank one) for all irregular pairs (p, k) with $p < 10^5$, and $(547, 486)$ is the only exception, with cuspidal Eisenstein Hida–Hecke algebra of rank 2. The project's own screen of the 487 irregular pairs with $\ell < 8000$ (an Eisenstein generalised eigenspace of dimension ≥ 3 only at $\ell = 547$,

$k = 486$) is consistent with theirs and smaller. What this section adds, as far as the search saw, is the reading on cohomology: the ramified conjugacy, the full ordinary block $\mathbb{F}[x]/x^3$, the module $\mathbb{F}[x]/x^3 \oplus \mathbb{F}[x]/x^2$ with $\delta = 0$ — relevant to their Gorenstein hypotheses [BP19, §1.2] — and so the first ordinary level-one block of Loewy length 3 ($g = 2$ on the forms side: [EX, ex:families]).
Sources: `docs/implementation/slot22/ordinary_loewy_investigation.md`, `ordinary_loewy_review.md`, `census34_lowl_ingest_report.md` §5; the 4.0 counts by `docs/latex/tools/pg_census_check.py` (the ordinarity join of `docs/implementation/slot22/ordloewy/ord_join.py`, on the census records of the project).

26 Hida's control along $n \equiv 10 \pmod{690}$ at $\ell = 691$, to $n = 6910$

The prediction. The checkable `[C, Hi11]` predicts the following along the Hida family of Δ glued to E_{12}:

- the ordinary part of $H^1(\Gamma; \mathrm{Sym}^n \mathbb{F}_{691}^2)$ has rank 3 at every $n \equiv 10 \pmod{690}$;
- the characteristic polynomials there are $(X - (1 + 2^{11}))^3$ for T_2, $(X - (1 + 3^{11}))^3$ for T_3, and $(X - 1)^3$ for U_{691}.

§21 is the same phenomenon at $\ell = 13$.

The evidence. Earlier runs held it at $n = 700, 1390, 2080, 2770, 4150$ [EX, nc:hi11]. The second-cluster job 3720162 (2026-10-18) adds four cells, each with no 691-torsion in H^1 ($e_{\max} = 0$) and `verdict=HELD` ([CMPD]):

n	$\dim H^1$	peak RSS (GiB)
4840	807	59.8
5530	923	77.6
6220	1037	97.9
6910	1153	120.5

So the ordinary rank stays 3 while $\dim H^1$ grows past 1100.

What is a theorem here. The ordinary rank 3 and the three characteristic polynomials along the family follow from Hida's control theorem mod ℓ at level one ($\ell \geq 5$, $k \geq 3$; §21) together with the base weight $n = 10$, where Δ is 691-ordinary and Ramanujan's congruence holds. [C] tags (Hi11) as a conjecture because it has not imported that theorem; the runs are a check of it and of the probe. Not part of the theorem is "no 691-torsion", and it fails once: $n = 3460 = 5 \cdot 692$ is a Dickson degree at 691 (the only one in the family up to $n = 6910$), so H^1 has 691-torsion there; the probe refused that cell by name (job 3464192, $e_{\max} = 1$) [EX, nc:hi11], and the table above has the other members only.
Source: the ingest report of the second-cluster job 3720162 (run data, out of tree).

Part VI
Integral cohomology and torsion

Over $\mathbb{Z}$ the cohomology $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2)$ is a finitely generated abelian group: a free part, whose rank is the count of §3, and torsion. The torsion lives at 2 and 3 (the orders of the finite stabilisers) and at the primes $\ell \geq 5$ for which n is a Dickson degree, and it is exactly what made the mod-ℓ dimensions of §3 jump. The project computes these groups, the integral Hecke operators on them and the boundary pair by the *integral descent*: a fracture square that glues the cohomology over $\mathbb{Z}[1/6]$ with the 2- and 3-adic pieces `[C, sec:integral]`, `[C, sec:descent]`,

certified by [C, thm:descentcert] (the groups and the Hecke data) and [C, prop:descentpair] (the boundary pair). This part shows the groups and their exponent law, the large-scale check of that law at $\ell = 2, 3$, the Hecke operators at $\ell = 2, 3$, and how the Eisenstein congruence primes of Part III reappear in the integral Hecke operators.

27 H^1 over $\mathbb{Z}$: torsion primes and the exponent law

The verb `descent(N, n)` (under `set coefficient_ring = "Z"`; $N \le 300$, $n \le 60$ and $\psi(N)(n+1) \le 300$, the Hecke primes summing to at most 100; larger cells by a campaign, [CLI, sec:campaign-z]) returns the certified descent record of $H^q(\Gamma_0(N); \mathrm{Sym}^n \mathbb{Z}^2)$, $q \le 3$ (higher degrees are periodic) [C, thm:descentcert]. Two statements make its torsion predictable from n alone.

- *Which primes* ([PROVED], [C, prop:badprimes], with [C, cl:bockstein] for the isomorphism; at level N, [C, prop:badprimesN]). At level one and for $\ell \ge 5$, $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2)[\ell] \cong H^0(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell^2)$, so the ℓ-primary torsion is non-zero exactly when n is a Dickson degree at ℓ, $n \in (\ell+1)\mathbb{Z}_{\ge 0} + \ell(\ell-1)\mathbb{Z}_{\ge 0}$, and it has as many cyclic summands as there are Dickson monomials of degree n; it lies in H^1 only. Every $\ell > n+1$ carries no torsion. The record lists these primes as `bad_primes`, with the count.
- *Which exponent* ([LIT], Callegaro–Cohen–Salvetti [CCS13, Thm. 4.5]). The largest ℓ-power order in the torsion is $\beta(n, \ell) = \max\{j+1 : n = \ell^j m,\ m \text{ a Dickson degree}\}$; the lower bound is [C, rmk:bocksteinexp](2). At $\ell = 2$ every $m \ge 2$ is a Dickson degree, at $\ell = 3$ the Dickson degrees are $4\mathbb{Z}_{\ge 0} + 6\mathbb{Z}_{\ge 0}$; so $\beta(n, 2) = v_2(n) + 1$ except at powers of 2, and $\beta(n, 3) = v_3(n) + 1$ except at $n = 2 \cdot 3^j$, where it is one less.

The snippet prints H^1 for seven values of n and, next to it, both predictions computed in the script from n. The run results that check the exponent law on ≈ 3000 cells up to $n = 10^4$ are §28.

The snippet `notebook/examples/descent_torsion_beta.oc`:

```
# descent_torsion_beta.oc -- the torsion of H^1(SL_2(Z); Sym^n Z^2) over Z, and its exponent
↪  law beta(n, ell).
# Phenomena gallery [PG] sec:pg-descent-torsion.  The integral descent ([C] sec:descent,
↪  thm:descentcert) at level 1, n = 4 .. 60.
# For each n the snippet prints H^1 as an abelian group, the torsion primes, and two
↪  predictions made here from n alone:
#   * the primes ell >= 5 at which n is a Dickson degree (n in (ell+1) Z_{>=0} + ell(ell-1)
↪  Z_{>=0}): the ell >= 5 torsion primes;
#   * beta(n, ell) = max{ j + 1 : n = ell^j m, m a Dickson degree } for ell = 2, 3: the largest
↪  exponent of ell in the torsion
#     (Callegaro--Cohen--Salvetti, Thm 4.5).
# Read_only verbs only (notebook-safe; descent needs coefficient_ring = "Z").  Cost: a few
↪  seconds, < 0.2 GB.
set coefficient_ring = "Z";
for n in [4, 10, 12, 16, 18, 54, 60] {
    d = descent(1, n);
    print("n =", n);
    print("  H^1 =", d.groups[1]);
    print("  torsion primes", d.torsion_primes, " bad primes (ell >= 5, with count)",
    ↪  d.bad_primes);
    # (a) the primes 5 <= ell <= n + 1 at which n is a Dickson degree
    for ell in range(5, n + 2) {
        isprime = true;
        for t in range(2, ell) { if (ell % t == 0) { isprime = false; } }
        if (isprime) {
            hit = false;
            for b in range(0, n + 1) {
                r = n - b * ell * (ell - 1);
                if (r >= 0 and r % (ell + 1) == 0) { hit = true; }
```

```
            }
            if (hit) { print("  predicted: n is a Dickson degree at ell =", ell, " so ell is a
            ↪  torsion prime"); }
        }
    }
    # (b) beta(n, ell) for ell = 2, 3
    for ell in [2, 3] {
        beta = 0;
        q = 1;
        for j in range(0, 7) {
            if (n % q == 0) {
                m = n / q;
                dick = false;
                for b in range(0, m + 1) {
                    r = m - b * ell * (ell - 1);
                    if (r >= 0 and r % (ell + 1) == 0) { dick = true; }
                }
                if (dick) { beta = j + 1; }
            }
            q = q * ell;
        }
        print("  beta(n, ell) for ell =", ell, ":", beta, " (the largest ell-primary summand
        ↪  should be Z/ell^beta)");
    }
}
```

Expected output (generated file `phenomena_gallery/descent_torsion_beta.out`):

```
n = 4
  H^1 = Z + Z/2^2 + Z/3
  torsion primes [2, 3]  bad primes (ell >= 5, with count) []
  beta(n, ell) for ell = 2 : 2  (the largest ell-primary summand should be Z/ell^beta)
  beta(n, ell) for ell = 3 : 1  (the largest ell-primary summand should be Z/ell^beta)
n = 10
  H^1 = Z^3 + Z/2 + Z/2^2 + Z/3
  torsion primes [2, 3]  bad primes (ell >= 5, with count) []
  beta(n, ell) for ell = 2 : 2  (the largest ell-primary summand should be Z/ell^beta)
  beta(n, ell) for ell = 3 : 1  (the largest ell-primary summand should be Z/ell^beta)
n = 12
  H^1 = Z + Z/2 + Z/2^2 + Z/2^3 + Z/3 + Z/3^2 + Z/5 + Z/11
  torsion primes [2, 3, 5, 11]  bad primes (ell >= 5, with count) [[5, 1], [11, 1]]
  predicted: n is a Dickson degree at ell = 5  so ell is a torsion prime
  predicted: n is a Dickson degree at ell = 11  so ell is a torsion prime
  beta(n, ell) for ell = 2 : 3  (the largest ell-primary summand should be Z/ell^beta)
  beta(n, ell) for ell = 3 : 2  (the largest ell-primary summand should be Z/ell^beta)
n = 16
  H^1 = Z^3 + Z/2 + Z/2^2 + Z/2^4 + Z/3 + Z/3 + Z/7
  torsion primes [2, 3, 7]  bad primes (ell >= 5, with count) [[7, 1]]
  predicted: n is a Dickson degree at ell = 7  so ell is a torsion prime
  beta(n, ell) for ell = 2 : 4  (the largest ell-primary summand should be Z/ell^beta)
  beta(n, ell) for ell = 3 : 1  (the largest ell-primary summand should be Z/ell^beta)
n = 18
  H^1 = Z^3 + Z/2 + Z/2 + Z/2^2 + Z/2^2 + Z/3 + Z/3^2 + Z/5 + Z/17
  torsion primes [2, 3, 5, 17]  bad primes (ell >= 5, with count) [[5, 1], [17, 1]]
  predicted: n is a Dickson degree at ell = 5  so ell is a torsion prime
  predicted: n is a Dickson degree at ell = 17  so ell is a torsion prime
  beta(n, ell) for ell = 2 : 2  (the largest ell-primary summand should be Z/ell^beta)
  beta(n, ell) for ell = 3 : 2  (the largest ell-primary summand should be Z/ell^beta)
n = 54
  H^1 = Z^9 + Z/2 + Z/2 + Z/2 + Z/2 + Z/2 + Z/2^2 + Z/2^2 + Z/2^2 + Z/2^2 + Z/2^2 + Z/3 + Z/3 +
  ↪  Z/3 + Z/3^2 + Z/3^3 + Z/5 + Z/17 + Z/53
  torsion primes [2, 3, 5, 17, 53]  bad primes (ell >= 5, with count) [[5, 1], [17, 1], [53, 1]]
  predicted: n is a Dickson degree at ell = 5  so ell is a torsion prime
  predicted: n is a Dickson degree at ell = 17  so ell is a torsion prime
```

```
  predicted: n is a Dickson degree at ell = 53  so ell is a torsion prime
  beta(n, ell) for ell = 2 : 2  (the largest ell-primary summand should be Z/ell^beta)
  beta(n, ell) for ell = 3 : 3  (the largest ell-primary summand should be Z/ell^beta)
n = 60
  H^1 = Z^9 + Z/2 + Z/2 + Z/2 + Z/2 + Z/2 + Z/2^2 + Z/2^2 + Z/2^2 + Z/2^3 + Z/2^3 + Z/2^3 + Z/3
  ↪  + Z/3 + Z/3 + Z/3 + Z/3^2 + Z/3^2 + Z/5 + Z/5^2 + Z/11 + Z/19 + Z/29 + Z/59
  torsion primes [2, 3, 5, 11, 19, 29, 59]  bad primes (ell >= 5, with count) [[5, 2], [11, 1],
  ↪  [19, 1], [29, 1], [59, 1]]
  predicted: n is a Dickson degree at ell = 5  so ell is a torsion prime
  predicted: n is a Dickson degree at ell = 11  so ell is a torsion prime
  predicted: n is a Dickson degree at ell = 19  so ell is a torsion prime
  predicted: n is a Dickson degree at ell = 29  so ell is a torsion prime
  predicted: n is a Dickson degree at ell = 59  so ell is a torsion prime
  beta(n, ell) for ell = 2 : 3  (the largest ell-primary summand should be Z/ell^beta)
  beta(n, ell) for ell = 3 : 2  (the largest ell-primary summand should be Z/ell^beta)
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 1.94 s, 543 MiB.

What to look for.

- `bad primes` equals the list of predicted primes on every line: $5, 11$ at $n = 12$ ($12 = 2 \cdot (5+1) = 11+1$), 7 at $n = 16$, $5, 17, 53$ at $n = 54$;
- the largest 2-power: $\mathbb{Z}/2^4$ at $n = 16 = 2^4$ ($v_2 + 1 = 5$ would overshoot), $\mathbb{Z}/2^3$ at $n = 12$ and 60;
- the largest 3-power: $\mathbb{Z}/3^2$ at $n = 18 = 2 \cdot 3^2$ and $\mathbb{Z}/3^3$ at $n = 54 = 2 \cdot 3^3$, one less than $v_3(n) + 1$ — the exception of β;
- $n = 60$: two 5-primary summands, $\mathbb{Z}/5 \oplus \mathbb{Z}/5^2$, because degree 60 has two Dickson monomials at 5, P^{10} and Q^3 ($\deg P = 6$, $\deg Q = 20$), with exponents $v_5(\gcd(10,0)) + 1 = 2$ and $v_5(\gcd(0,3)) + 1 = 1$ [CCS13, Thm. 4.5]; compare `dim H^0` at $\ell = 5$ in §3.

28 The torsion exponent is the Frobenius bound (Callegaro–Cohen–Salvetti), checked at $\ell = 2$ and $\ell = 3$ (runs)

Write the ℓ-primary torsion of $H^1(\Gamma; V_n)$ as $\bigoplus_t \mathbb{Z}/\ell^{a_t}$ and put $e_1 = \max_t a_t$.

The bound β. Call $m > 0$ a *Dickson degree* at ℓ if $H^0(\Gamma; M_m) \neq 0$. The ring of $\mathrm{SL}_2(\mathbb{F}_\ell)$-invariants is generated in degrees $\ell + 1$ and $\ell(\ell-1)$, and $\Gamma \to \mathrm{SL}_2(\mathbb{F}_\ell)$ is onto, so the Dickson degrees are the positive elements of $3\mathbb{Z}_{\geq 0} + 2\mathbb{Z}_{\geq 0}$ at $\ell = 2$ (every $m \geq 2$) and of $4\mathbb{Z}_{\geq 0} + 6\mathbb{Z}_{\geq 0}$ at $\ell = 3$. Put

$$\beta(n, \ell) := \max\{\, j + 1 : \ n = \ell^j m, \ m \text{ a Dickson degree} \,\}.$$

Since 1 is not a Dickson degree at 2 and 2 is not one at 3,

$$\beta(n,2) = v_2(n) + 1 - [n = 2^j], \qquad \beta(n,3) = v_3(n) + 1 - [n = 2 \cdot 3^j].$$

In words: $v_2(n) + 1$ alone overshoots by one at the powers of 2, and $v_3(n) + 1$ alone at $n = 2 \cdot 3^j$, where the best factorisation is $n = 3^{j-1} \cdot 6$.

The theorem ([LIT]). Callegaro, Cohen and Salvetti [CCS13, Thm. 4.5] (equivalently their Thm. 3.7(2)) determine the whole ℓ-primary torsion of $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^n \mathbb{Z}^2)$, for every prime ℓ and every $n > 0$: each Dickson monomial $P^k Q^h$ of degree n (with P, Q the invariants of degrees $\ell + 1$ and $\ell(\ell-1)$) gives one summand $\mathbb{Z}/\ell^{v_\ell(\gcd(k,h))+1}$. Maximising over (k, h) gives exactly

$$e_1 = \beta(n, \ell).$$

What [C] proves. The lower bound, at every prime: `[C, rmk:bocksteinexp]`(2) shows that if m is a Dickson degree, then $H^1(\Gamma; V_{\ell^j m})$ has an element of order ℓ^{j+1}; so $e_1 \geq \beta(n, \ell)$. The equality is used from the literature, not proved in [C].

The data agree with it ([CMPD], 0 exceptions). $e_1 = \beta(n, \ell)$ on every cell computed; the data are the L23 section of census 4.0 (a pre-release addendum, §2; table `deep1_l23_torsion40.tsv`):

- the L23 records, even $n = 4, \ldots, 700$: 349/349 cells at $\ell = 2$, 349/349 at $\ell = 3$;
- the DEEP-1 run, even n: 2158/2158 cells at $\ell = 2$ ($n = 600, \ldots, 10^4$; 2572 computations, since 414 cells were run in both modes `r3` and `full`), 128/128 cells at $\ell = 3$ ($n = 600, \ldots, 3840$; 146 computations).

Examples: $e_1 = 13$ at $(8192, 2)$; $e_1 = 6$ at $(972, 3)$; and $e_1 = 5$ at $(486, 3)$, where $486 = 2 \cdot 3^5$. For $\ell \geq 5$ the same equality was seen on the $\mathbb{Z}/\ell^8$ probe of `[C, rmk:bocksteinexp]`(3); the ℓ^2-torsion of [EX, ex:depth] is an instance of the same theorem. The groups printed by the descent snippet of §27 agree with the theorem's multiset exactly, which supports that the two Sym^n conventions match. So the runs are a large-scale check of [CCS13] (and of the code), not a new law.

The cells where β and $v_3(n) + 1$ differ. The DEEP-1 grid at $\ell = 3$ contains no $n = 2 \cdot 3^j$, so on it $\beta(n, 3) = v_3(n) + 1$. The table checks the exception itself, beyond that grid, with a control cell where the two agree:

cell	$v_3(n) + 1$	$\beta(n, 3)$ ([CCS13, Thm. 4.5])	observed e_1
$(1458 = 2 \cdot 3^6, 3)$	7	6	6
$(2916 = 4 \cdot 3^6, 3)$ (control)	7	7	7
$(4374 = 2 \cdot 3^7, 3)$	8	7	7 (sum rules resolved)

At $(1458, 3)$ the law $v_3(n) + 1$ overshoots by one, exactly as β says. Source: `docs/implementation/slot24/L23S14_report.md`.

Two companion observations at $\ell = 2$ ([EMP]). Let c_p be the least elementary-divisor exponent of $T_p - (1 + p)$ on $\Lambda^{\mathrm{fr}} = H^1(\Gamma; V_n \otimes \mathbb{Z}_2)/\mathrm{tors}$ (the free quotient, not localised: at $\ell = 2$ there is one block).

- *Constant floors.* $c_p = \min(v_2(p^n - 1), \nu_p)$ is [PROVED] `[C, lem:cprule]`, where ν_p is the least elementary-divisor exponent of $T_p - (1 + p^{n+1})$. The data show ν_3, ν_5, ν_7 constant, equal to 7, 9, 8, so

$$c_3 = \min(v_2(n) + 2, 7), \qquad c_5 = \min(v_2(n) + 2, 9), \qquad c_7 = \min(v_2(n) + 3, 8)$$

  on all 2572 DEEP-1 computations at even n to $n = 10^4$. The floor ≥ 7 at $p = 3$ (and ≥ 7 at $p = 5$) is Rustom's theorem [Rus, Thm. 8.1, Prop. 8.4], an eigenvalue congruence $a_p \equiv 1 + p^{k-1} \pmod{128}$ for every level-one eigenform; the sharpness and the values 9, 8 go beyond what our search found (2026-10-19).
- *Linear growth.* The largest finite elementary-divisor exponent of T_ℓ grows linearly, close to $(n + 1)/(\ell + 1)$. For example, it is 334 at $(1000, 2)$ and 249 at $(1000, 3)$.

Sources: the L23 section of census 4.0 (`deep1_l23_torsion40.tsv`, `deep1_l23_depths40.tsv`, `l23_torsion_law40.txt`; pre-release), recomputed by `pg_census_check.py`; `docs/implementation/slot24/L23S14_report.md`; `docs/implementation/slot22/census_l23_plan.md` §4.2–4.3; `slot23/DEEP0_report.md` §2; `slot23/DEEP1_final_report.md`; `slot23/DEEP1_ell3_torsion_law.md`; the catalogue entry PH-01; `docs/phenomenology/literature_search_B.md` §1, §3.

29 $\ell = 2, 3$: what the Hecke operators see (runs)

At $\ell \in \{2, 3\}$ the seven-vertex model computes *Bredon* cohomology, not group cohomology, for Γ itself (`[C, prop:modeltorsion]`: it computes group cohomology only for subgroups without elements of order divisible by ℓ). The statements below are about group cohomology: the L23 census computes it on the isotropy model C_I, cross-checked against the amalgam resolution of $\Gamma = \mathbb{Z}/4 *_{\mathbb{Z}/2} \mathbb{Z}/6$ (`[C, cl:amalgam]`), and the integral descent on its tree model.

1. [PROVED] **Prime-to-ℓ operators are scalars.** $T_p = (1+p)\cdot\mathrm{id}$ as a matrix on $H^q(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell^2)$ for every $p \neq \ell$ and every q `[C, prop:ltwothreescalar]`. So mod ℓ the Hecke algebra is $\mathbb{F}_\ell[T_\ell]$, and the arithmetic is ℓ-adic. The mechanism at level $\Gamma(\ell)$ is known: Calegari–Emerton [CE, Lemma 3.1], who credit Hatada (Math. Ann. 239 (1979), as quoted in [Rus]); the level-one form for every q was not found. It is consistent with the large mod-2 Hecke algebra on forms (Nicolas–Serre) only through a large saturation defect: on the lattice the image of $\mathbb{T}/\ell$ is $\mathbb{F}_\ell[T_\ell]$, so $\dim(\mathbb{T}^{\mathrm{sat}}/\mathbb{T})[\ell] = \dim M_k - 2$. The L23 records show the hypothesis on the lattice ($T_p \equiv 1+p$ and one unit divisor of T_ℓ) on 349/349 even cells at $\ell = 2$ and at $\ell = 3$ (the L23 section of 4.0, pre-release).

2. [EMP] **T_ℓ is a rank-one idempotent.** On $H^1(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell^2)$, n even, the characteristic polynomial of T_ℓ is $x^{h^1-1}(x-1)$. This holds on all DEEP-1 cells that compute it: 9303 at $\ell = 2$ ($n \le 10^4$, odd n included) and 196 at $\ell = 3$ ($n \le 7500$). For example, at $(10\,000, 2)$, $h^1 = 5001$, $\mathrm{rank}\, T_2 = 1$ and $\mathrm{rank}(T_2 - 1) = 5000$. Only $\mathrm{rank}\, T_\ell \le 2$ is proved `[C, lem:tellrank]`. The eigenvalue shadow is Hatada's ($a_2 \equiv 0 \bmod 8$, $a_3 \equiv 0 \bmod 3$, via [Rus]); for the matrix statement no prior was found in our search (2026-10-19).

3. [LEAN] **Closed dimensions.**

   - $\dim_{\mathbb{F}_2} H^q(\Gamma; \mathrm{Sym}^n \mathbb{F}_2^2) = \lfloor n/2 \rfloor + 1$ for every $q \ge 1$;
   - $\dim_{\mathbb{F}_3} H^{q+2}(\Gamma; \mathrm{Sym}^n \mathbb{F}_3^2) = [n \equiv 0, 4 \bmod 6]$ for even n and $q \ge 0$;
   - at $\ell = 3$, odd n, all groups vanish.

   The rank formulas behind these values ($h^q = 2d - a - s$ at $\ell = 2$, and their analogues at $\ell = 3$) are Lean theorems for every n; the values themselves are kernel-checked cell by cell for $n \le 700$ ([LEAN] evidence for $n \le 64$ and a few larger n, data beyond) `[C, rmk:lvdimsltwothree]`. The closed forms above are read off those values, not formalised for every n. Odd n at $\ell = 2$ is `[C, prop:oddnltwo]`.

4. [EMP] **Cusp classes are divisible by ℓ.** $H^1_!(\Gamma; V_n) \subset \ell \cdot H^1(\Gamma; V_n)$ on every computed cell, $n \le 700$.

5. [EMP] **A $\pm$-law at $\ell = 3$.** For $n \equiv 4 \pmod 6$, T_3 has the same elementary divisors on $\Lambda^{\mathrm{fr},+}_!$ and on $\Lambda^{\mathrm{fr},-}$, the ι-eigenlattices of the free quotient over $\mathbb{Z}_3$ (33/33 cells, $n \le 200$). There is no such law for $n \equiv 0 \pmod 6$.

6. [CMPD] **Integral examples.** The groups $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2)$ themselves, with their 2- and 3-primary parts, are printed by the snippet of §27 (the integral descent, $n \le 60$); for instance $H^1(\Gamma; \mathrm{Sym}^{12} \mathbb{Z}^2) \cong \mathbb{Z} \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/2^2 \oplus \mathbb{Z}/2^3 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/3^2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/11$ there. The number of ℓ-primary cyclic summands is $\dim H^0(\Gamma; M_n)$ `[C, cl:bockstein]`; their exponents are the subject of §28.

Sources: the L23 section of census 4.0 (`docs/matrix-engine/results/congruence_census/edition3/4.0/README_l23.md`: `deep1_l23_h01_40.tsv`, `deep1_l23_side40.tsv`, `l23_section40_v2.tsv`; pre-release); `docs/implementation/slot22/census_l23_classification_review.md`, `L23_3_report.md`, `BIN3_report.md`, `slot23/DEEP1_final_report.md`.

30 Eisenstein congruence primes over $\mathbb{Z}$, and the boundary pair

Ramanujan's congruence (§6) is invisible in the abelian group $H^1(\Gamma;\mathrm{Sym}^{10}\mathbb{Z}^2) = \mathbb{Z}^3 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/2^2 \oplus \mathbb{Z}/3$: the prime 691 is not torsion. It is in the Hecke operators. On the free quotient, T_2 has the characteristic polynomial $\chi_{\mathbb{Z}}(T_2) = (x - (1 + 2^{11}))(x - \tau(2))^2$ `[C, prop:crtcharpolyN]`: one Eisenstein root, and the root of Δ twice (one class in each ι-eigenspace, §5). Two roots of an integral characteristic polynomial are congruent modulo exactly the primes dividing their difference, and $1 + 2^{11} - \tau(2) = 2049 + 24 = 2073 = 3 \cdot 691$. So the Eisenstein congruence prime appears as a resultant of the integral Hecke data, and its reduction is the non-semisimple block of §6. (The factor 3 is the universal congruence at $\ell = 3$, where every T_p acts as $1 + p$ (§29).)

The same at level 11 and weight 2 ($n = 0$), Mazur's original setting: $H^1(\Gamma_0(11);\mathbb{Z}) = \mathbb{Z}^3$, the Eisenstein class with $T_2 = 1 + 2 = 3$ and the elliptic curve $11a$ with $a_2 = -2$; the difference is 5, the numerator of $(N - 1)/12 = 10/12$, Mazur's Eisenstein prime ([LIT]). Modulo 5 the three roots collapse: `level_n(11, 0, 5)` has $\chi(T_2) \equiv (x - 3)^3$. The integral Hecke data are read on the descent's certified groups `[C, sec:descenthecke]`.

Last, the boundary pair over $\mathbb{Z}$ `[C, sec:descentpair]`: the cohomology of $\Gamma_0(11)$, of the boundary of its Borel–Serre compactification (two cusps, so $H^0_\partial = \mathbb{Z}^2$), and the compactly supported cohomology H^*_c as the cone of the restriction — the integral version of the sequence of §5. $H^2_c = \mathbb{Z}$ is the fundamental class of the compactified modular curve, and the rank-2 image of $H^1_c \to H^1$ is the interior cohomology of the genus-one curve $X_0(11)$.

The snippet `notebook/examples/descent_congruence_primes.oc`:

```
# descent_congruence_primes.oc -- Eisenstein congruence primes read from the integral Hecke
↪  operators, and the boundary pair over Z.
# Phenomena gallery [PG] sec:pg-descent-congruence.  The integral descent ([C] sec:descent,
↪  sec:descenthecke, sec:descentpair).
#   (1) level 1, n = 10 (weight 12): char_Z(T_2 | H^1/tors) = (x - (1 + 2^11)) (x - tau(2))^2,
↪  and 1 + 2^11 - tau(2) = 3 * 691;
#   (2) level 11, n = 0 (weight 2): char_Z(T_2) = (x - 3) (x - a_2(11a))^2 with a_2 = -2, and 3
↪  - (-2) = 5 = numerator of (11 - 1)/12;
#   (3) the same congruence mod 5 on the mod-5 cohomology of Gamma_0(11) (level_n), and the
↪  boundary pair over Z.
# The expansions on the "expected" lines are computed here from the Eisenstein eigenvalue 1 +
↪  p^(k-1) and the cusp eigenvalue a_2.
# Read_only verbs only (notebook-safe).  Cost: about 1 s, < 0.2 GB.
set coefficient_ring = "Z";

# (1) Ramanujan's 691 over Z
d = descent(1, 10);
print("level 1, n = 10:  H^1 =", d.groups[1]);
print("  char_Z(T_2) coefficients (high -> low):", d.charpoly[0].coefficients);
a = 1;
for i in range(0, 11) { a = a * 2; }
a = a + 1;
t = -24;
print("  expected (x - a)(x - t)^2 with a = 1 + 2^11 =", a, " t = tau(2) =", t, ":", [1, -(a +
↪  2 * t), 2 * a * t + t * t, -(a * t * t)]);
print("  a - t =", a - t, " = 3 * 691:", a - t == 3 * 691);

# (2) Mazur's Eisenstein ideal at N = 11, weight 2
e = descent(11, 0);
print("level 11, n = 0:  H^q =", e.groups);
print("  char_Z(T_2) coefficients (high -> low):", e.charpoly[0].coefficients);
a = 3;
t = -2;
print("  expected (x - a)(x - t)^2 with a = 1 + 2 =", a, " t = a_2(11a) =", t, ":", [1, -(a + 2
↪  * t), 2 * a * t + t * t, -(a * t * t)]);
print("  a - t =", a - t, " = numerator of (11 - 1)/12 = 5/6");

# (3) The same congruence mod 5, and the boundary pair
```

```
set coefficient_ring = "F_ell";
L = level_n(11, 0, 5);
print("mod 5, level 11: h1 =", L.h1, " T_2 charpoly (low -> high)", L.ops[0].charpoly, " = (x -
↪  3)^3 mod 5");
set coefficient_ring = "Z";
P = e.pair;
print("pair at (11, 0): bulk     H^q =", P.bulk.groups);
print("                 boundary H^q =", P.boundary.groups);
print("                 compact  H^q =", P.compact.groups);
```

Expected output (generated file `phenomena_gallery/descent_congruence_primes.out`):

```
level 1, n = 10:  H^1 = Z^3 + Z/2 + Z/2^2 + Z/3
  char_Z(T_2) coefficients (high -> low): ["1", "-2001", "-97776", "-1180224"]
  expected (x - a)(x - t)^2 with a = 1 + 2^11 = 2049  t = tau(2) = -24 : [1, -2001, -97776,
  ↪  -1180224]
  a - t = 2073  = 3 * 691: true
level 11, n = 0:  H^q = ["Z", "Z^3", "Z/2", "Z/2 + Z/2 + Z/2"]
  char_Z(T_2) coefficients (high -> low): ["1", "1", "-8", "-12"]
  expected (x - a)(x - t)^2 with a = 1 + 2 = 3  t = a_2(11a) = -2 : [1, 1, -8, -12]
  a - t = 5  = numerator of (11 - 1)/12 = 5/6
mod 5, level 11: h1 = 3  T_2 charpoly (low -> high) [3, 2, 1, 1]  = (x - 3)^3 mod 5
pair at (11, 0): bulk     H^q = ["Z", "Z^3", "Z/2", "Z/2 + Z/2 + Z/2", "Z/2"]
                 boundary H^q = ["Z^2", "Z^2", "Z/2 + Z/2", "Z/2 + Z/2", "Z/2 + Z/2"]
                 compact  H^q = ["0", "Z^3", "Z", "Z/2 + Z/2 + Z/2", "Z/2"]
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.17 s, 31 MiB.

What to look for.

- the computed coefficients of $\chi_{\mathbb{Z}}(T_2)$ equal the `expected` expansion, at level 1 and at level 11;
- `a - t = 2073 = 3 * 691` and `a - t = 5`: the congruence primes;
- the mod-5 characteristic polynomial `[3, 2, 1, 1]` $= (x-3)^3$ (coefficients low to high);
- the pair: `boundary H^0 = Z^2`, `compact H^0 = 0`, `compact H^2 = Z`, and torsion only at 2 (the torsion primes at $n = 0$ are $2, 3$, and here only 2 occurs).

The same congruence as a certified record over $\mathbb{Z}$

The resultant above uses one operator, T_2. The congruence is also a statement about the whole Hecke algebra, and the HeckeZ record certifies it. The record is the output of `hecke_structure(N, n)` (`[C, sec:recordschema]`, item 8). Let T^{fr} be the image of the Hecke algebra on the free quotient $L = H^1/\mathrm{tors}$. At $(N, n) = (1, 10)$, L has two components, the Eisenstein one ($T_p \mapsto 1 + p^{11}$) and the cuspidal one ($T_p \mapsto \tau(p)$), and the *congruence module* is $C_T = T^{\mathrm{fr}}/(I_{\mathrm{Eis}} + I_{\mathrm{cusp}})$ (`[C, def:heckeZcong]`). The record prints `C_T: certified:0x2b3`, and `0x2b3` $= 691$: $C_T = \mathbb{Z}/691$, certified because the lower and upper bounds meet at every ℓ dividing the upper bound (`[C, prop:heckeZcong]`(4)). Since C_T is cyclic, $T^{\mathrm{fr}} = \mathbb{Z} \times_{\mathbb{Z}/691} \mathbb{Z} = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \bmod 691\}$ (`[C, prop:heckeZcong]`(3),(4)), the pair of T_n being $(\sigma_{11}(n), \tau(n))$. This one ring statement is Ramanujan's $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ for all n, and for T_2 it is $2049 + 24 = 3 \cdot 691$ again. A second, independent route reaches the same prime: $v_{691}(B_{12}/12) = 1$, the numerator of the Eisenstein constant (the record's second route for C_T). The level-11, $n = 0$ companion (Mazur) reads the same way: the anemic record has $C_T = \mathbb{Z}/5$, `certified:0x5`.

Why "over $\mathbb{Z}$" says more than "modulo 691*".* Modulo 691 one sees only that two maximal ideals coincide: the mod-691 Hecke algebra has one local factor where the characteristic-0 algebra has two. The congruence module records *by how much* the two factors fail to split: T^{fr} is the subring of $\mathbb{Z} \times \mathbb{Z}$ of index exactly 691, so C_T is the precise integral obstruction to writing the Hecke algebra as a product, with the exponent (691^1, not 691^2) visible. It is a statement about the whole algebra, not about one eigenvalue, and the status `certified` says that nothing in it is an unverified bound.

The snippet `notebook/examples/ramanujan_691_over_Z.oc`:

```
# ramanujan_691_over_Z.oc -- Ramanujan's congruence at 691 as a certified statement over Z, and
↪  its level-11 companion.
# Phenomena gallery [PG] sec:pg-descent-congruence.  The HeckeZ record hecke_structure(N, n)
↪  ([C] sec:recordschema, item 8, sec:heckeZ).
#   (1) level 1, n = 10 (weight 12): the congruence module C_T between the Eisenstein and the
↪  cuspidal part of the Hecke algebra
#       over Z is Z/691, "certified" (lower and upper bound meet, [C] prop:heckeZcong); 0x2b3 =
↪  691;
#   (2) the second route, not computed here: v_691(B_12/12) = 1, since B_12/12 = -691/32760 and
↪  32760 is prime to 691;
#   (3) level 11, n = 0 (weight 2), the anemic algebra: C_T = Z/5, Mazur's Eisenstein prime.
# Read_only verbs only (notebook-safe).  Cost: about 1 s, < 0.3 GB.
set coefficient_ring = "Z";

R = hecke_structure(1, 10);
print(R.schema, " (", R.N, ",", R.n, ",", R.algebra, ")");
print("  generation per ell:", R.generation);
print("  C_T:", R.C_T, "  (0x2b3 = 691:", 691 == 2 * 256 + 11 * 16 + 3, ")");
assert(R.C_T == "certified:0x2b3", "the 691 congruence module is exact");

E = hecke_structure(11, 0, "anemic");
print("level 11, n = 0 (anemic): generation", E.generation, " C_T:", E.C_T);
assert(E.C_T == "certified:0x5", "Mazur's congruence module at level 11 is Z/5");
```

Expected output (generated file `phenomena_gallery/ramanujan_691_over_Z.out`):

```
integral_hecke_structure/v1  ( 1 , 10 , full )
  generation per ell: ["2:certified(saturation)", "3:certified(saturation)",
  ↪  "691:certified(saturation)"]
  C_T: certified:0x2b3   (0x2b3 = 691: true )
level 11, n = 0 (anemic): generation ["2:certified(saturation)", "5:certified(saturation)"]
↪  C_T: certified:0x5
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.35 s, 34 MiB.

What to look for.

- `C_T: certified:0x2b3`, i.e. 691, and `certified:0x5` at level 11;
- `generation per ell` lists 2, 3, 691, each `certified(saturation)`: $T \otimes \mathbb{Z}_\ell$ is generated by the computed operators;
- the script's `asserts`: a failing certificate stops it with exit code 4.

The full record, with the local type at 691 (monogenic of rank 2, Gorenstein) and the torsion data, is printed by `src/hacs/Console/scripts/060_s25_hecke_structure.oc` and by tour 10 of the notebook.

31 When the torsion does not split off: non-split torsion in H^1 over $\mathbb{Z}$, and what Galois sees (runs)

The integral descent of §27 writes $H := H^1(\Gamma_0(N); V)$, $V = \mathrm{Sym}^n \mathbb{Z}^2$, $n > 0$, as $\mathcal{T} \oplus L$ with $\mathcal{T} = \mathrm{tors}\, H$ and $L = H/\mathcal{T}$ free. As an abelian group the sum is trivial. The Hecke operators need not respect it: in the basis torsion first, then free, $T_p = \left(\begin{smallmatrix} \tau_p & M_p \\ 0 & F_p \end{smallmatrix}\right)$ `[C, prop:torsionhecke]`, and the question is whether the sequence

$$0 \to \mathcal{T}_\ell \to H_\ell \to L_\ell \to 0 \qquad (\varepsilon)$$

splits as a sequence of modules over the Hecke algebra. This section is about the first examples where it does not, and about what that means. Tags: [PROVED] (in [C] or proved here), [COND] (proved here under a named standard hypothesis), [CMPD] (exact computation, command given), [LIT] (cited from the literature), [EMP] (a regularity seen in the stated range, not proved), [SPEC] (a consistent reading, not proved).

1. The test

Write $\mathbb{T}'$ for the algebra of the T_p with $p \nmid \ell N$, $c_p := \tau_p|_{\mathcal{T}_\ell}$, $B_p := F_p|_{L_\ell}$. A complement of $\mathcal{T}_\ell$ is the graph of some $h \in \mathrm{Hom}(L, \mathcal{T}_\ell)$, and it is T_p-stable iff $hB_p - c_p h = M_p$ `[C, prop:heckeZglue]`(2)(b). No solution for the computed T_p certifies that ε is non-split for the whole Hecke algebra (the record's word `non_split`). A solution is weaker: it gives a complement stable under $\mathbb{T}'_{\mathcal{P}} := \mathbb{Z}[T_p : p \in \mathcal{P}]$ only, for the finite set $\mathcal{P}$ of primes computed (the field `P=[...]` of the `.out` files). The record's word for it is `split_on_P`, and `[C, sec:recordschema]` says explicitly that this is not `split` for $\mathbb{T}$: that would need $\mathcal{P}$ to generate the image of $\mathbb{T}$ on H^1_ℓ/ℓ^K, torsion included (`[C, rmk:heckeZcbthree]`). *Convention for this section.* A computed verdict "split", written split$_{\mathcal{P}}$, always means `split_on_P`, and so does the bare word `split` printed by the tool; every "exactly" built on such a verdict is a statement on the computed primes. The word "non-split" is a certificate and needs no qualifier. A solution on a saturated $\mathbb{T}'$-stable sublattice restricts to any smaller such sublattice, so split$_{\mathcal{P}}$ passes to sublattices ([PROVED]). The non-split test, again, does *not* give an element of the glue J: split implies $J = 0$, and the converse fails (2)(c). Nor does a non-zero M_p matter by itself: M_p changes with the basis, by $hB_p - c_p h$ (`[C, sec:recordschema]`, item 8).

2. What the torsion is, and how T_p acts on it

Proposition ([PROVED] here)**.** *For $n > 0$ and $e \geq 1$ put $M_e := (V/\ell^e)^{\Gamma_0(N)}$. Then $\mathcal{T}_\ell = \varinjlim M_e$, and for $p \nmid N\ell$, $v \in M_e$,*

$$T_p v = (1+p)\,\sigma_p v, \qquad \sigma_a := \rho(\mathrm{diag}(1,a))\big|_{M_e}, \quad \sigma_a\sigma_b = \sigma_{ab}, \quad \sigma_a^2 = a^n \qquad (a \in (\mathbb{Z}/\ell^e)^\times).$$

Proof. The first claim is the coefficient sequence for $0 \to V \to V_{\mathbb{Q}} \to V_{\mathbb{Q}/\mathbb{Z}} \to 0$ and $V_{\mathbb{Q}}^\Gamma = 0$, equivariantly by `[C, lem:bocksteinallell]`'s argument. Let g have integer entries, $\det g = p$, $c(g) \equiv 0 \bmod N$. In $\mathrm{GL}_2(\mathbb{Z}/\ell^e)$ the element $\bar{g}\,\mathrm{diag}(1,p)^{-1}$ has determinant 1 and $c \equiv 0$ modulo the ℓ-part of N; $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\ell^e)$ is onto, so it is the image of some $\gamma \in \Gamma$. Hence $\rho(g)v = \rho(\gamma)\rho(\mathrm{diag}(1,p))v = \sigma_p v$, $\sigma_p v$ being Γ-invariant because $\mathrm{diag}(1,p)$ normalises the image of Γ. Summing over the $1+p$ cosets of $\Gamma\,\mathrm{diag}(1,p)\Gamma$ gives the formula. Multiplicativity is clear. Since $\mathrm{diag}(a,1) = \mathrm{diag}(a,a^{-1})\mathrm{diag}(1,a)$ and $\mathrm{diag}(a,a^{-1})$ is in the image of Γ, $\rho(\mathrm{diag}(a,1))$ acts as σ_a on M_e; applying this to $\sigma_a v$ gives $\sigma_a^2 = \rho(aI) = a^n$. □

At level 1, mod ℓ, this is `[C, prop:heckeH0]`(1). *Example*: the vector x^n lies in M_e iff $a^n \equiv 1 \pmod{\ell^e}$ for all units a and $\binom{n}{j}c^j \equiv 0 \pmod{\ell^e}$ for $c \equiv 0 \bmod N$, $1 \leq j \leq n$ (with [C]'s convention $\gamma \cdot x = ax + cy$); then $\sigma_a x^n = x^n$ and T_p is exactly $1+p$ on the cyclic subgroup $\langle x^n \rangle \subseteq \mathcal{T}_\ell$ of order ℓ^e. At $\ell \mid N$, ℓ odd, the largest such e is $v_\ell(n)+1$ (for $\ell - 1 \mid n$) and at $\ell = 2$ it is $v_2(n)+2$ (n even) except that the term $j = 1$ of the binomial condition intervenes, exactly when $\ell = 2$ and $2 \,\|\, N$, where it lowers the bound to $v_2(n)+1$: $e = 4$ at $(8,4)$ and $(16,4)$, 3 at $(8,10)$ and $(16,2)$, 1 at $(27,2)$, $(27,4)$ ([PROVED]; the table below shows it is the full exponent of $\mathcal{T}_\ell$ on the census cells, [CMPD]).

The operators σ_a need not be trivial, and T_p does not notice. At $(9,4)$, $\ell = 3$, the space M_1 contains besides x^4 the vector $xy^3 - x^3y$, and $\sigma_a(xy^3 - x^3y) = \omega(a)(xy^3 - x^3y)$ with ω the quadratic character modulo 3. On it $T_p = (1+p)\,\omega(p)$, which is $\equiv 1 + p \pmod 3$ for every p, because $1 + p \equiv 0$ when $\omega(p) = -1$; so the flag `scal=True` of `nonsplit_census.out` cannot detect it. Computing M_e and σ_a directly from Sym^n, independently of the descent (mode `sigma`, the `SIGMA` rows of `nonsplit_census.out`, whose orders agree with the torsion of the record on all 15 lines computed), one finds $\sigma_a = 1$ on all of $\mathcal{T}_\ell$ at the $\ell = 2$ lines $(8,4)$, $(8,10)$, $(16,2)$, $(16,4)$, $(32,2)$,

$(4,4)$, $(1,6)$ and at $(27,2)$, $(9,2)$, $(25,4)$; but $\mathcal{T}_3 = \mathbb{Z}/3(1) \oplus \mathbb{Z}/3(\omega)$ at $(27,4)$, $(9,4)$, $(3,4)$, and $\mathcal{T}_3 = (\mathbb{Z}/3 \oplus \mathbb{Z}/9)(1) \oplus \mathbb{Z}/3(\omega)$ at $(9,6)$, $\mathcal{T}_3 = (\mathbb{Z}/3)^2(1) \oplus (\mathbb{Z}/3)^2(\omega)$ at $(9,10)$ ([CMPD]). Here (ψ) means that σ_a acts by $\psi(a)$ on the summand; §5 returns to this.

3. The census: where ε is non-split

HeckeZ edition 1 (`docs/integral_census/results/hecke_ed1/torsion_points.tsv`) has 12 lines with `split=non_split`, one for each of the anemic and the full algebra of the six cells below; at $\ell \mid N$ and none at level 1. The anemic algebra is reduced and its records are *final*, except $(8,10)$, where two fields wait for an exact export word (`refused(export_word)`, EXPORT-Z). The full algebra, which adds the U_p at $p \mid N$, is non-reduced (`redQ` = `non_reduced_certified`) and its records were written *pending* in edition 1, before package Z2 (U_p over $\mathbb{Z}_{(\ell)}$); with the binary `cf93a542`, which has Z2, the six full records read `status=final`, generation `open(sandwich)` at each prime dividing N with $\ell \mid N$ and C_T `not_run` (`C1=refused(non_reduced)`), and the edition's rows are not re-issued. The verdict `non_split` never waited: it uses only T_p with $p \nmid N$, and it is the same on the anemic and on the full line of each cell. The columns Eis_1, Eis_χ, cusp repeat the test on the saturated $\mathbb{T}'$-stable sublattices of L: the Eisenstein classes of trivial character, the twisted ones (eigenvalues $\chi(p)(1 + p^{n+1})$, χ quadratic of conductor 4 or 3, $\mathrm{cond}^2 \mid N$), and the cuspidal part; $S := \mathcal{T}_\ell \cap \sum_p (T_p - 1 - p)H$ is explained in §4. ([CMPD] exact over $\mathbb{Z}$: `docs/latex/tools/pg_nonsplit_tors.py <binary> census` and `tbt`; output `phenomena_gallery/nonsplit_census.out`. The verdict on L reproduces the record on every line. The columns other than L are ours: the record states no verdict for them, and every entry of the table other than "non-split" is $\mathrm{split}_{\mathcal{P}}$ in the sense of §1. Eis_χ is the Eisenstein part with χ of conductor 4 at $N = 16$ and of conductor 3 at $N = 9, 27$.)

(N,n,ℓ)	$\mathcal{T}_\ell$	$\mathrm{rk}\, L$	L	Eis_1	Eis_χ	cusp	$\lvert S\rvert$
$(8,4,2)$	$\mathbb{Z}/2, \mathbb{Z}/4, \mathbb{Z}/16$	10	non-split	$\mathrm{split}_{\mathcal{P}}$	–	non-split	2
$(8,10,2)$	$(\mathbb{Z}/2)^3, \mathbb{Z}/4, (\mathbb{Z}/8)^2$	22	non-split	$\mathrm{split}_{\mathcal{P}}$	–	non-split	2
$(16,2,2)$	$\mathbb{Z}/2, \mathbb{Z}/8$	12	non-split	$\mathrm{split}_{\mathcal{P}}$	non-split	non-split	2
$(16,4,2)$	$\mathbb{Z}/2, \mathbb{Z}/4, \mathbb{Z}/16$	20	non-split	$\mathrm{split}_{\mathcal{P}}$	non-split	non-split	4
$(27,2,3)$	$\mathbb{Z}/3$	18	non-split	$\mathrm{split}_{\mathcal{P}}$	non-split	non-split	3
$(27,4,3)$	$(\mathbb{Z}/3)^2$	30	non-split	$\mathrm{split}_{\mathcal{P}}$	non-split	non-split	3

Beyond the census. The family of mode `scan` consists of the prime-power levels and level 36, $N \in \{2,3,4,5,7,8,9,16,25,27,32,36,49,64\}$ with $n \le 12$ even, and the composite levels $N \in \{12,18,20,24,28,40,45,48,50,63\}$ with $n \le 6$ even, in both cases subject to the console's bound $\psi(N)(n+1) \le 300$: 78 cells, by no means all cells under that bound (`phenomena_gallery/nonsplit_family.out`). It has 83 lines with $\ell \mid N$. On them the verdict is non-split *on the* 35 *lines* with $\ell = 2$, $8 \mid N$, or ℓ odd, $\ell^2 \mid N$, $(\ell - 1) \mid n$, and on the other 48 lines the test on $\mathcal{P}$ finds a complement, $\mathrm{split}_{\mathcal{P}}$ ([CMPD]; mode `law`, `LAW set=family`, 0 violations). This is an *empirical law* in this range, [EMP]: $(N,\ell) = (4,2), (12,2), (20,2), (28,2), (3,3), (5,5), (7,7)$ and $(N,n,\ell) = (25,6,5)$ read $\mathrm{split}_{\mathcal{P}}$. The clause $(\ell - 1) \mid n$ rests on the single line $(25,6,5)$ (`scal=False`; the next test, $(49,6)$, is refused by the console's bound). Five further cells, not in the family and within the bound, confirm the law ([CMPD], `XCELL` and `XKZERO` rows of `nonsplit_census.out`, `LAW set=extra`, 0 violations): $(56,2)$ at $\ell = 2$ (non-split, $K_0 = 2$ in the sense of §7), $(8,14)$ (non-split, $K_0 = 2$), $(9,14)$ at $\ell = 3$ (non-split, $K_0 = 1$), $(12,8)$ at 2 and 3 and $(5,16)$ at 5 ($\mathrm{split}_{\mathcal{P}}$). On every line of the family Eis_1 reads $\mathrm{split}_{\mathcal{P}}$; at $N = 64$, and at $N = 49$ on the line $\ell = 3$ (where $p_0 = 2$), the computed lattice Eis_1 contains twisted classes as well (a quadratic χ with $\chi(p_0) = 1$ at the prime p_0 used to cut it); on the $\ell = 2$ lines of $N = 49$ the cut at $p_0 = 3$ is the trivial-character part, so the verdict there is the one for the larger lattice, and it passes to the trivial-character part by §1 ([PROVED]). The smallest non-split line is $(8,2,2)$, where L is non-split while both Eis_1 and the cuspidal lattice read $\mathrm{split}_{\mathcal{P}}$: the non-splitting there lives in how L_{Eis} and L_{cusp} are glued inside L, not on either of them ([CMPD]).

Note the pattern of the table: *the trivial-character Eisenstein lattice reads* $\mathrm{split}_{\mathcal{P}}$ *on every line*, and the twisted one is non-split on every line of the table where such a χ exists; Eis_χ is absent at

$N = 8$ (no conductor 4 or 3 character with square dividing 8) and those are the cells where the whole Eisenstein part is split$_{\mathcal{P}}$. (At $(8, 2)$ this is not enough, as just said.)

4. A canonical certificate: $S = \mathcal{T}_\ell \cap \mathbb{I}H$

Assume that every T_p, $p \nmid \ell N$, acts on $\mathcal{T}_\ell$ by the scalar $1 + p$; `scal=True` of `nonsplit_census.out` records this, on every census line (it fails on some other cells, e.g. $(9, 12)$, $\ell = 3$, and at level one for $\ell \geq 5$, and we then have no S below). Let $\mathbb{I}$ be the ideal generated by the $T_p - 1 - p$. It kills $\mathcal{T}_\ell$, so $\mathbb{I}H \subseteq H$ maps onto $\mathbb{I}L$ with kernel $S := \mathcal{T}_\ell \cap \mathbb{I}H$, a subgroup of $\mathcal{T}_\ell$, canonical once the scalar hypothesis holds. The tool computes it with the primes of $\mathcal{P}$; that is a subgroup $S_{\mathcal{P}} \subseteq S$, and so a non-zero value is still a certificate.

Proposition ([PROVED] here)**.** *If ε splits for $\mathbb{T}'$ then $S = 0$.*

Proof. If $H = \mathcal{T}_\ell \oplus L'$ with L' stable, then $\mathbb{I}H = \mathbb{I}L' \subseteq L'$, so $\mathbb{I}H \cap \mathcal{T}_\ell = 0$. □

So a non-zero S is a *witness*: an explicit combination $\sum_p (T_p - 1 - p)x_p$ of classes that lands in the torsion and is not zero. On the census lines $S \neq 0$ exactly where the verdict is non-split, on $\mathcal{P}$ ([CMPD]; an exact form with S and purity is §7). On the six census lines S is cyclic (the field `S_inv`: $\mathbb{Z}/2$, $\mathbb{Z}/2$, $\mathbb{Z}/2$, $\mathbb{Z}/4$, $\mathbb{Z}/3$, $\mathbb{Z}/3$); in the family it is not always so, e.g. $(\mathbb{Z}/3)^2$ at $(9, 10, 3)$ and at the extra cell $(9, 14, 3)$. In the exported generators, one cyclic summand of $\mathcal{T}_\ell$ carries the whole class ε ([CMPD]). That this summand is the one generated by x^n is what the exponents of §2 suggest ([SPEC]: the export's torsion generator is not compared with x^n).

5. The Galois structure, and Eichler–Shimura

Etale cohomology. By Artin's comparison theorem (SGA 4, XI; cited from memory, not re-read for this section) applied to a finite-level cover with Hochschild–Serre, $H_\ell \cong H^1_{\text{et}}(Y_0(N)_{\bar{\mathbb{Q}}}; \mathrm{Sym}^n R^1\pi_*\mathbb{Z}_\ell)$ as $\mathbb{Z}_\ell[T_p]$-modules, on the Deligne–Mumford stack $Y_0(N)$, so that $G_{\mathbb{Q}}$ acts on H_ℓ, with $\mathcal{T}_\ell$ and L_ℓ stable and with $G_{\mathbb{Q}}$ commuting with every T_p, $p \nmid N\ell$. Everything below uses this comparison.

Proposition ([COND] comparison)**.** *$G_{\mathbb{Q}}$ acts on $\mathcal{T}_\ell$ through the cyclotomic character: the geometric Frobenius at $p \nmid N\ell$ acts as σ_p.*

Proof. On $M_e = H^0(\bar{Y}; \mathrm{Sym}^n R^1\pi_*\mathbb{Z}/\ell^e)$, $G_{\mathbb{Q}}$ acts through $\pi_1(Y_{\mathbb{Q}})$ modulo the geometric part. Its image in $\mathrm{GL}_2(\mathbb{Z}/\ell^e)$ is the Borel-type group $\{c \equiv 0 \bmod \ell^{v_\ell(N)}\}$ with all determinants, and the geometric image is its determinant-one subgroup. Two elements of the same determinant differ by the geometric image, which fixes M_e; and σ_p is the action of one with determinant p. □

The torsion is therefore Galois-theoretically simple: it is unramified outside ℓ, and $\sigma_p^2 = p^n$. It need not be trivial: by §2 the action on $\mathcal{T}_3$ at $(27, 4)$ is $1 \oplus \omega$. The essential input for the extension is the integral Eichler–Shimura relation.

Hypothesis (ES$_{\mathbb{Z}}$). On H_ℓ, for $p \nmid N\ell$, with F_p the geometric Frobenius: $T_p = F_p + p^{n+1}F_p^{-1}$.
[LIT] Deligne proves this for $\mathbb{Z}_\ell$-coefficients on the interior cohomology of level-n moduli schemes ([Del71, Prop. 4.3, 4.4(iii), 4.8]: the correspondence T_p on the normalisation is $F + I_p^*V$, with F the geometric Frobenius and $FV = p^{n+1}$; the diamond operator I_p^* is 1 on the trivial-character part). The extension to the open curve and to the stack needs only the restriction $H^1(Y_{\bar{\mathbb{F}}_p}, \mathcal{F}) \hookrightarrow H^1(U_{\bar{\mathbb{F}}_p}, \mathcal{F})$ to the ordinary locus (injective by Gysin, since a point has no H^1 of local cohomology) where the correspondence splits as a disjoint union, the argument being Deligne's on an open curve with $\ell \neq p$ (a tame sheaf on a smooth curve). We flag this extension as *sketched, not a citation of a literal statement*; the integral form on open cohomology is hypothesis ES$_{\mathbb{Z}}$ here ([COND] below).

Check on the torsion: there $F_p = \sigma_p$, and $\sigma_p + p^{n+1}\sigma_p^{-1} = \sigma_p + p\,\sigma_p = (1+p)\sigma_p = T_p$ by $\sigma_p^2 = p^n$. ([PROVED] this consistency; it does not test the sign: on a summand where $\sigma_p = \pm 1$ one has $p^n \equiv 1$ there and the identity holds for σ_p^{-1} as well; $T_p = F_p + p^{n+1}F_p^{-1}$ is symmetric under $F_p \leftrightarrow p^{n+1}F_p^{-1}$, and the sign enters below only through the weight hypothesis of (1). The other common normalisation $T_p = \sum \rho(g^*)$, $g^* = p\,g^{-1}$, gives the same formula for T_p, since $p^n\sigma_p^{-1} = \sigma_p$.)

Theorem ([COND] $\mathrm{ES}_{\mathbb{Z}}$)**.** *Assume* $\mathrm{ES}_{\mathbb{Z}}$*. If* ε *is non-split for* $\mathbb{T}'$*, then* ε *is non-split as a sequence of* $\mathbb{Z}_\ell[G_{\mathbb{Q}}]$*-modules. Its class is a non-zero element of* $H^1(G_{\mathbb{Q},N\ell\infty}, \mathrm{Hom}(L_\ell, \mathcal{T}_\ell))$*, with finite coefficients. Moreover, on a line* $L' \subseteq L$ *with* $F_p = \lambda_p$ *scalar, writing* $F_p = \left(\begin{smallmatrix}\sigma_p & b_p \\ 0 & \lambda_p\end{smallmatrix}\right)$ *on* H'*, one has* $T_p = \left(\begin{smallmatrix}(1+p)\sigma_p & m_p \\ 0 & a_p\end{smallmatrix}\right)$ *with*

$$m_p = \left(1 - p^{n+1}\sigma_p^{-1}\lambda_p^{-1}\right) \circ b_p,$$

Proof. A $G_{\mathbb{Q}}$-stable complement is stable under F_p and F_p^{-1}, hence by $\mathrm{ES}_{\mathbb{Z}}$ under T_p: non-split for $\mathbb{T}'$ forces non-split for $G_{\mathbb{Q}}$. The formula is the $(1,2)$ entry of $T_p = F_p + p^{n+1}F_p^{-1}$, with σ_p^{-1} acting on the values of b_p. □

So the Hecke module sees the Galois class only through the factor $\kappa_p := p^{n+1}\sigma_p^{-1}\lambda_p^{-1}$, which depends only on the pair (summand of $\mathcal{T}_\ell$, line): $m_p = (1-\kappa_p)\circ b_p$. On a line with $\lambda_p = \chi(p)\,p^{n+1}$ and a summand on which $\sigma_a = \psi(a)$, ψ of order at most 2 (which covers everything below), $\kappa_p = (\psi\chi)^{-1}(p) = \sigma_p\lambda_p^{-1}p^{n+1}$, and the Hecke algebra is *blind* to b on that pair exactly when $\psi\chi = 1$. The cases are organised by this character.

1. [COND] ($\mathrm{ES}_{\mathbb{Z}}$ and the weight of the Eisenstein part: Frobenius on the Eisenstein part of H^1 is p^{n+1} times a root of unity.) *Trivial character on both sides,* $\psi = 1$, $\chi = 1$: $\kappa_p = 1$, so $m_p = 0$ for every p, whatever b is: the pair is blind. This explains the column $\mathrm{Eis}_1 = \mathrm{split}_{\mathcal{P}}$ *on the summand of* $\mathcal{T}_\ell$ *where* $\sigma = 1$ (all of $\mathcal{T}_\ell$ on the lines of §2 with $\sigma_a = 1$; the x^n-line in general). It says nothing about the Galois class on these lines: there b may be non-zero while $\mathbb{T}'$ splits.

2. [COND] *Twisted line,* $\sigma = 1$ *summand,* $\psi = 1$: $\lambda_p = \chi(p)p^{n+1}$, where the Galois action on the (non-rational) cusps supplies the character χ, so

$$m_p = (1 - \chi(p))\, b_p \qquad (\chi \text{ quadratic}),$$

   and $m_p = 0$ whenever $\chi(p) = 1$ ([CMPD], the `ELINE` rows of `nonsplit_census.out`). (A cusp of denominator $d \mid N$ is defined over $\mathbb{Q}(\zeta_{\gcd(d,N/d)})$ [LIT]; so non-rational cusps occur exactly when some $\gcd(d, N/d) > 2$: $N = 16$ and $N = 9, 27$ in the table, not $N = 8$. Standard; not re-derived here. $N = 25$ also has non-rational cusps and a twisted line that the tool does not isolate; no claim is made about it.)

3. [COND] *The summand where* $\sigma = \omega$, $\ell = 3$ (it exists at $(27,4)$, $(9,4)$, $(9,6)$, $(3,4)$, $(9,10)$). On the trivial-character line $\psi\chi = \omega \neq 1$: $m_p = (1-\omega(p))\,b_p$, *not* blind, so the explanation in case (1) does not apply here. There $\mathrm{Eis}_1 = \mathrm{split}_{\mathcal{P}}$ is a computed fact: $M_p|_{\mathrm{Eis}_1} = 0$ on the whole of $\mathcal{T}_3$ at all 17 primes $p \le 61$ prime to $3N$, at $(27,4)$, $(9,4)$, $(9,6)$ ([CMPD], the `EIS1M` rows). Under $\mathrm{ES}_{\mathbb{Z}}$ this says that the ω-component of b on these lines, a homomorphism of $G_{\mathbb{Q}}$ into $\mathbb{Z}/3$ because $\omega \cdot \chi_{\mathrm{cyc}}^{-(n+1)} \bmod 3$ is trivial, vanishes at the Frobenius of each such p with $\omega(p) = -1$; if it did so at all p, it would vanish, by Chebotarev. On the *twisted* line of those cells, with $\chi = \omega$ (quadratic of conductor 3), the pair (summand ω, line) *is* blind: $\psi\chi = 1$. So m_p on the twisted lines takes values in the $\sigma = 1$ summand of $\mathcal{T}_3$, and this is the summand the verdict Eis_χ non-split speaks about.

4. [CMPD] *The value of* b *on the twisted lines, on the* $\sigma = 1$ *summand.* On $(16,2)$, $(16,4)$, $(9,4)$, $(27,2)$, for the 17 primes $p \le 61$ prime to N (line `ELINE_CHECK`), $m_p = (1-\chi(p))\,t\,\log(p)$ on the one exported generator that carries the class, of order ℓ^f, where log is the discrete logarithm ($\ell = 3$: $\log p$ taken base 2 in $(\mathbb{Z}/9)^\times/\{\pm 1\} \cong \mathbb{Z}/3$; $\ell = 2$: $\log p = k$ for $p \equiv \pm 5^k$), and $t = 2$ at

$\ell = 2$, t a unit at $\ell = 3$ (the scale of the line's generator). As a function of p the right side depends only on p modulo 2^f at $\ell = 2$, modulo 3^{f+1} at $\ell = 3$.[1]

What b is on these lines. $\mathrm{Hom}(L', \mathcal{T}_\ell)$ is not a trivial $G_\mathbb{Q}$-module: on the $\sigma = 1$ summand A, of exponent ℓ^f, the class lies in $H^1(G_{\mathbb{Q},N\ell\infty}, A(\psi_b))$ with $\psi_b(\mathrm{Frob}_p) = \sigma_p \lambda_p^{-1} = \chi(p)\, p^{-(n+1)}$, that is $\psi_b = \chi\, \chi_{\mathrm{cyc}}^{-(n+1)}$ ([COND]). At $\ell = 3$, $f = 1$, this is trivial ($\chi_{\mathrm{cyc}} \bmod 3 = \omega = \chi$ and n is even), so $b \in \mathrm{Hom}(G_{\mathbb{Q},\{3,\infty\}}, \mathbb{Z}/3)$, a group spanned by the logarithm of the first layer of the cyclotomic $\mathbb{Z}_3$-extension (Kronecker–Weber, [LIT]): at $(9,4)$, $(27,2)$ the non-zero b is a unit multiple of it, which is the formula above. At $\ell = 2$, $f \geq 2$, the twist is not trivial: at $(16,2)$ it is $\psi_b(p) \equiv \chi_{-4}(p)\, p \pmod 8$, equal to 5 at $p = 3$. So b is a cocycle, not a homomorphism of $\mathbb{Z}_2^\times$, and there is no statement to make inside $\mathrm{Hom}(\mathbb{Z}_2^\times, \mathbb{Z}/2^f)$. What the data say is weaker: the dependence on p just stated, and at $(16,2)$, where $m_p \in \{0,4\}$: for the primes with $\chi(p) = -1$ the value $m_p/4 \in \mathbb{Z}/2$ is the Kummer character of 2, that is $m_p = 4$ exactly when $p \equiv 3 \pmod 8$ ([CMPD], the `KUMMER2` row, on the 17 primes).

Proposition ([COND] $\mathrm{ES}_\mathbb{Z}$, the weight hypothesis, and the computed verdict Eis_χ = non-split)**.** *At $(16,2)$, $(9,4)$, $(27,2)$ the class b of the twisted Eisenstein lattice is ramified at ℓ.*

Proof. Let A be the cyclic summand carrying the class ($\mathbb{Z}/8$ at $(16,2)$; at $\ell = 3$ the summand $\mathbb{Z}/3$ with $\sigma = 1$, the other summand of $\mathcal{T}_3$ being blind by case (3)), and K the field cut out by ψ_b on A: $K = \mathbb{Q}$ at $\ell = 3$, $K = \mathbb{Q}(\sqrt{2})$ at $(16,2)$ (ψ_b is 1 on ± 1 and 5 on ± 3 modulo 8). Suppose the class of the A-component is unramified at ℓ. Since N is a power of ℓ, the restriction to G_K is a homomorphism to the finite ℓ-group A unramified at every finite place of K, hence zero: K has narrow class number 1 ([LIT]; for $\mathbb{Q}(\sqrt{2})$ the unit $1 + \sqrt{2}$ has norm -1). So the class comes from $\Delta = \mathrm{Gal}(K/\mathbb{Q})$. For $\ell = 3$, $\Delta = 1$ and it is zero; for $(16,2)$, $\Delta = \mathbb{Z}/2$ acts on $\mathbb{Z}/8$ by 5, and $\widehat{H}^1(\Delta, \mathbb{Z}/8(5)) = \ker(1+5)/(5-1)\mathbb{Z}/8 = \{0,4\}/\{0,4\} = 0$. Thus $b_p = (\sigma_p - \lambda_p)c$ for a fixed $c \in A$, and changing the complement by c kills it, so $m_p = 0$ on A. The rest of the torsion contributes nothing: at $(16,2)$ the summand $\mathbb{Z}/2$ has $m_p = (1 - \chi(p))b_p$, even, so 0; at $\ell = 3$ the other summand is blind. So ε would split on the lattice, against the certificate. □

At $(16,4)$ the field K is the real quartic subfield of $\mathbb{Q}(\zeta_{16})$ and the same argument would need Weber's parity theorem with signs; we have not written it out.

On the cuspidal lattice the theorem gives, *on the $\sigma = 1$ summand*, $M_p = b_p \circ (1 + \rho_L(F_p) - T_p|_L)$ (use $p^{n+1}F^{-1} = T - F$ on L; on a summand with $\sigma = \psi$ one has $M_p = b_p - \psi(p)^{-1} b_p \circ (T_p - F_p)$), so again the Hecke class only sees b through this factor. Beyond this we have no closed form for the class on L_{cusp}; on the $\sigma = 1$ summand the residual system is $T_p \equiv 1 + p$, i.e. $\bar{\rho}^{\mathrm{ss}} = 1 \oplus \omega$ ([COND] $\mathrm{ES}_\mathbb{Z}$, with Brauer–Nesbitt on L), and the torsion carries only the direction "$F_p = \sigma_p$", not its companion $p\sigma_p$. Beyond these two facts the cuspidal class is not identified.

What the class is not, and what is unknown. [LIT] The mod-ℓ phenomenon of Harder and Ribet (§6, §7) lives in the reduction of the free lattice and is tied to L-values: 691 is not torsion at $(1,10)$, and these extension classes are not of that type. There is no theorem connecting ε to a Selmer group H^1_f. A Ribet class is unramified at ℓ; the classes on the twisted Eisenstein lines above are *ramified* at ℓ at $(16,2)$, $(9,4)$, $(27,2)$ (the Proposition above, [COND], with its computed input), and they have finite coefficients. No motivic interpretation is known, and we do not imply one ([SPEC]). At $\ell \mid N$ the curve has bad reduction at ℓ, so no local condition at ℓ is forced by what we compute. To be proved: (i) $\mathrm{ES}_\mathbb{Z}$ in the form above on open integral cohomology; (ii) the value of b on the twisted Eisenstein lines, conceptually; (iii) anything about the cuspidal class. [SPEC]: the table suggests that the Eisenstein part of the class is controlled by the Galois action on the non-rational cusps; we have no proof.

[1] Computed only on the listed lines and primes ≤ 61; at $\ell = 2$ the identification of b with a named class is [SPEC] (see below).

6. Torsion by torsion

What should non-split *torsion-by-torsion* mean? As in the torsion-by-free case, an extension that splits as abelian groups but not as Hecke modules; the natural reading is: $\mathcal{T}_\ell$ is a direct sum of $\mathbb{T}$-stable *cyclic* groups, or it is not.

Proposition ([PROVED] here)**.** *Let ℓ be odd and $\mathbb{T}'$ the prime-to-ℓN algebra. Then $\mathbb{T}'$ acts on $\mathcal{T}_\ell$ through scalars on the $\mu_{\ell-1}$-eigenspaces $M_{[\psi]}$: every abelian-group decomposition of $M_{[\psi]}$ is $\mathbb{T}'$-stable. Thus prime-to-ℓN operators produce no torsion-by-torsion non-splitting at odd ℓ.*

Proof. n is even (for odd n, $-I$ acts by -1, so $\mathcal{T}_\ell = 0$ for odd ℓ). For $a \in 1+\ell\mathbb{Z}_\ell$ write $a = b^2$ with $b \in 1+\ell\mathbb{Z}_\ell$; then $\sigma_a = \sigma_b^2 = b^n = a^{n/2}$, a scalar. The prime-to-ℓ part of $\mathbb{Z}_\ell^\times$ acts semisimply, so $\mathcal{T}_\ell$ decomposes over its characters, and on each eigenspace $T_p = (1+p)\sigma_p$ is a scalar. □

For $\ell = 2$ the group $\mathbb{Z}_2^\times$ has a piece (±1, and 5 not a square) the argument does not cover. Of the 136 lines (cell, ℓ) of the census cells, the extra cells $(4,4)$, $(9,4)$, $(9,6)$, $(3,4)$ and level 1, $n \le 60$, the 117 with $|\mathcal{T}_\ell| \le 1024$ were searched exhaustively and the other 19 were not searched (`TBT_SUMMARY`): at each of the 117 the group $\mathcal{T}_\ell$ is a direct sum of $\mathbb{T}'$-stable cyclic groups ([CMPD]). Also at $\ell = 5$, $(1,60)$, where τ_p is not diagonal in the exported generators: it diagonalises by the change of basis $g_1 \mapsto g_1 + 15g_2$.

The answer changes as soon as T_ℓ (or U_ℓ) is in the algebra. *There* torsion-by-torsion non-splitting exists, and it is `[C, prop:heckeH0]`(3),(4) in the integral setting: [CMPD] at level 1, $\ell = 2$, $n = 6$ the torsion is $\mathbb{Z}/2 \oplus \mathbb{Z}/4$ with $T_2 = \left(\begin{smallmatrix} 0 & 0 \\ 2 & 2 \end{smallmatrix}\right)$ (rows modulo the generators' orders) and every T_p, p odd, the scalar $1+p$, and the module is not a direct sum of T_2-stable cyclic groups (exhaustive). Of the 117 lines searched, 15 are non-decomposable for the algebra with T_ℓ (`TBT` rows, `nonsplit_census.out`): $\ell = 2$ at $(1,6)$, $(1,10)$, …, $\ell = 3$ at $(1,24)$, $(1,42)$, and $(9,6)$ at $\ell = 2$; exhaustive search under the bound ([CMPD]). The first $\ell \ge 5$ instance is $(120,5)$, by `[C, prop:heckeH0]`(4), outside the console's limits. T_ℓ is not in the image of $\mathbb{Z}_\ell[G_\mathbb{Q}]$, so Eichler–Shimura does not transfer this kind of non-splitting to Galois; and since $G_\mathbb{Q}$ acts on $\mathcal{T}_\ell$ through the operators σ_a of §2 ($a \in \mathbb{Z}_\ell^\times$), the proof of the Proposition shows that *at odd ℓ the Galois action on $\mathcal{T}_\ell$ is by scalars on each $\mu_{\ell-1}$-eigenspace* (for instance the characters 1 and ω at $(27,4)$): no non-split torsion-by-torsion comes from Galois ([COND] comparison). At $\ell = 2$ this is open.

Between degrees one can also ask about $0 \to H^1(V)/\ell \to H^1(V/\ell) \to H^2(V)[\ell] \to 0$ of Hecke modules (at $\ell \in \{2,3\}$ the two ends are non-zero, `[C, sec:descent]`). The shipped binary does not expose T_p on H^2 or on $H^1(V/\ell)$ with a basis compatible with this sequence; we did not test it.

7. What the computations modulo ℓ see

The extension is a statement about $\mathbb{Z}$-modules. The owner asked whether one can find it from the much cheaper $\mathbb{F}_\ell$-computations of the census. Put $H_K := H \otimes \mathbb{Z}/\ell^K$, $A_K := \mathcal{T}_\ell/\ell^K$ and let $\varepsilon_K \colon 0 \to A_K \to H_K \to L/\ell^K \to 0$, exact because L is free.

Proposition ([PROVED] here; (2) uses `[C, prop:heckeZglue]`(2)(a))**.** (1) *If ε_{K+1} splits for $\mathbb{T}'$ so does ε_K; so non-split modulo ℓ implies non-split over $\mathbb{Z}$.* (2) *For $K \ge e_\ell$, ε_K splits iff ε does: if the exponent of $\mathcal{T}_\ell$ is $e_\ell = 1$, the test modulo ℓ is exact.* (3) Exact test. *Suppose every T_p, $p \nmid \ell N$, acts on A_K by a scalar c_p, and let $I_K := \sum_p (T_p - c_p)H_K$. Then ε_K splits iff $A_K \to H_K/I_K$ is injective with image a direct summand of the group H_K/I_K. For $K = 1$: iff $A_1 \cap I_1 = 0$.*

Proof. (1) A complement reduces. (3) A complement C contains I_K (the T_p act by c_p on $H_K/C \cong A_K$), so $H_K/I_K = A_K \oplus C/I_K$. Conversely, the T_p act on H_K/I_K by the scalars c_p; so the preimage of an abelian-group complement of A_K in H_K/I_K is $\mathbb{T}'$-stable, and it is a complement of A_K in H_K. □

So $\mathbb{F}_\ell$ sees everything at $e_\ell = 1$, and the counterexamples for $e_\ell > 1$ exist. Let K_0 be the least K with ε_K non-split. On the 35 non-split lines of the family of §3 ([CMPD], the `KZERO` rows; all

35 are non-split over $\mathbb{Z}$ by the exact test of `pg_nonsplit_tors.py`, and `hecke_structure` reads `non_split` in `G6` at the cells we also ran through it: $(8,2)$, $(9,2)$, $(9,4)$, $(25,4)$, $(25,8)$, $(32,2)$, $(36,2)$, $(64,2)$), $K_0 = 1$ except where the 2-part of N is exactly 8: $K_0 = 2$ at $(8,4),\dots,(8,12)$, $(24,2)$, $(24,4)$, $(40,2)$, and $K_0 = 3$ at $(8,2)$. *At these levels the class dies modulo* 2: ε_1 reads split$_{\mathcal{P}}$ while ε is non-split, a counterexample to the converse of (1) for $\mathbb{T}'_{\mathcal{P}}$, which is what is computed. The same verdicts are read from the shipped mod-ℓ^K route `D.module(ell, K)`, independently of `module_Z`, at $(8,4)$, $(8,2)$, $(16,2)$, $(27,2)$, $(25,4)$, and the control $(4,4)$ (`MODK` rows). On the lines where the scalar hypothesis holds, $\dim(A_1 \cap I_1) > 0$ exactly where $K_0 = 1$ (on $\mathcal{P}$).

Three warnings about what "modulo ℓ" means. (a) $H^1(\Gamma; V/\ell)$ is larger than H_1: the universal-coefficient sequence $0 \to H/\ell \to H^1(V/\ell) \to H^2(V)[\ell] \to 0$ has a non-zero right term at $\ell = 2$ (e.g. $H^2(\Gamma_0(16); \mathrm{Sym}^2 \mathbb{Z}^2) = (\mathbb{Z}/2)^2$, printed by `descent(16, 2)`); it vanishes for $\ell \geq 5$ and for $\ell = 3$ when $\Gamma_0(N)$ has no element of order 3 (e.g. $9 \mid N$), because then H^2 has no ℓ-part (`[C, prop:badprimesN]`; the examples 9, 16, 27 happen to have no element of order 2 in $\Gamma_0(N)/\pm 1$ either). (b) In these calculations A_1 is the image of the torsion, which a pure Hecke computation over $\mathbb{F}_\ell$ does not mark: without `D.module` one knows only the block. (c) The verb `level_n(N, n, ell)` ($\ell \geq 5$) prints characteristic polynomials: it cannot see any of this.

What a block alone can decide. If $\mathfrak{m}M = 0$ on an $\mathbb{F}_\ell$-block M containing the torsion, then M is a module over the field $\mathbb{T}/\mathfrak{m}$, so the image of A_1 in M is a direct summand and the block splits, for $\mathbb{T}$ and a fortiori for $\mathbb{T}'$; no scalar hypothesis is used. This is the one place in the section where a split is certified, and it is not a `split_on_P`. Supports do the rest: `[C, prop:heckeZsplit]`. (The census's H^0-block is $\mathcal{T}_\ell[\ell]$, while $A_1 = \mathcal{T}_\ell/\ell$. They differ as $\mathbb{T}$-modules in general, but multiplication by ℓ on $\mathcal{T}_\ell$ shows they have the same composition factors, hence the same support, which is all the search below uses.) This gives a *search* of the census, by `pg_nonsplit_tors.py modl`: every row of the signed census 4.0 (main and b1; level 1, $\ell \geq 5$; 448 978 rows, $n \leq 1000$, 46 primes) whose H^1-ideal is matched by an H^0-ideal. There are 3236 such collisions, and all of them have Loewy length 1 ($\mathfrak{m}M = 0$; in 2718 of them L/ℓ is non-zero on the block, the rows with a `sturm` object, field `free_on_block`). Of the 3236, 3080 are certified and 156 are heuristic in the census's sense (`certified=heuristic`, `final=false`, at $\ell = 17, 19, 23, 29, 31$ (53, 51, 38, 11, 3 rows), $n \geq 552$; by `[E, sec:census]`, column rule (d7), their module columns, the Loewy length included, are heuristic). So no level-one torsion point in the census range is non-split *modulo* ℓ: ε_1 splits on the 3236 rows, certified on 3080 ([CMPD], from the census's data; the `MODL_CERT` row). By (2), at the 2952 rows with $\beta(n,\ell) = 1$ (the exponent, [CCS13, Thm. 4.5]) the extension ε itself is split over $\mathbb{Z}$ (2801 certified, 151 heuristic); at the remaining 284 rows (152 cells, $\beta > 1$) only ε_1 is split, and the census does not decide ε. At $\ell \in \{2, 3\}$, level 1, the prime-to-ℓ operators are scalars on the whole of $H^1(\mathbb{F}_\ell)$ `[C, prop:ltwothreescalar]`, so ε_1 splits for $\mathbb{T}'$ at every n ([PROVED]). HeckeZ edition 1, which tests over $\mathbb{Z}$ on $\mathbb{T}'$ to $n \leq 268$, reads `split_on_P` at every level-one point it tests, consistent with this.

For *torsion by torsion* the same census row gives a criterion directly: if the block of $H^0(\Gamma; V/\ell) \cong \mathcal{T}_\ell[\ell]$ is not semisimple for $\mathbb{T}$ (Loewy length ≥ 2), then $\mathcal{T}_\ell$ is not a direct sum of $\mathbb{T}$-stable cyclic groups ([PROVED]: a $\mathbb{T}$-stable cyclic decomposition would restrict to one of $\mathcal{T}_\ell[\ell]$). Of the 3236 H^0-rows at $n > 0$ (the census has 5 more H^0-rows, at $n = 0$, where $H^0 = \mathbb{Z}$ is not torsion), 30 are non-semisimple, at level 1, $\ell \geq 5$: the cells $(n,\ell) = (120,5), (220,5), (240,5), (320,5), (336,7), \dots, (980,5)$ (`MODL_H0` row; the first cells are the $\ell^3 - \ell$ of `[C, prop:heckeH0]`(4)). At $e_\ell = 1$ this reading is exact, in general one-way. None of these n is within reach of the descent over $\mathbb{Z}$ (the console's $n \leq 60$), and T_ℓ does not come from $G_{\mathbb{Q}}$. For the prime-to-ℓN algebra (§6) there is nothing to search: H^0 carries $(1+p)p^a$, semisimple.

In short: the prime-to-ℓN torsion-by-free obstruction is *exactly* visible modulo ℓ when $e_\ell = 1$; modulo ℓ it is sufficient but not necessary (level 8), and it needs the torsion to be marked: the census alone can only exclude it (Loewy length 1), not find it. At level 1 in the census range no torsion point is non-split modulo ℓ (certified on 3080 of the 3236 rows); at 284 rows with $\beta > 1$ the extension over $\mathbb{Z}$ stays undecided. At level N one needs `D.module` at $K = e_\ell$ (the console's limits), the only route that marks the torsion. At level N the law of §3 (with $\ell = 2$ the level must

be divisible by 8; with ℓ odd, by ℓ^2, and $\ell - 1 \mid n$) is the empirical statement to test next, on $\mathcal{P}$; it is not proved, and any proof must explain why $\ell^2 \mid N$ (resp. $8 \mid N$) is what matters.

An open direction for the odd-ℓ half of the law ([SPEC]). For ℓ odd, $\ell^2 \mid N$ is exactly the condition for a cusp with $\ell \mid \gcd(d, N/d)$, hence for the twisted Eisenstein line $E_{n+2}^{\chi_\ell, \chi_\ell}$ with χ_ℓ quadratic of conductor ℓ in L; and $(\ell - 1) \mid n$ is exactly the condition $x^n \in M_1$, the $\sigma = 1$ summand of §2. Under $\mathrm{ES}_{\mathbb{Z}}$ one has $m_p = (1 - \chi_\ell(p))\, b_p$ on that pair, and b restricted to $G_{\mathbb{Q}(\zeta_\ell)}$ is a homomorphism to $\mathbb{Z}/\ell$ unramified outside ℓ, so it is the cyclotomic logarithm up to a unit if it is not zero. The odd-ℓ law would then reduce to the non-vanishing of one coefficient of the Eisenstein cocycle of E^{χ_ℓ, χ_ℓ} on x^n, a Stevens-type computation possibly within reach for $\ell = 3$. At $\ell = 2$ the mechanism must differ ($N = 8$ has only rational cusps and the non-splitting is cuspidal), so "$8 \mid N$" needs another idea.

8. Run it

The snippet `notebook/examples/nonsplit_torsion_16_2.oc`:

```
# nonsplit_torsion_16_2.oc -- a torsion-by-free extension in H^1 that does not split, at (N, n)
↪  = (16, 2).
# Phenomena gallery [PG] sec:pg-nonsplit.  H^1(Gamma_0(16); Sym^2 Z^2) = Z^12 (+) Z/2 (+) Z/8
↪  ([C] thm:descentcert), and T_p = [[tau, M], [0, F]]
# on tors (+) free ([C] prop:torsionhecke).  G6 of the record is the verdict of the split test
↪  ([C] prop:heckeZglue(2)(b)): non_split at 2.
# Read_only verbs only (notebook-safe; descent needs coefficient_ring = "Z").  Cost: under two
↪  seconds, 0.2 GB.
set coefficient_ring = "Z";
R = hecke_structure(16, 2);
print("status:", R.status);
print("signature:", R.signature);
D = descent(16, 2, [3, 5, 7]);
print("H^1 =", D.groups[1]);
Z = D.module_Z();
print("torsion generators:", Z.torsion_generators);
for o in Z.ops {
    print("T_" + str(o.p) + " on the torsion (row i mod d_i), 1 + p =", 1 + o.p, ":", o.tau);
    print("  M (the 2-part of tors against the free basis):", o.M[0].matrix);
}
```

Expected output (generated file `phenomena_gallery/nonsplit_torsion_16_2.out`):

```
status: final
signature: (16,2,full) C1=refused(non_reduced) redQ=non_reduced_certified C2=not_run
↪  gen=[2:open(sandwich)] C_T=not_run fp=not_certified G6=[2.pt[0]=non_split] status=final
H^1 = Z^12 + Z/2 + Z/2^3
torsion generators: [{ell: 2, exponent: 1, d: 2}, {ell: 2, exponent: 3, d: 8}]
T_3 on the torsion (row i mod d_i), 1 + p = 4 : [[0, 0], [0, 4]]
  M (the 2-part of tors against the free basis): [[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [5, 7,
  ↪  7, 0, 7, 7, 4, 1, 5, 0, 2, 2]]
T_5 on the torsion (row i mod d_i), 1 + p = 6 : [[0, 0], [0, 6]]
  M (the 2-part of tors against the free basis): [[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [2, 0,
  ↪  6, 0, 0, 6, 0, 0, 2, 0, 0, 0]]
T_7 on the torsion (row i mod d_i), 1 + p = 8 : [[0, 0], [0, 0]]
  M (the 2-part of tors against the free basis): [[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [2, 6,
  ↪  6, 0, 6, 6, 0, 2, 2, 0, 4, 4]]
```

Cost (wall time, peak RSS; reference box, binary named in `costs.tex`): 0.67 s, 152 MiB.

What to look for.

- `status: final`, `gen=[2:open(sandwich)]`, `C_T=not_run` (this record is non-reduced, `C1=refused(non_reduced)`; the transcript is that of the binary with sha256 prefix `cf93a542`, which has U_2, U_3 over $\mathbb{Z}_{(2)}, \mathbb{Z}_{(3)}$), and `G6=[2.pt[0]=non_split]`, the verdict, does not depend on that;

- T_3, T_5, T_7 act on the $\mathbb{Z}/2$ and $\mathbb{Z}/8$ torsion generators as 0, 4; 0, 6; 0, 0: the scalars $1+p \bmod d_i$;
- the row M of the $\mathbb{Z}/8$ generator is not zero on the free basis (but M alone proves nothing, §1);
- the verdict, the sublattice columns, S and the torsion-by-torsion lines come from `docs/latex/tools/pg_nonsplit_tors.py`, not from this snippet.

A What the console cannot show yet, and reading $\Sigma(691)$

Every phenomenon of this gallery has a `read_only` verb, shown once in §22–24; the one exception is the $\Sigma(691)$ atlas store, which is read outside the console (next paragraph).

phenomenon	verb or reader
level N, old forms	`level_n(N, n, ell)`, §18
nebentypus characters	`nebentypus(N, m, n, ell)` (m = Conrey label), §19
p-adic L-values, the measure congruence [E, sec:lvalue]	`lfunction`
the Λ-adic Hecke algebra [U, sec:u-lambda]	`hida_lambda(ell, k, W)`, §24
Hida towers	`hida_tower`, §23
Kummer generators [E, sec:lvnf]	`kummer_unit`, §8
the forms side	`forms_algebra(k, ell, primes, terms)`, §4
a matrix power	`pow`, §22
the $\Sigma(691)$ atlas store	not a verb: the notebook's atlas channel, or `hacs -atlas -query <store.json> '<request>'` (below)

Reading $\Sigma(691)$. A path from the user is file I/O, so no `read_only` verb opens a store and `-serve` does not grow one. Two readers exist, both on the same request handler (`info`, `cells`, `cell`, `system`, `systems`; the store is opened read-only, its FNV-1a-64 is in every reply, and the path comes from the command line only): the notebook's atlas channel (`-serve-atlas`, protocol `oc-serve-atlas/1`, behind the gateway), and the one-shot

```
hacs --atlas --query <atlas dir>/ell691/hecke_atlas_ell691.json \
    '{"op":"cell","id":"c1","n":690}'
```

which prints one `atlas_reply` frame on standard output (exit 0 for `ok`, 3 for an `error` or `rejected` reply, 1 for a usage error or a refused store) and is built only with `OC_BUILD_NOTEBOOK=ON`. On the real store of $\Sigma(691)$ (FNV `0x73c7210047a81b97`) a query takes about a second and 0.5 GB, dominated by the open (the scan of the 452 MB file); several queries should share one `-serve-atlas` session. The reply of `cell` with `n = 690` names 13 ideals and the dimension 115 of H^1.

A per-block eigenvalue reader over $\mathbb{Z}/\ell^K$ (the ordinary characteristic polynomial of §21 in one call) is still a candidate; `hida_tower` gives it for level one.

B Version history

- Version 1.5.5 (slot 26, 2026-11-12): §30 gains a subsection on the certified HeckeZ record at $(N, n) = (1, 10)$ (the congruence module $C_T = \mathbb{Z}/691$ over $\mathbb{Z}$, $T^{\mathrm{fr}} = \mathbb{Z} \times_{\mathbb{Z}/691} \mathbb{Z}$, `[C, prop:heckeZcong]`) with its level-11 companion, and a new snippet `ramanujan_691_over_Z` (twenty-two snippets). Documentation only; no earlier statement changes.
- Version 1.5.4 (2026-11-09, release audit fixes): the program is called `hacs` in the snippets (`cohomology_app` remains a link to it); the build instructions are cited as `BUILDING.md`; a source report whose path named a machine is described instead, and a cluster kit's path is dropped. No change of content.

- Version 1.5.3 (slot 26, 2026-11-09): machine names and private paths removed for the public release (owner ruling 2026-11-09); no change of content.
- Version 1.5.2 (slot 26, PG-NONSPLIT, 2026-11-09): referee (1.5.0) and confirm-read (1.5.1, `docs/implementation/slot26/PG_nonsplit_confirm.md`) passed, with two one-sentence repairs applied and nothing else changed: the sign check after hypothesis $\mathrm{ES}_{\mathbb{Z}}$ no longer claims to test the sign ($F_p = \sigma_p^{-1}$ satisfies the identity too), and the twisted classes in the computed Eis_1 occur at $N = 64$ and at $N = 49$, $\ell = 3$ only. The status text about the non-reduced full records, written before package Z2 merged, now agrees with the binary `cf93a542` (status `final`, generation `open(sandwich)`, $\mathrm{C_T}$ `not_run`); snippet `nonsplit_torsion_16_2` regenerated; no mathematics changed.
- Version 1.5.1 (slot 26, PG-NONSPLIT referee fixes, 2026-11-09): §31 revised on the report of `docs/implementation/slot26/PG_nonsplit_referee.md`: computed splits are `split_on_P` on the printed $\mathcal{P}$; the operators σ_a on $\mathcal{T}_\ell$ are computed and need not be trivial, the Eisenstein discussion is organised by the character $\psi\chi$; the coefficients of the class on the twisted lines and a proof of ramification under $\mathrm{ES}_{\mathbb{Z}}$; corrected counts (census certification, H^0-rows at $n > 0$, searched torsion-by-torsion lines, S not always cyclic). The tool gains the modes `sigma`, `extra`, `law` and the field `S_inv`.
- Version 1.5.0 (slot 26, PG-NONSPLIT, 2026-11-08; owner request; not yet refereed): new §31 (torsion against the free part, and torsion by torsion, over $\mathbb{Z}$, with a Galois reading conditional on an integral Eichler–Shimura hypothesis stated there, and the question of what the computations modulo ℓ see, with the exact test and the census search of its §7), new snippet `nonsplit_torsion_16_2` and tool `docs/latex/tools/pg_nonsplit_tors.py`. Regenerated outputs include the two snippet edits of da84747c (`descent_congruence_primes`, `hida_control_13`: comment lines only).
- Version 1.4.3 (slot 25, 2026-11-06; LaTeX polish stage 1): uniform author block; the abstract no longer quotes figures that change with every edition. Removed verbatim from the abstract, item III: "and what census edition 4.0, 448 978 local Hecke algebras, found;" and, from its last sentence, "(2026-10-19)" ("where a literature search (2026-10-19) found a source, it is cited"). Former author line: "The orbifold-cohomology-of-GL_2 project". The figures remain in §2. No number changed.
- Version 1.4.2 (slot 25, 2026-11-06): two parenthetical notes, in §15 and §16: the $\ell = 59$ candidate is verified (`docs/implementation/slot25/R59_verification.md`) and [E] 1.25.0-pre.1 restates (R) with "of one weight" for "not dihedral", by owner ruling of 2026-11-06. No number changed.
- Version 1.4.1 (slot 25, PG141, 2026-11-06; referee `docs/implementation/slot25/PG140_referee.md`, R1–R6). R1: the rows at $\ell = 59$ and 139 are no longer "not checked": 139 has a congruence in weight 138 ((R) silent), 59 is congruence-free and not dihedral, a candidate counterexample to [E, pr:latticesplit] (R) pending the owner (four-prime evidence). R2: the algebra $\mathbb{F}_q[x,y]/(x^3, xy, y^2)$ named. R3: the module route reads A, not A_1, on a torsion block. R4: the hidden flag cited to [LG, def:hiddencong], a notation concordance. R5: the 459 `not_stable` rows. R6: `pg_census_check.py` reads the printed literals from this file. No number changed.
- Version 1.4.0 (slot 25, PG140, 2026-11-06; owner request of 2026-11-06; report `docs/implementation/slot25/PG140_report.md`). *Notation:* the full localisation $\mathbb{L}_{\mathfrak{m}} = H^1(\Gamma; V_n \otimes \mathbb{Z}_\ell)_{\mathfrak{m}}$ (torsion included) and its free quotient $\Lambda_{\mathfrak{m}}^{\mathrm{fr}}$ are named apart (§2); 1.3.2 wrote $\Lambda_{\mathfrak{m}}$ for the free quotient in §2 and for the localisation without "/tors" in §9. Each statistic is assigned to its object as the code computes it: A on the full mod-ℓ block, the module route on the full $\mathbb{L}_{\mathfrak{m}}/\ell^K$ (free part separately), saturation and `hidden_congruence` on the free quotient of torsion-free blocks only. The lookfor of §9 misread `torsion_stable` (a pure-torsion block,

not one with a stable free part). *Census:* edition 3.4 replaced by the signed edition 4.0 (main + b1), with b2 and the L23 section marked as pre-release addenda and the integral census edition 1 listed; every census number re-derived from the rows files by the new script `docs/latex/tools/pg_census_check.py`, which also checks the shape table of §15 (now six algebras: the $(1,2,1)$ rows at $n > 700$ break "curvilinear or $\mathfrak{m}^2 = 0$"); new Loewy-length-≥ 3 rows at $\ell = 59$, 67, 139; the saturation defect to $\delta = 190$, and $\delta = 182$ final at $(700, 7)$; the ordinary rows re-joined to the records; the DEEP-1 counts as distinct cells (2158, 128) rather than computations (2575, 146). The section label `sec:pg-census34` is kept (alias `sec:pg-census40`). No snippet changed; all twenty outputs regenerated on the slot-25 binary are byte-identical (only `costs.tex` moved).

- Version 1.3.2 (slot 24, 2026-10-30). The $(4374, 3)$ cell is confirmed: the `-sumrules` cross-check resolved and gives `max_tors=7` (report `docs/implementation/slot24/CELL3_4374_report.md`); the P4 marker is removed.

- Version 1.3.1 (slot 24, DOCS-FIX, 2026-10-29; referee `docs/implementation/slot24/DOCS_referee_1029.md`, P1–P9). At $\ell = 2, 3$ the seven-vertex model computes Bredon cohomology for Γ (`[C, prop:modeltorsion]`); the L23 data are group cohomology on C_I (P1). The Lean status of the $\ell = 2, 3$ dimensions is stated as in `[C, rmk:lvdimsltwothree]`: rank formulas for every n, values cell by cell (P2). $n = 0$ excluded from the H^1 count (P3). The $(4374, 3)$ row reads "main reader only; cross-check pending" until the run's report is committed (P4). §28 restructured as definition, theorem, what [C] proves, data, the exceptional cells (P5); its history is below. The boundary pair is certified by `[C, prop:descentpair]` (P6); the third console limit $\psi(N)(n+1) \leq 300$ is named (P7); §A is a phenomenon→verb table (P8); the level-one Bockstein statement cites `[C, prop:badprimes]` (P9). No snippet or generated output changed; every label kept.

  History of §28. Version 1.2.0 stated $e_1 = \beta(n, \ell)$ as an empirical observation; it is Callegaro–Cohen–Salvetti's theorem. Two earlier forms were wrong, both refuted by the data: "$e_1 = v_2(n)+1$" fails at every power of 2 (10/10 up to 2048), and "exponent 1 at $\ell = 3$" fails at every n with $3 \mid n$ (49 of the 146 DEEP-1 cells). The note on the DEEP-1 data stated $e_1 = v_3(n)+1$, the restriction of β to its grid (no $n = 2 \cdot 3^j$); both datasets measure the same invariant and agree on the overlap $n \in \{600, 700\}$. The cells $(1458, 3)$, $(2916, 3)$, $(4374, 3)$ were run in slot 24 (L23S14, catalogue entry PH-01) to check the exception beyond DEEP-1. Version 1.2.0 of §16 named only $\ell = 59$ as the next onset; $\ell = 47$ had been omitted. The table of §A replaced, in slot 22, a list of candidate verbs (`level_n`, `nebentypus(N, chi, n, ell)`, `lvalue`, `hida_lambda(ell, class)`, `hida_tower`, `kummer_unit`, `forms_algebra(k, ell, K)`, `pow`, a store reader); the store reader became the atlas channel, not a verb (owner decision of slot 22).

- Version 1.3.0 (slot 24, DOCS-PG, 2026-10-29). Reorganised by mathematical phenomenon into six parts (owner direction of 2026-10-29); each part shows its snippets, then the run results on the same phenomenon; the former Part II "Phenomena from the runs" is distributed over Parts III–VI, its introduction is §2. Three new snippets: `cohomology_dimensions` (§3), `descent_torsion_beta` (§27) and `descent_congruence_primes` (§30), the integral descent of slot 24. The hand-typed Smith table of §29 is replaced by the generated output; §28 records the slot-24 cells $(1458, 3)$, $(2916, 3)$, $(4374, 3)$. All outputs regenerated (the seventeen old transcripts unchanged). Every label of 1.2.1 is kept.

- Version 1.2.1 (slot 23, phenomenology package, 2026-10-19, the literature verdicts): Part II cites the sources found by the literature searches (docs/phenomenology/literature_search_A.md, _B.md; bibliography phenomenology_refs.bib) and corrects the over-statements: the torsion exponent = Frobenius bound is Callegaro–Cohen–Salvetti's Thm 4.5 (KNOWN), (547,486) is Bellaiche–Pollack's exception, (21929,2) e_2 = 127 is Calegari–Emerton's example, the floor 7 is Rustom's Thm 8.1, (Hi11) rank 3 is Hida control and n = 3460 is the Dickson exception, the dihedral onset list gains ell = 47; cells written (n, ell) throughout. No snippet or generated

output changed.

- Version 1.2.0 (slot 23, phenomenology package, 2026-10-19): Part II "Phenomena from the runs" (§2–20): census 3.4 at a glance, the dihedral Loewy-3 classes, the ordinary Loewy-3 block (547,486), the saturation defect, ell = 2, 3 (L23, DEEP-1), the torsion exponent = Frobenius bound (with the ell = 3 reconciliation), (Hi11) to n = 6910 (second-cluster ACL2O), Mazur's Eisenstein ideal at p = 2, 3; tag [EMP]; no snippet and no generated output changed. Input: docs/phenomenology/phenomena_catalogue.md.
- Version 1.1.0 (slot 22, VERB-3, 2026-10-09): one snippet per verb added in slots 21–22 (`pow`, `kummer_unit`, `hida_tower`, `level_n`, `nebentypus`, `forms_algebra`, `hida_lambda`), §A rewritten.
- Version 1.0.0 (hecke-extensions slot 20, stretch package S5, 2026-10-04; owner request of 2026-10-04).

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[CCS13] F. Callegaro, F. R. Cohen, and M. Salvetti. The cohomology of the braid group B_3 and of $\mathrm{SL}_2(\mathbb{Z})$ with coefficients in a geometric representation. *Q. J. Math.*, 64(3):847–889, 2013. arXiv:1204.5390. Full arXiv text read (literature search B, 2026-10-19): Thm. 4.5 (= Thm. 3.7(2)), the p-torsion of $H^1(\mathrm{SL}_2(\mathbb{Z}); M_n)$ for every prime p. Journal data from the first author's CV.

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